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Shigematsu

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(45) **Date of Patent:** **Sep. 9, 2025**

(54) **ADAPTER DEVICE, MOUNT APPARATUS,
AND ACCESSORY**

(56) **References Cited**

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(71) Applicant: **CANON KABUSHIKI KAISHA**,
Tokyo (JP)

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(72) Inventor: **Masayasu Shigematsu**, Tokyo (JP)

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(73) Assignee: **Canon Kabushiki Kaisha**, Tokyo (JP)

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“Official statement: Nikon ‘currently developing’ new mirrorless cameras” article by DPReview, and associated comments <https://www.dpreview.com/news/9148147105/official-statement-nikon-currently-developing-new-mirrorless-cameras> (Year: 2018).*
Wikipedia entry for “Lens Mounts” dated Jul. 28, 2018 <https://en.wikipedia.org/w/index.php?title=Lens_mount&oldid=852378432> (Year: 2018).*

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 180 days.

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(22) Filed: **Jun. 1, 2023**

* cited by examiner

(65) **Prior Publication Data**

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Primary Examiner — Leon W Rhodes, Jr.

(74) *Attorney, Agent, or Firm* — Canon U.S.A., Inc. IP Division

Related U.S. Application Data

(60) Division of application No. 17/184,362, filed on Feb. 24, 2021, now Pat. No. 11,698,573, which is a continuation of application No. PCT/JP2018/031995, filed on Aug. 29, 2018.

(51) **Int. Cl.**
G03B 17/14 (2021.01)
G03B 17/56 (2021.01)

(52) **U.S. Cl.**
CPC **G03B 17/56** (2013.01); **G03B 17/14** (2013.01); **G03B 17/565** (2013.01); **G03B 2206/00** (2013.01)

(58) **Field of Classification Search**
CPC G03B 17/14; G03B 17/565
See application file for complete search history.

(57) **ABSTRACT**

An adapter device includes a first mount and a second mount. The first mount is fastened to the adapter device by multiple screws, and has first through third mount recesses and first through third mount claws disposed following a circumferential direction. The second mount is fastened to the device by multiple screws, and has fourth through sixth mount claws and fourth through sixth mount recesses disposed following the direction. The first through third mount claws are insertable to mount recesses on an imaging apparatus, the fourth through sixth mount claws are insertable to mount recesses on an accessory, and the smallest angle range of the first through third mount recesses in the direction of the first mount is not greater than the smallest angle range of the fourth through sixth mount claws in the direction of the second mount.

14 Claims, 26 Drawing Sheets

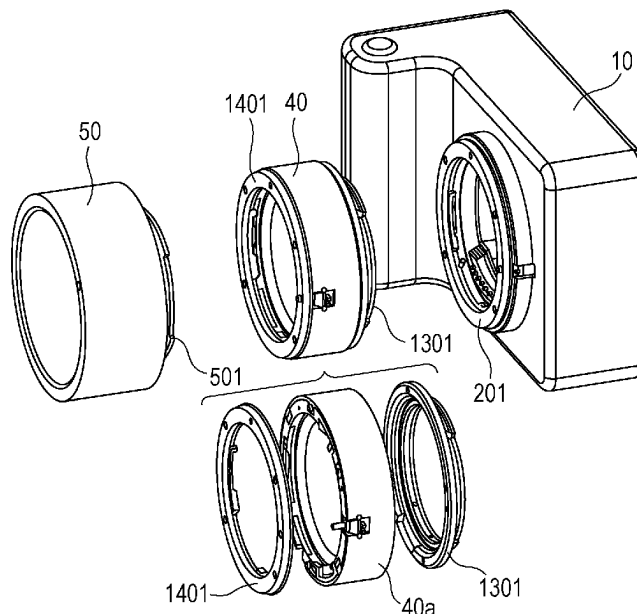


FIG. 1A

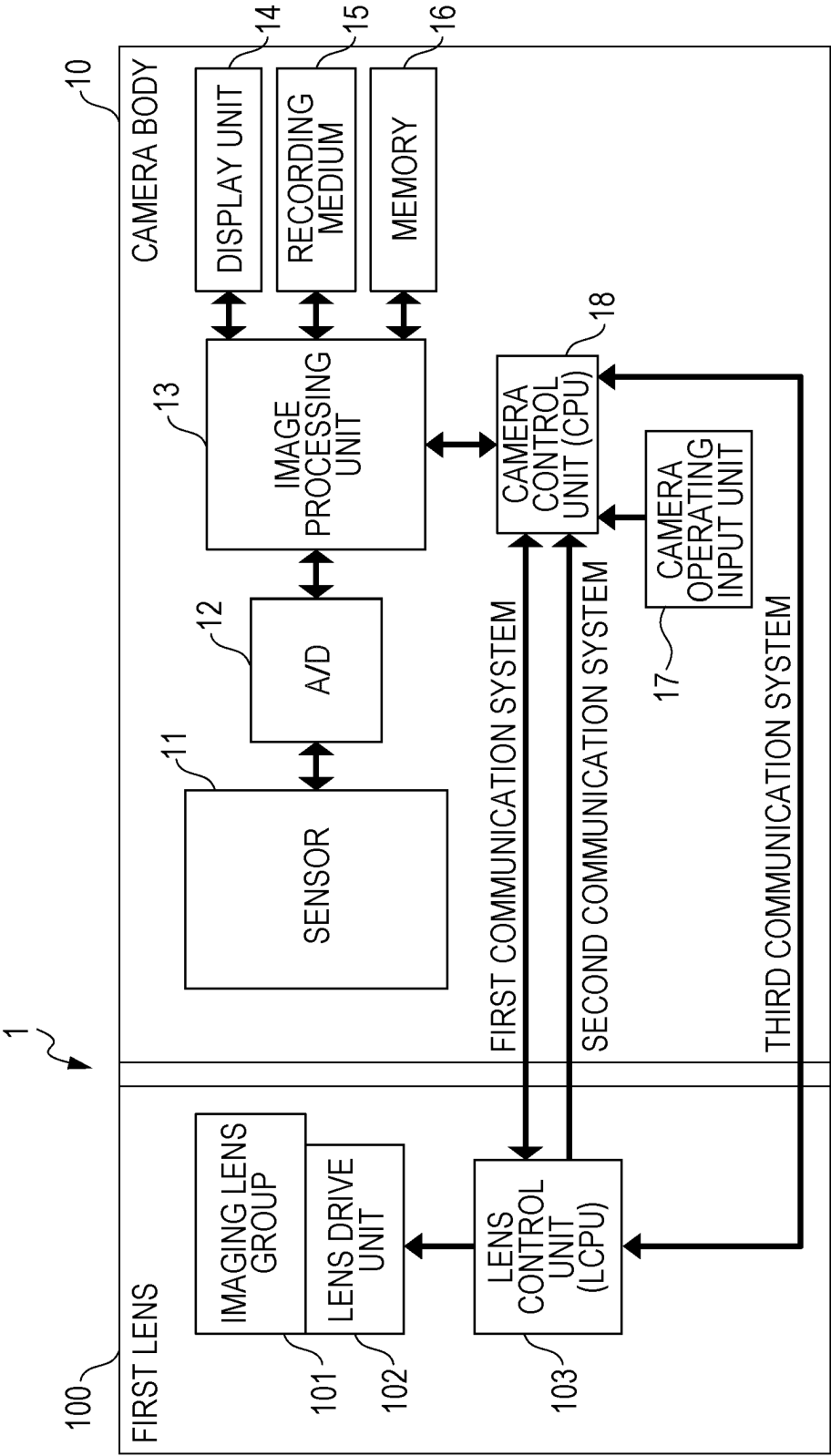


FIG. 1B

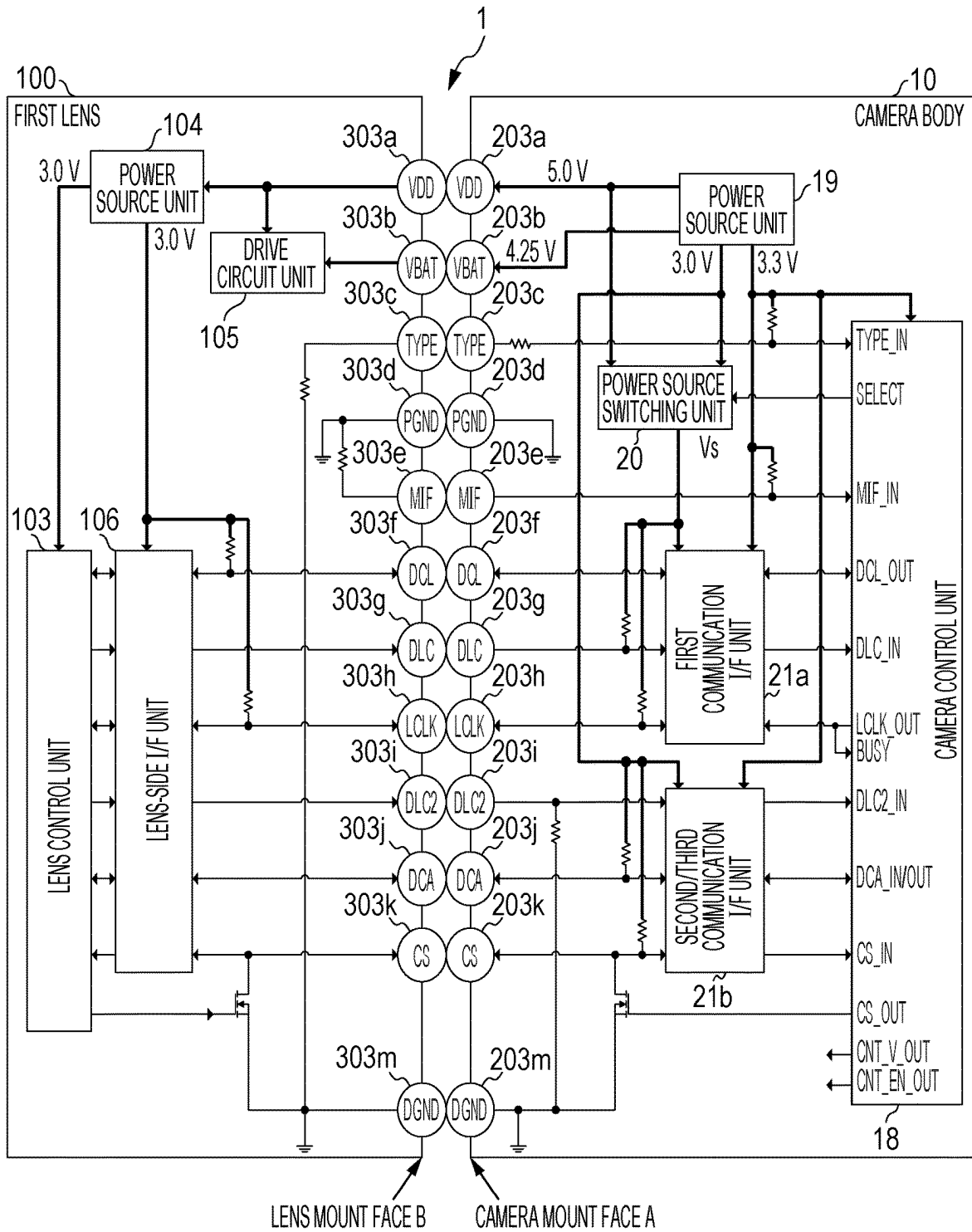


FIG. 2A

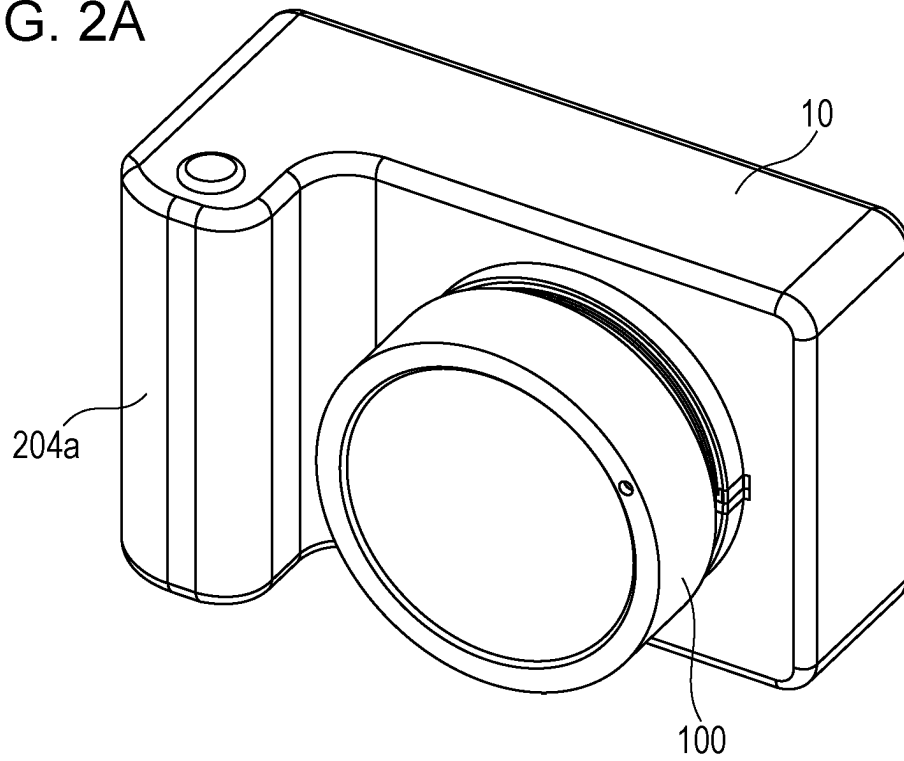


FIG. 2B

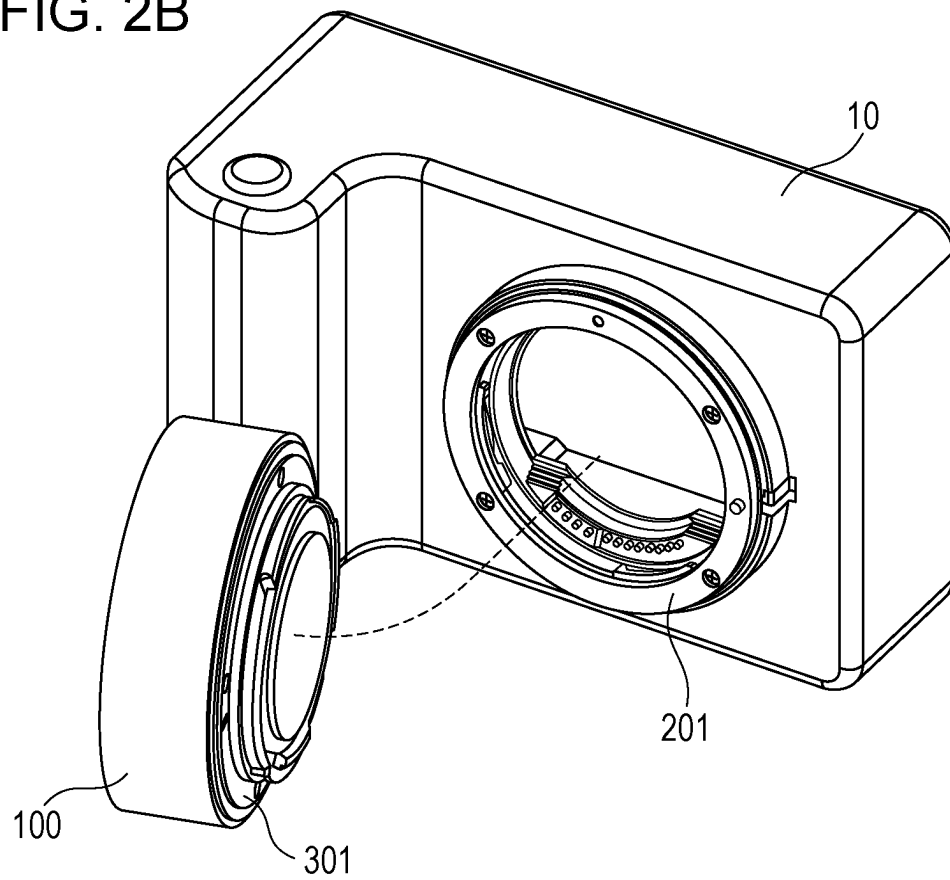


FIG. 3A

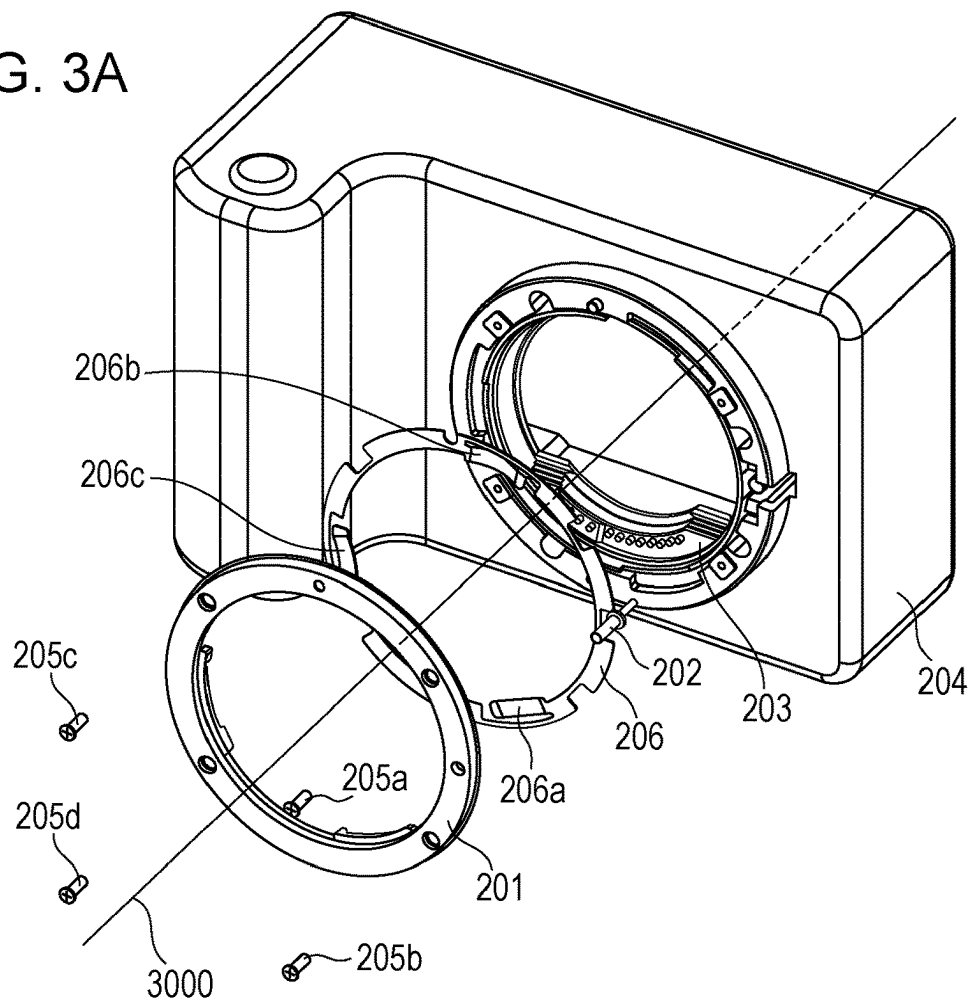


FIG. 3B

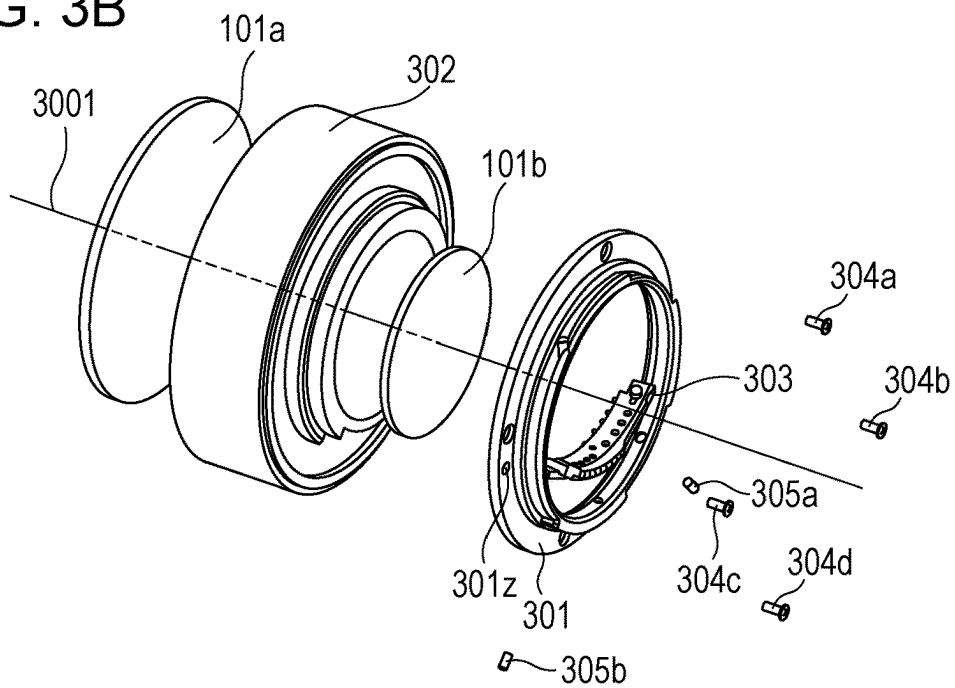


FIG. 4A

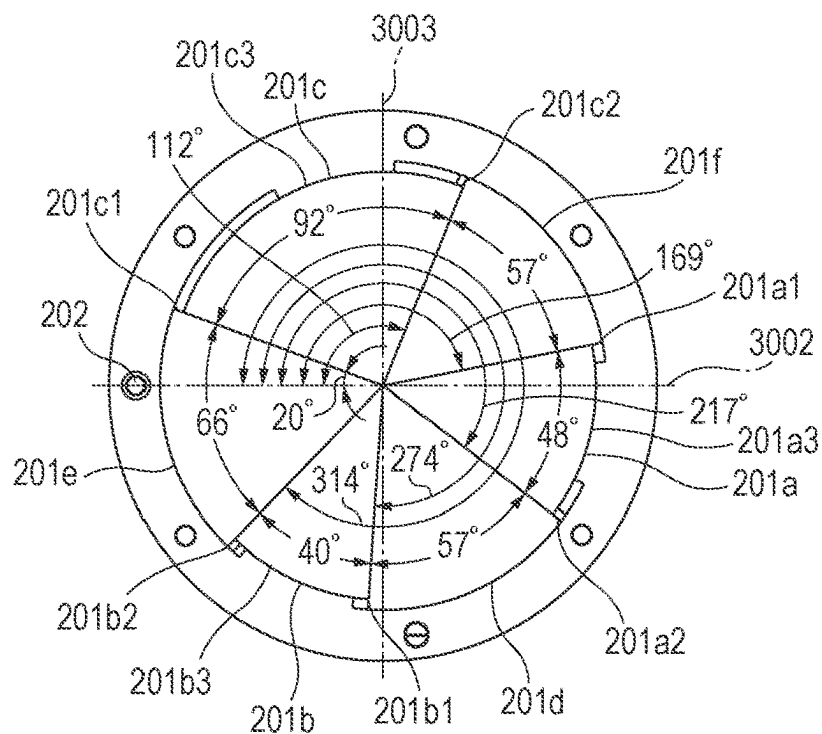


FIG. 4B

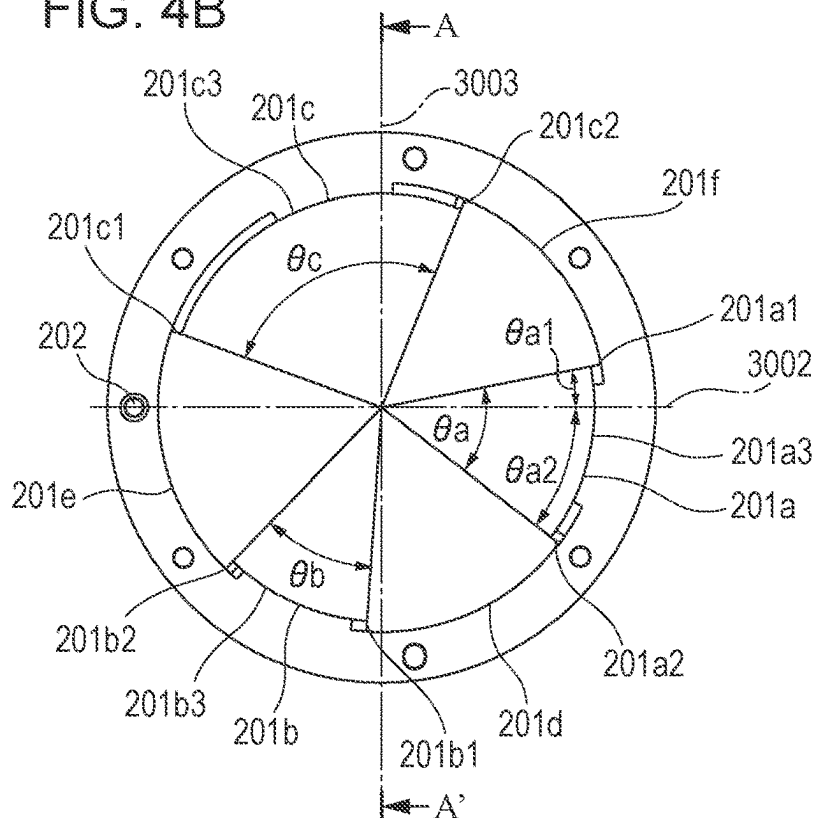


FIG. 4C

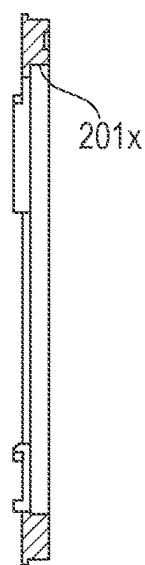


FIG. 5A

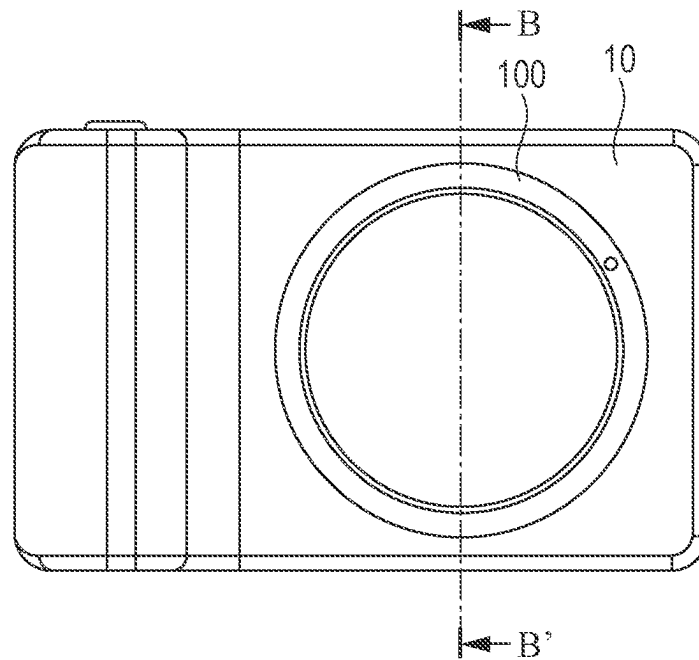


FIG. 5B

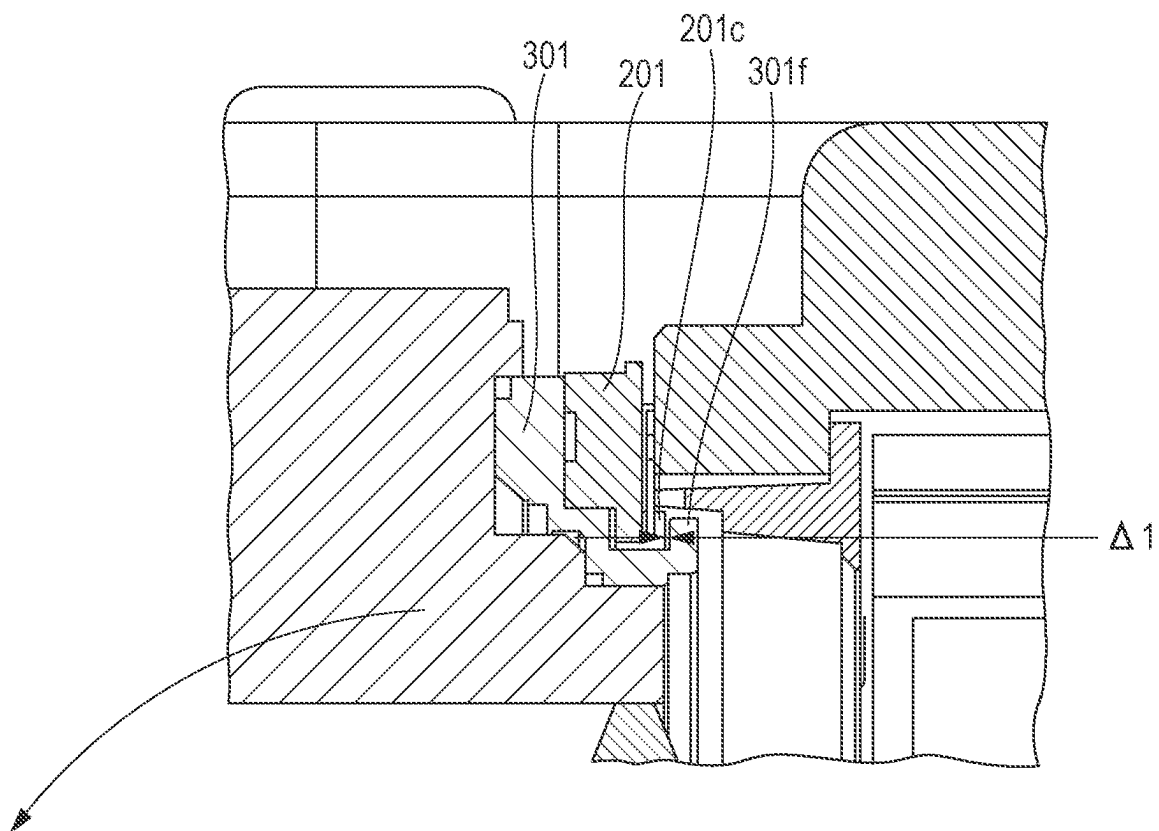


FIG. 6A

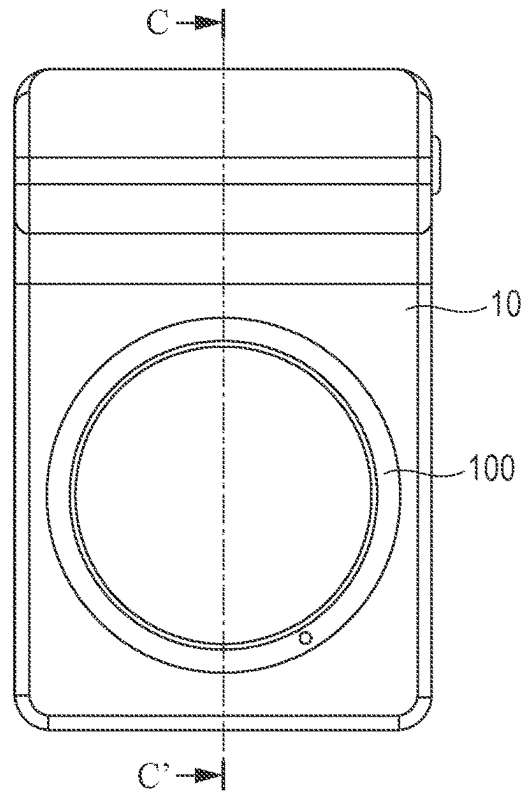


FIG. 6B

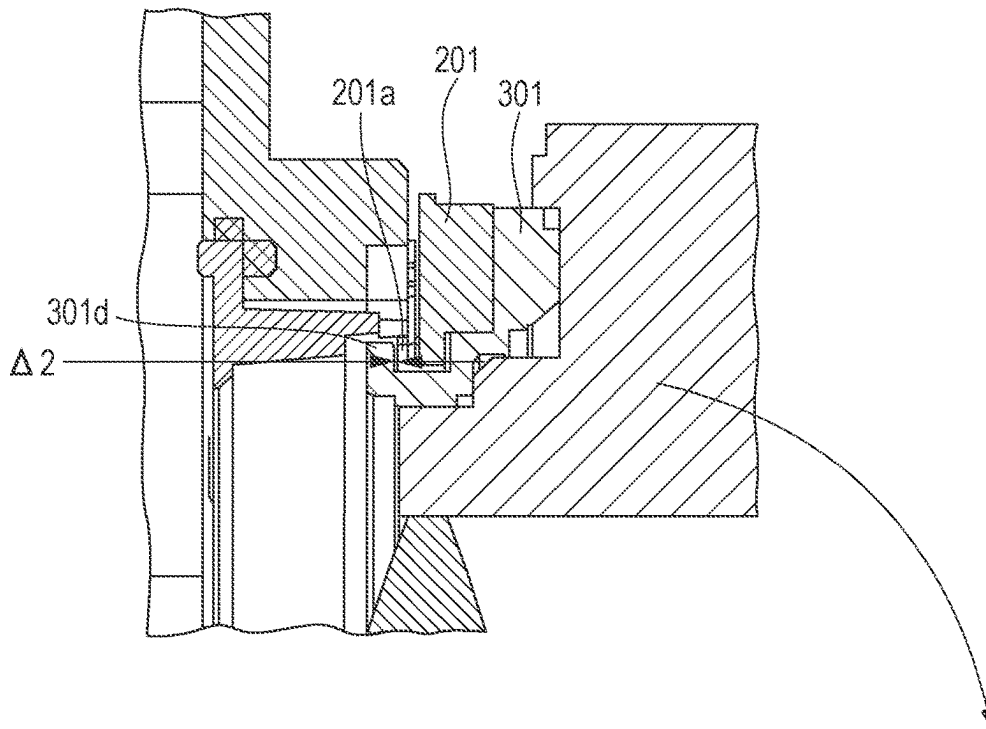


FIG. 7A

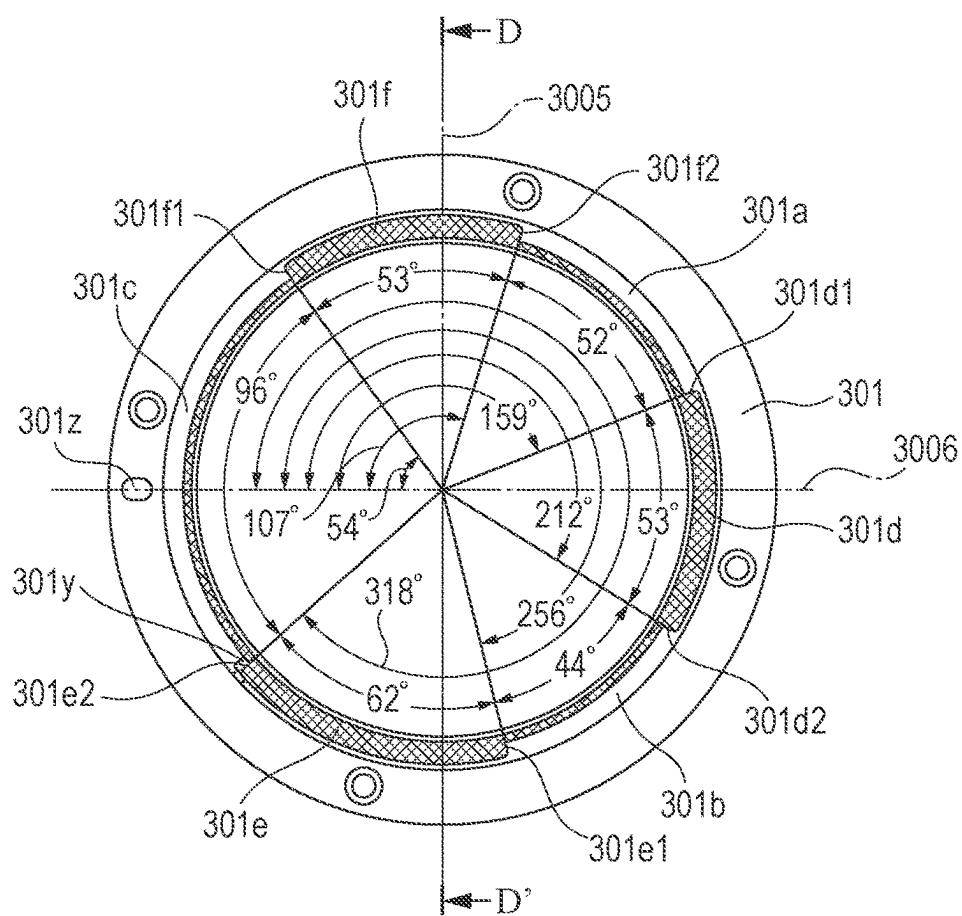


FIG. 7B

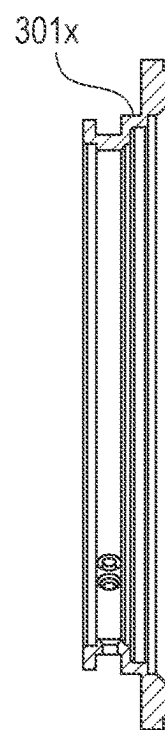


FIG. 8A

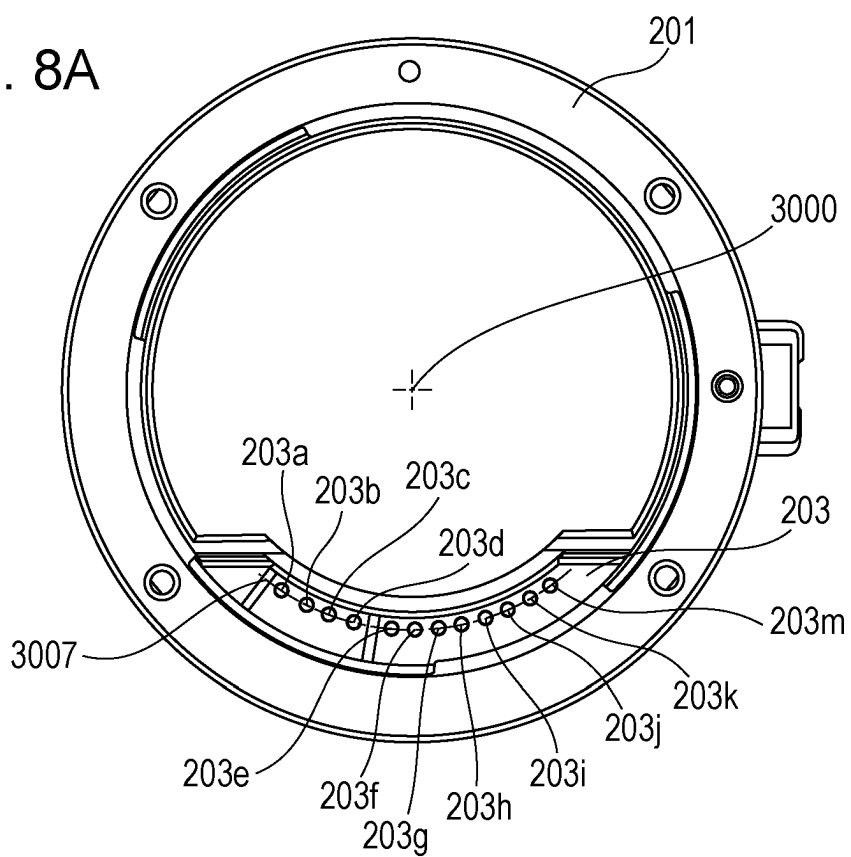


FIG. 8B

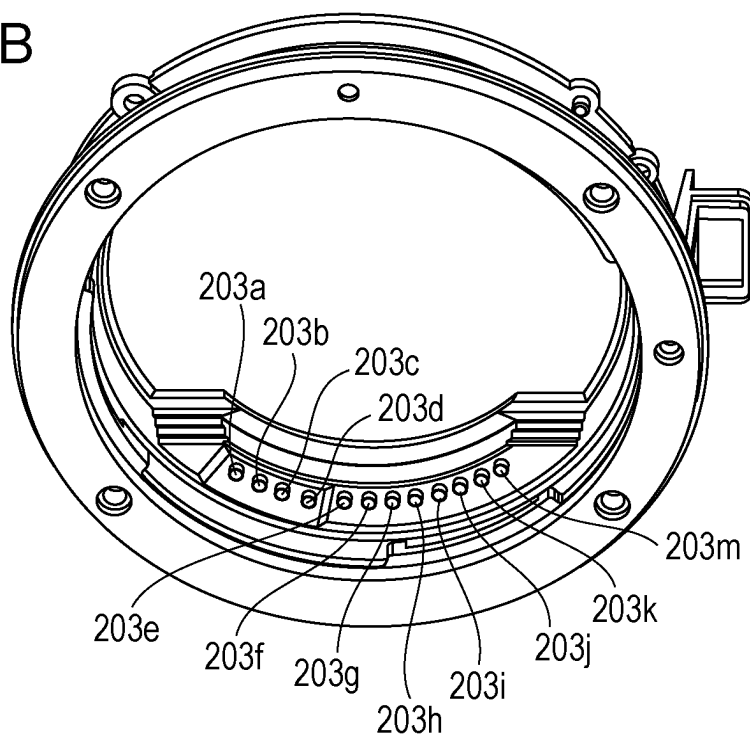


FIG. 9A

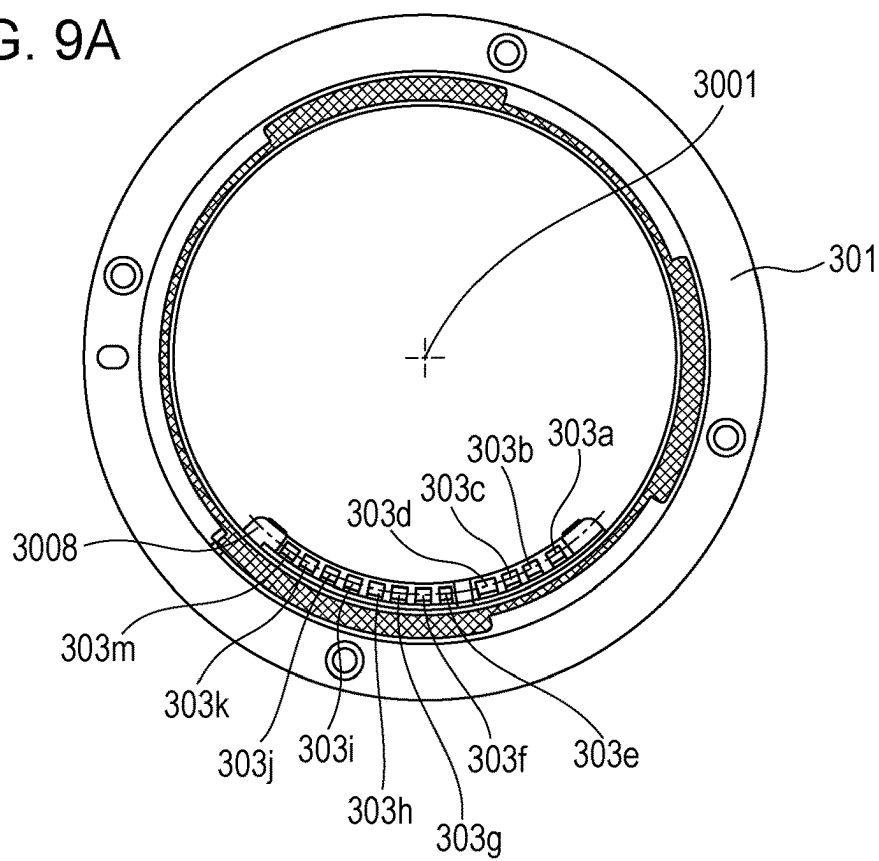
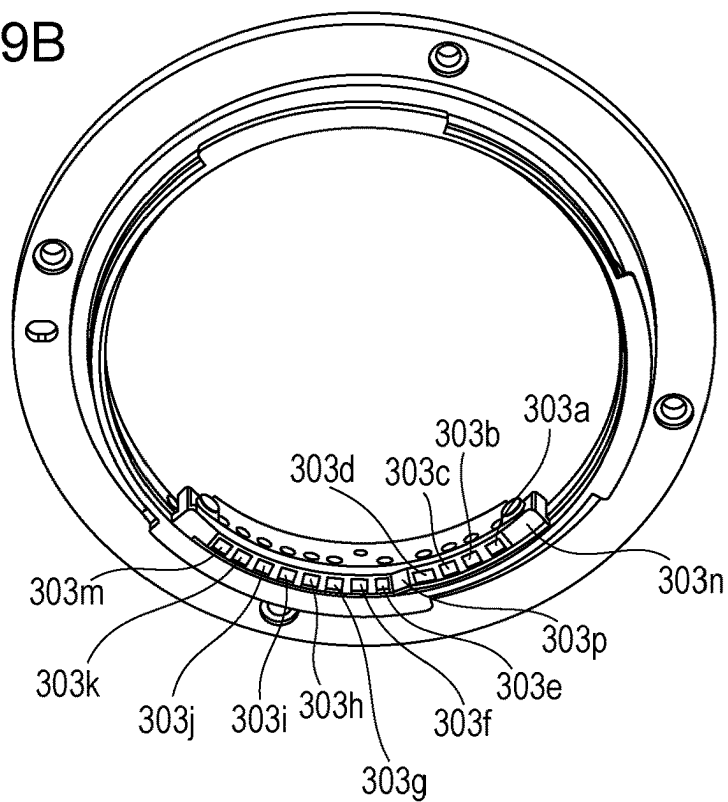


FIG. 9B



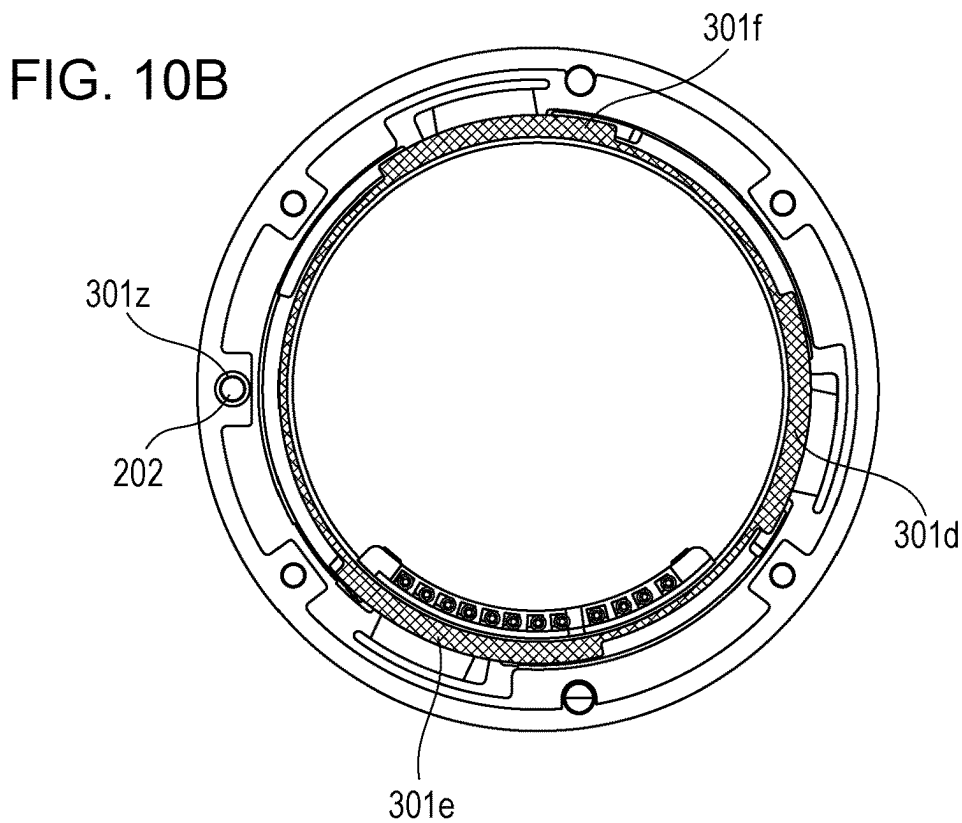
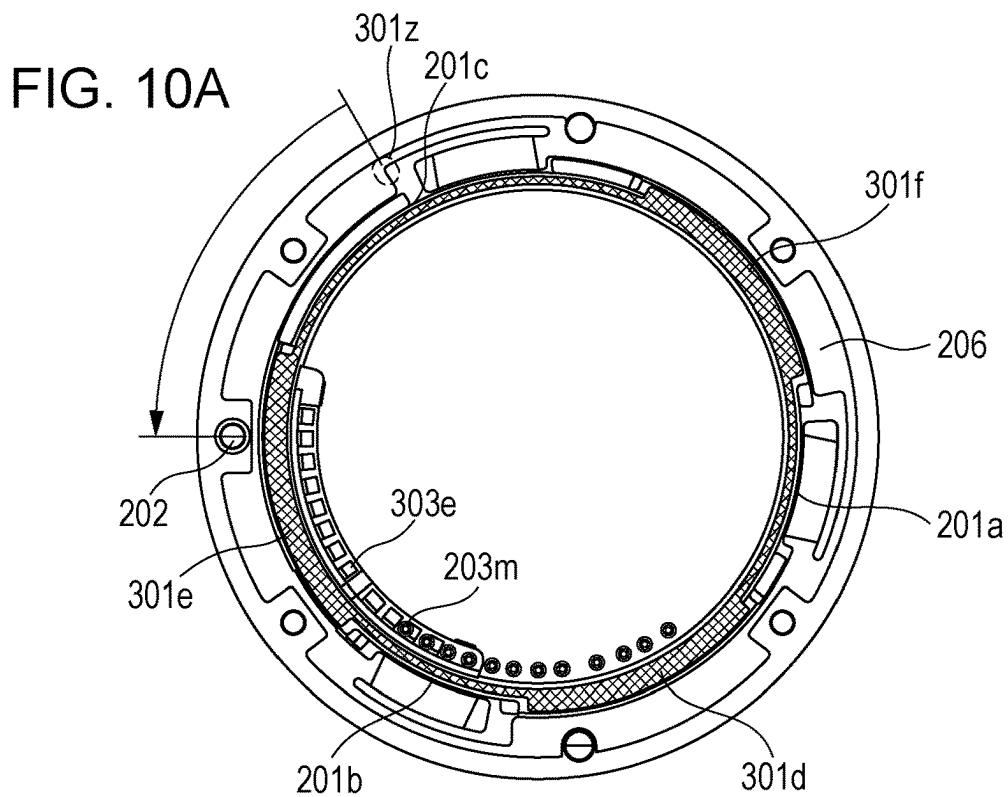


FIG. 11A

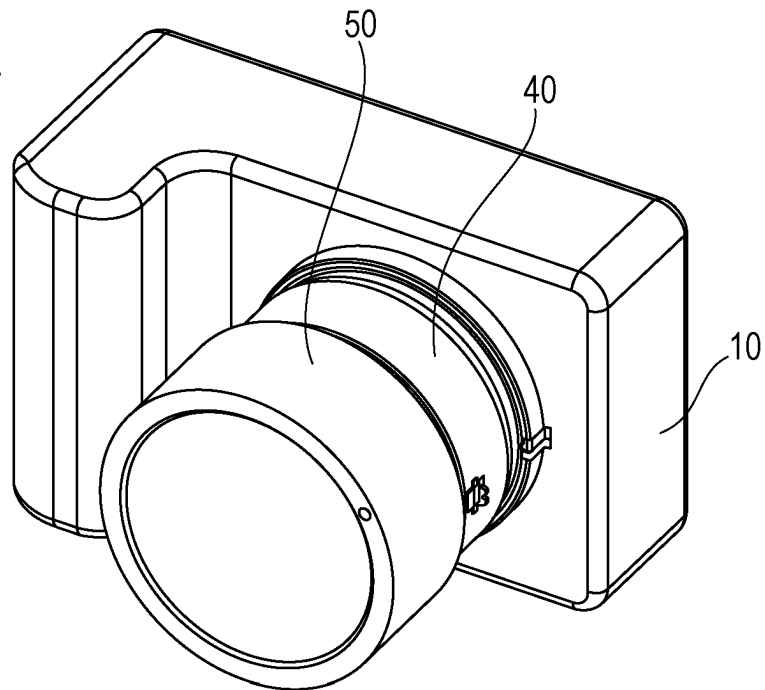


FIG. 11B

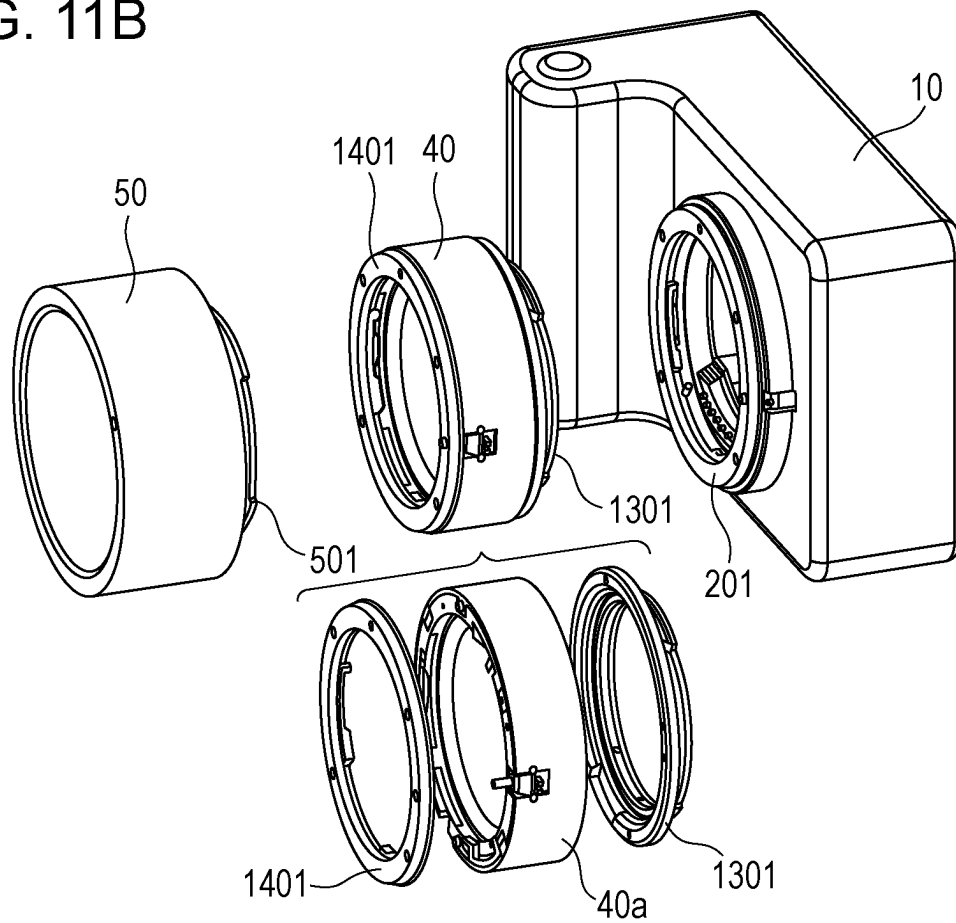


FIG. 12A

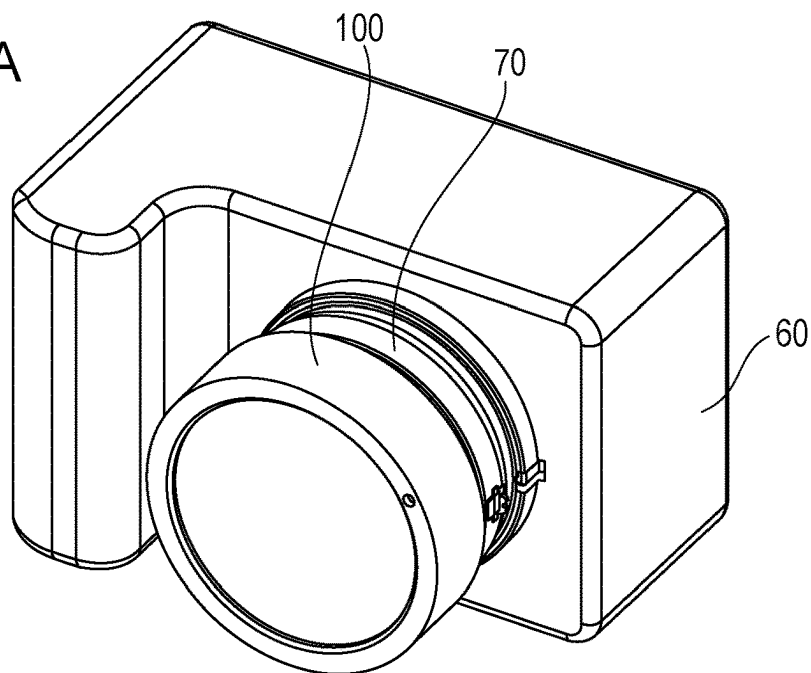


FIG. 12B

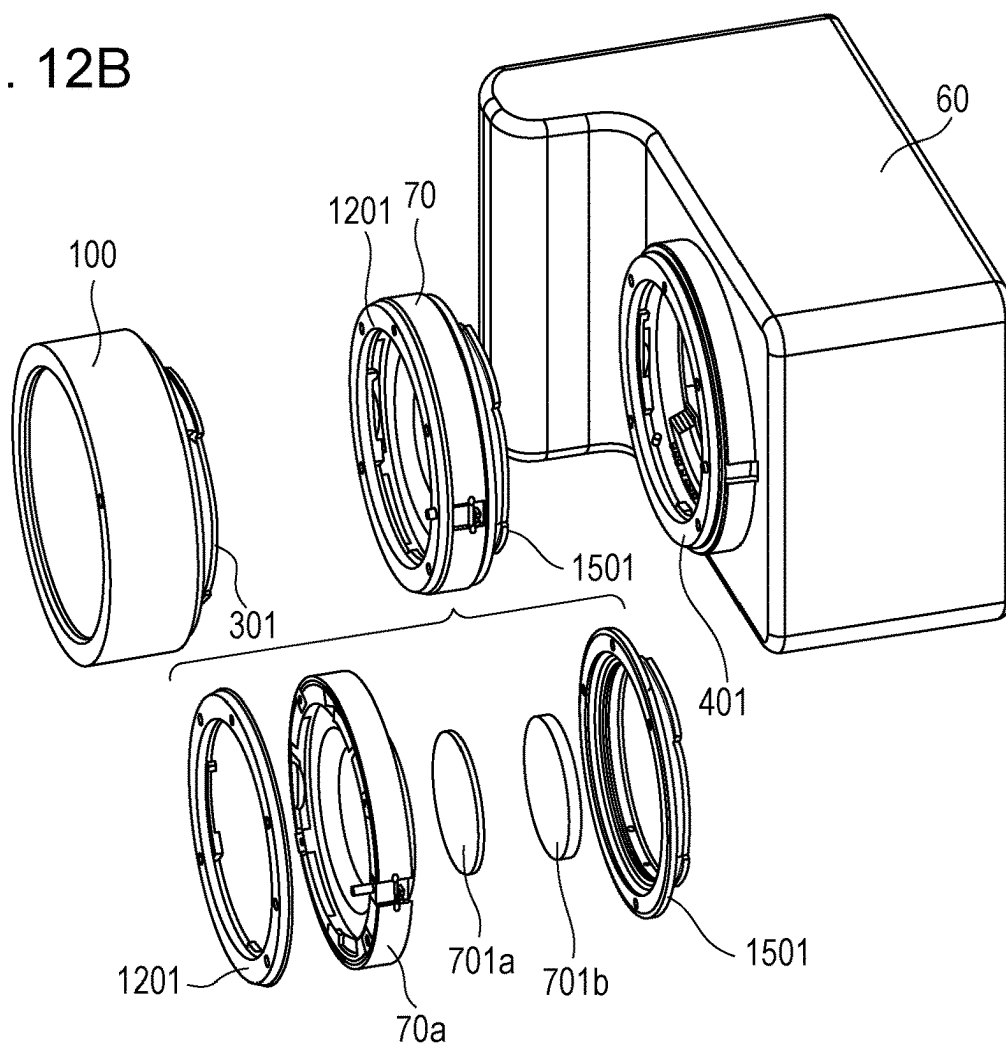


FIG. 13A

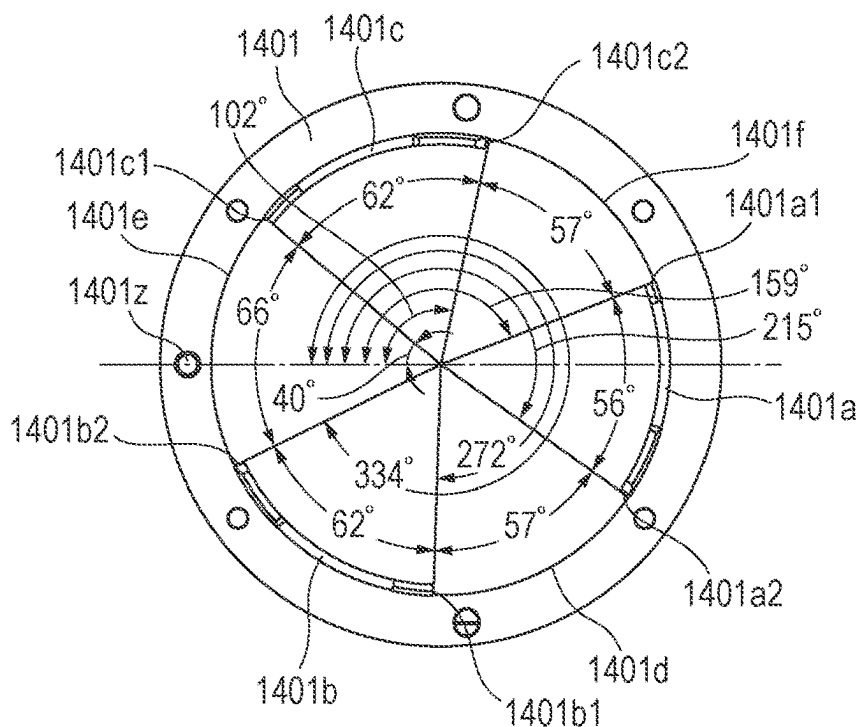


FIG. 13B

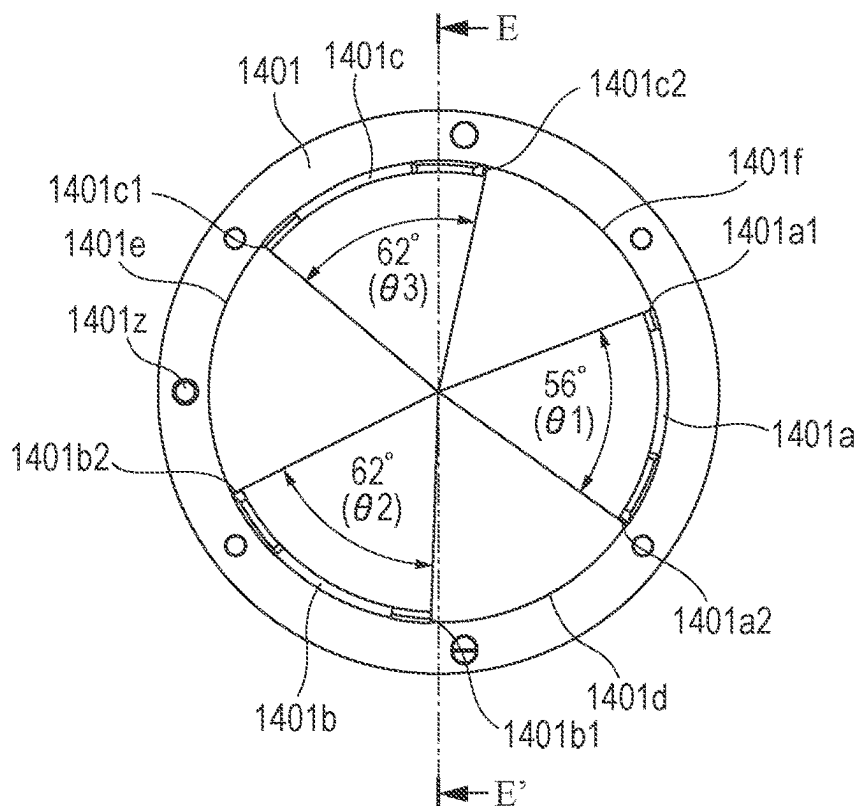


FIG. 13C

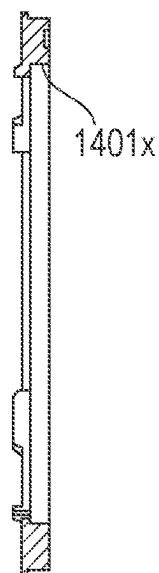


FIG. 14A

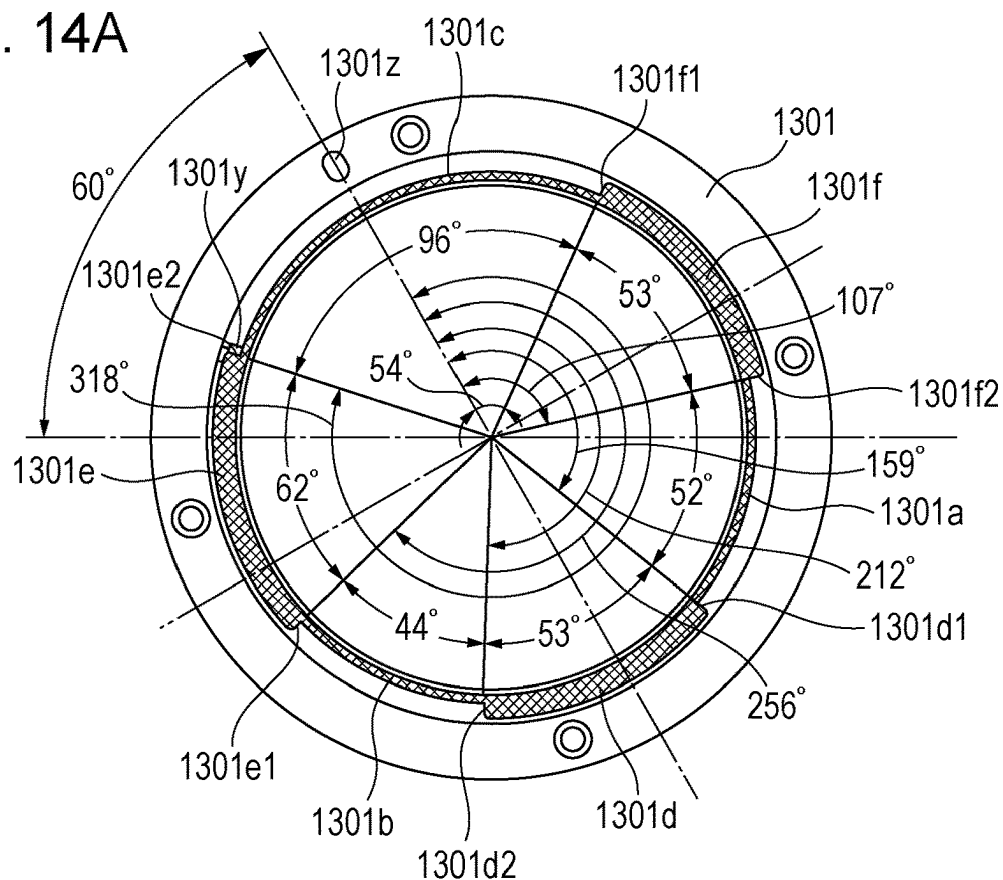


FIG. 14B

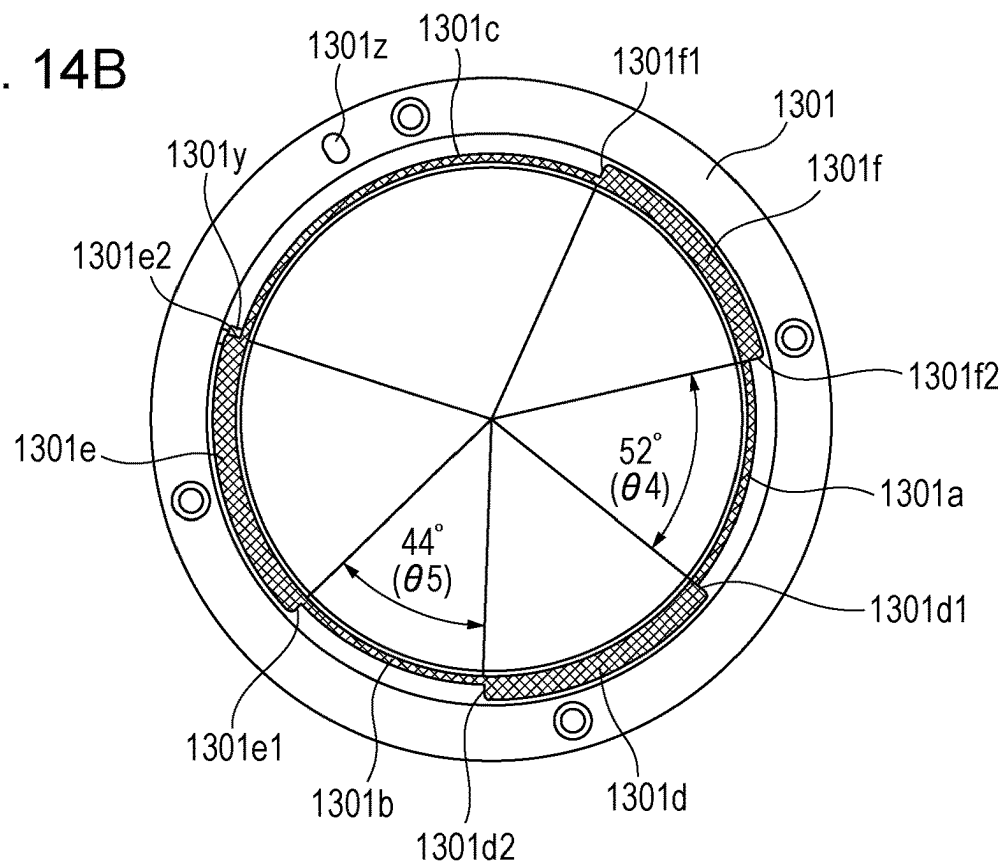


FIG. 15A

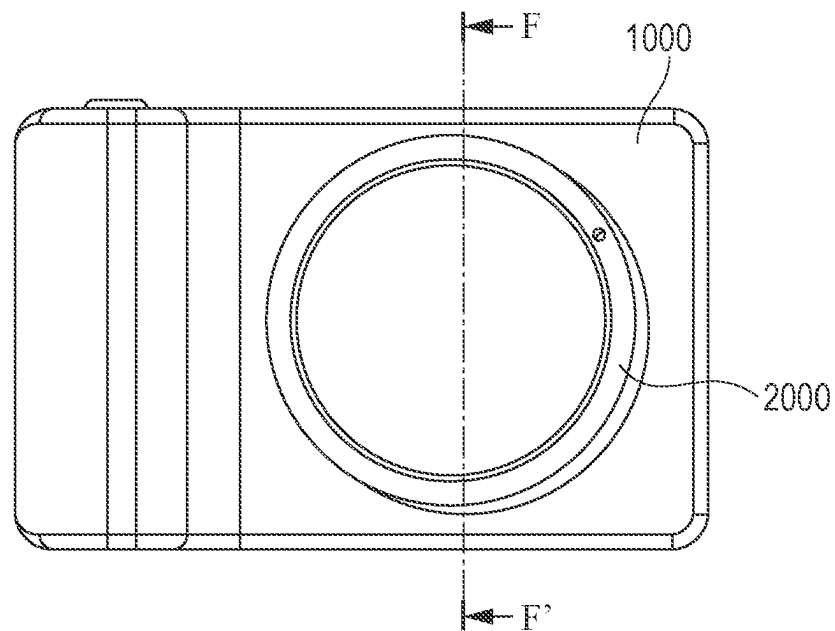


FIG. 15B

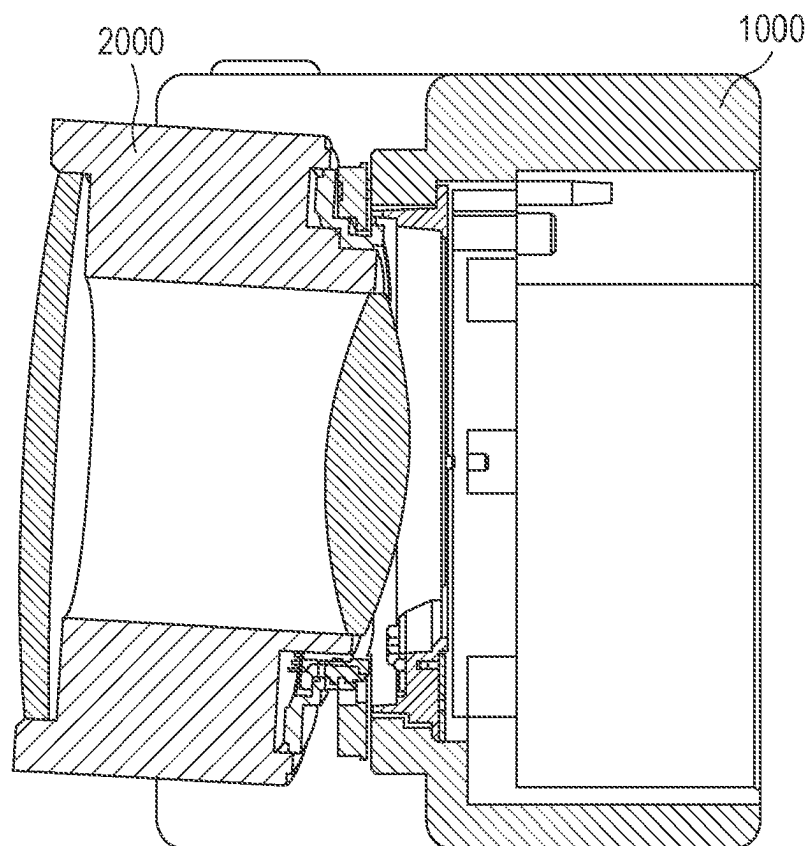


FIG. 16A

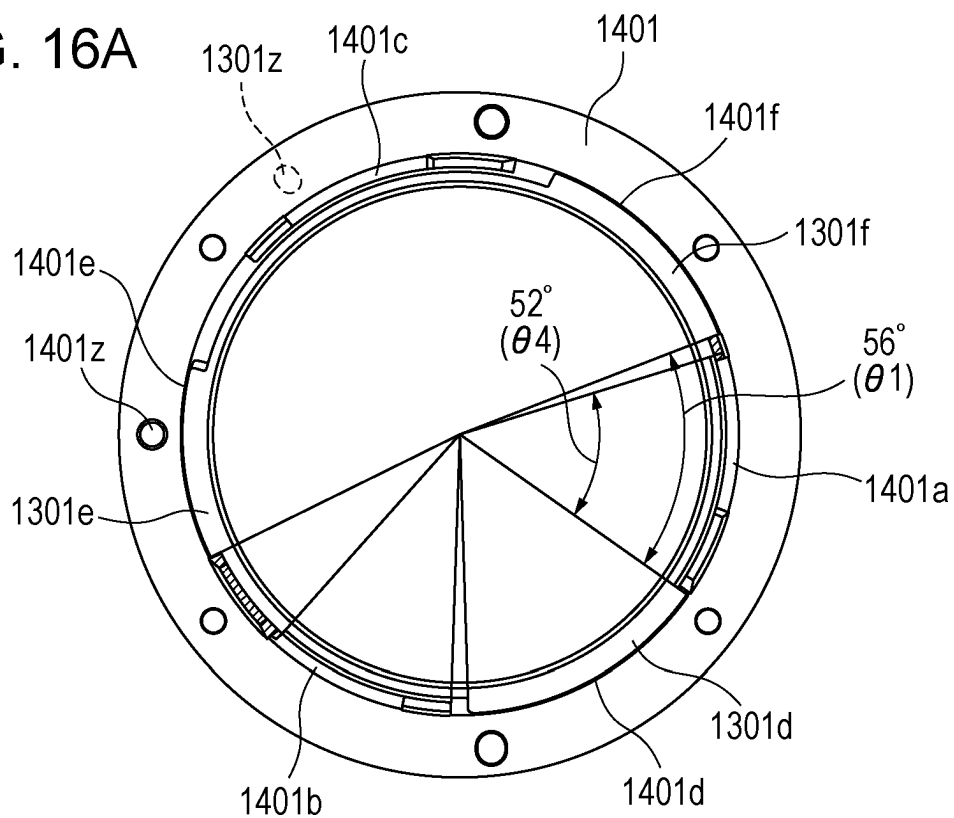


FIG. 16B

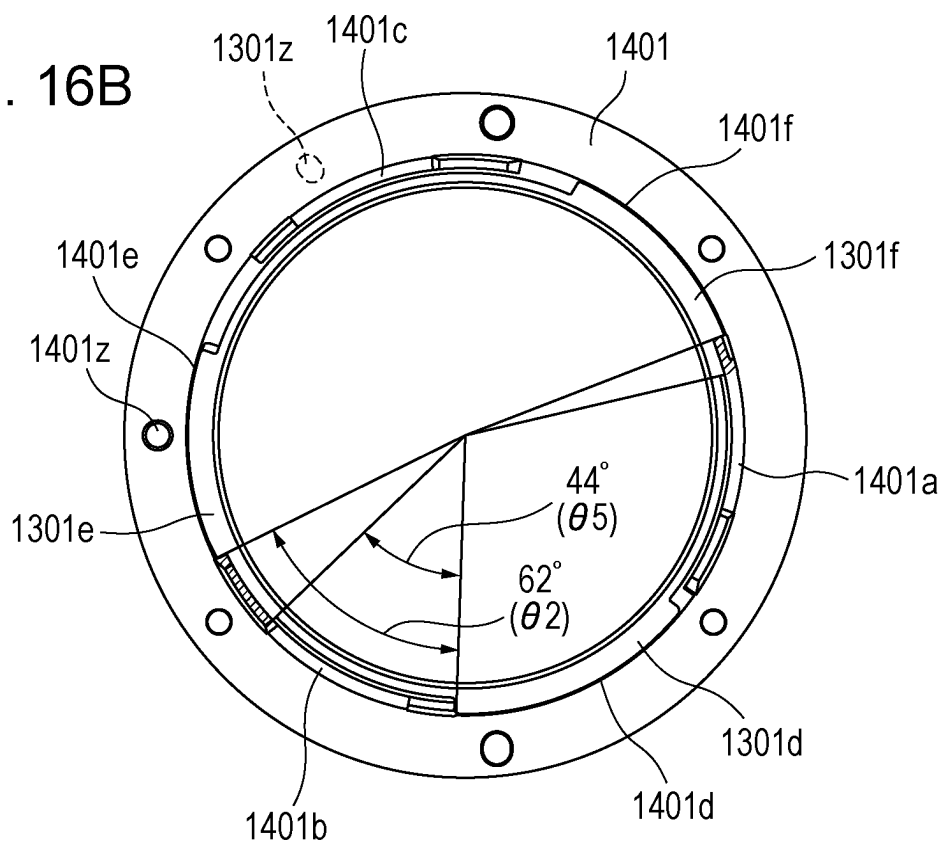


FIG. 17A

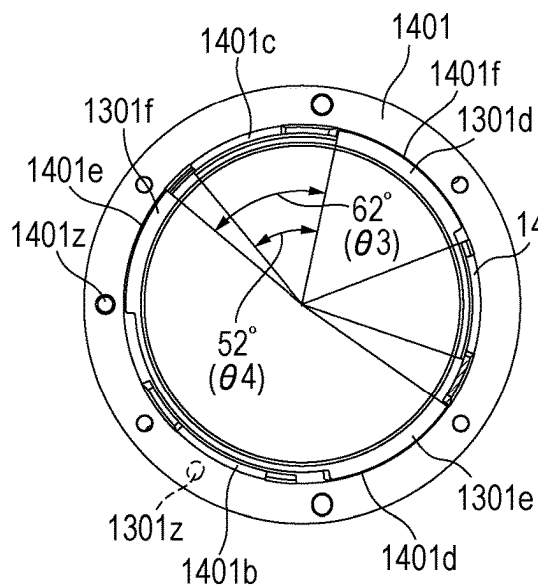


FIG. 17C

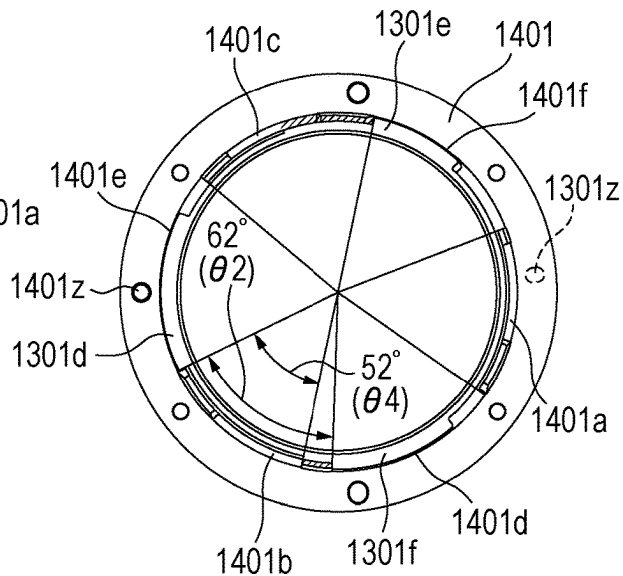


FIG. 17B

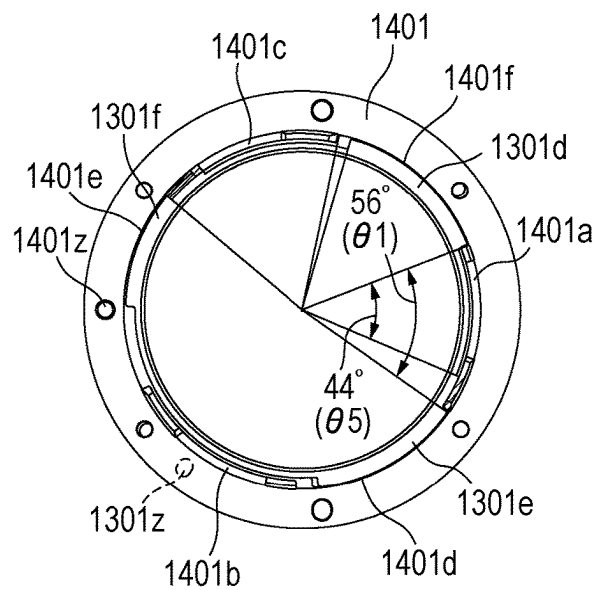


FIG. 17D

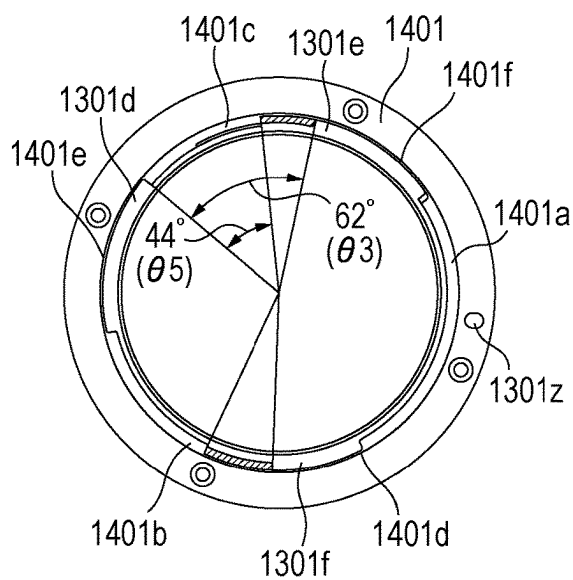


FIG. 18A

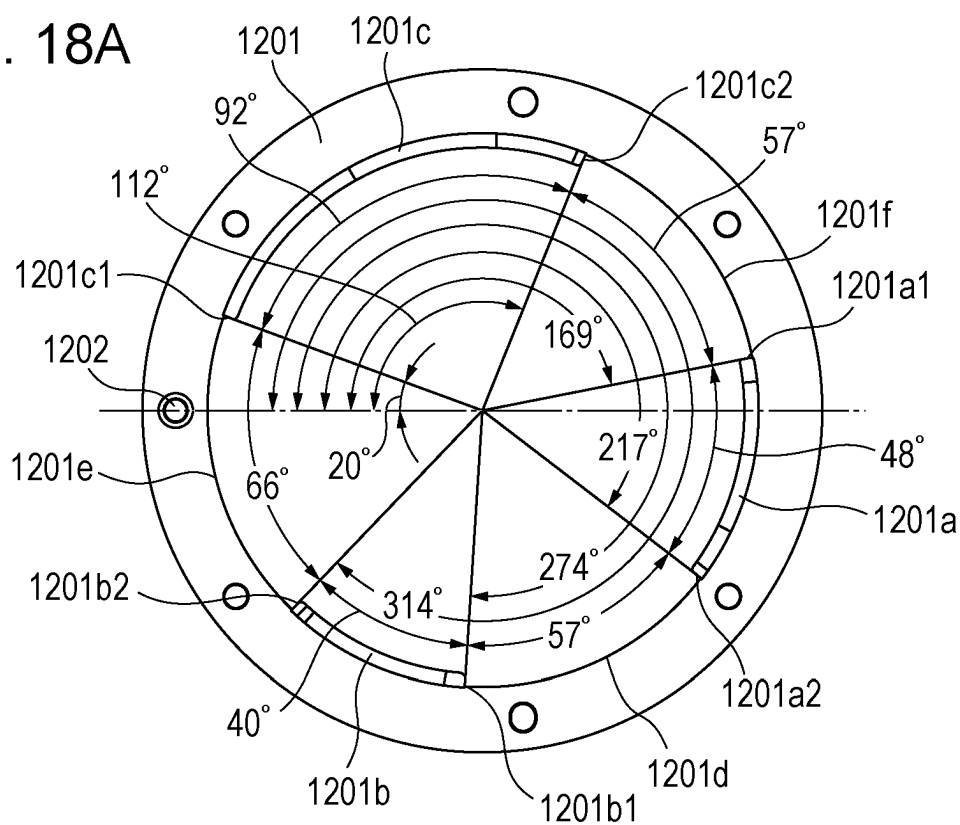


FIG. 18B

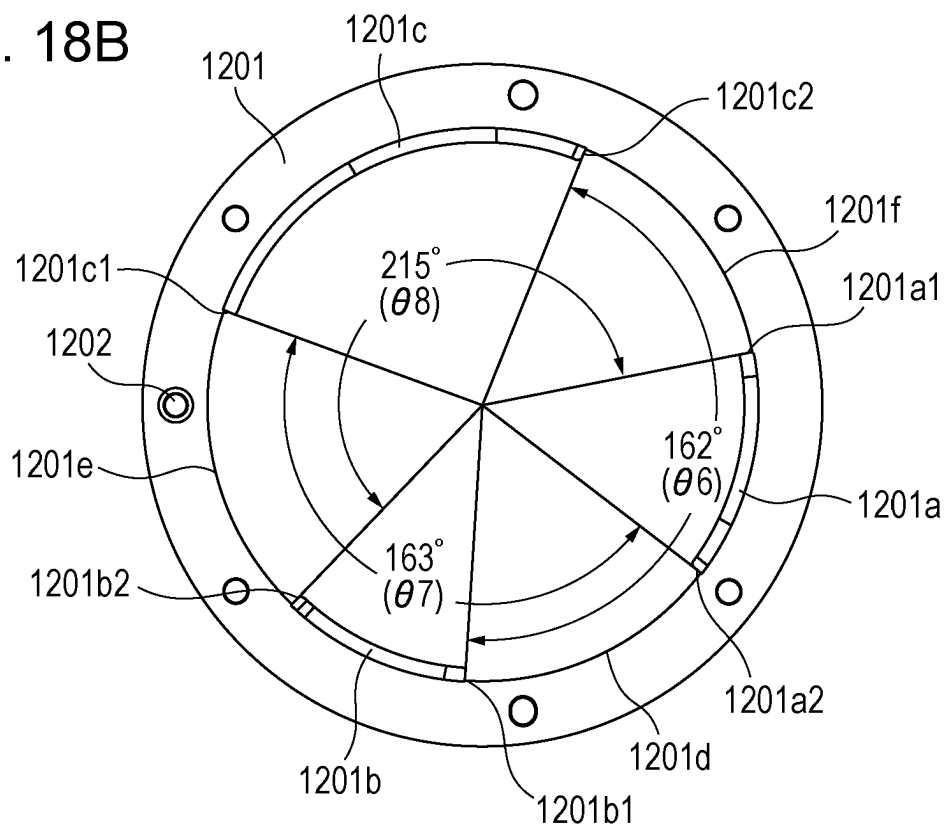


FIG. 19A

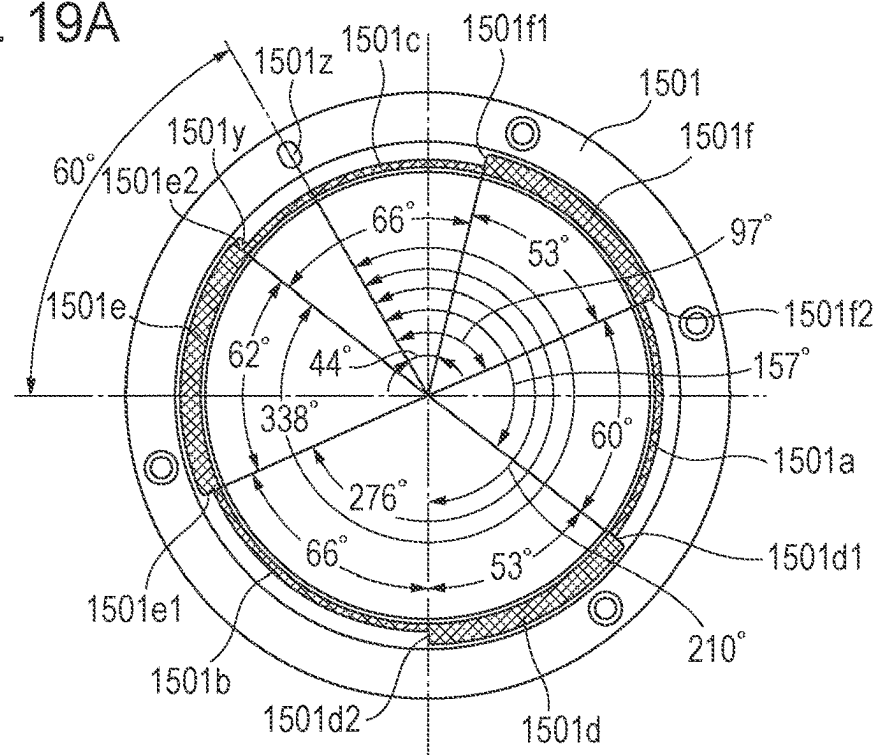


FIG. 19B

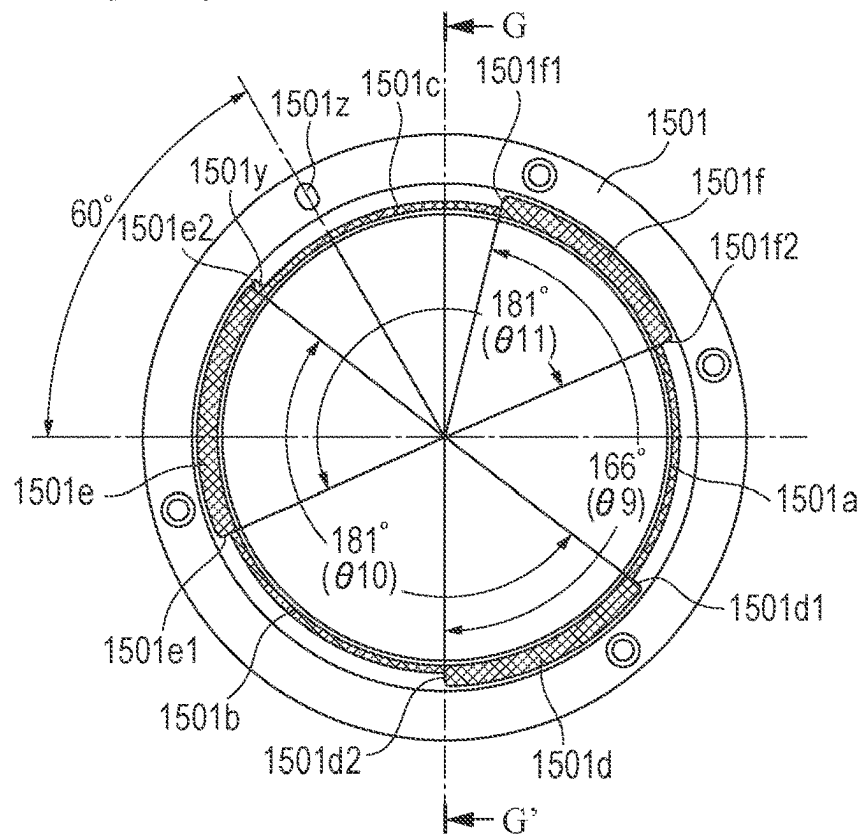


FIG. 19C

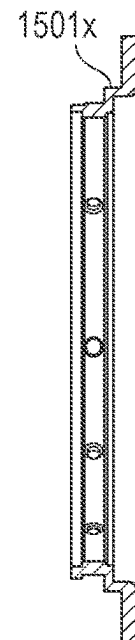


FIG. 20A

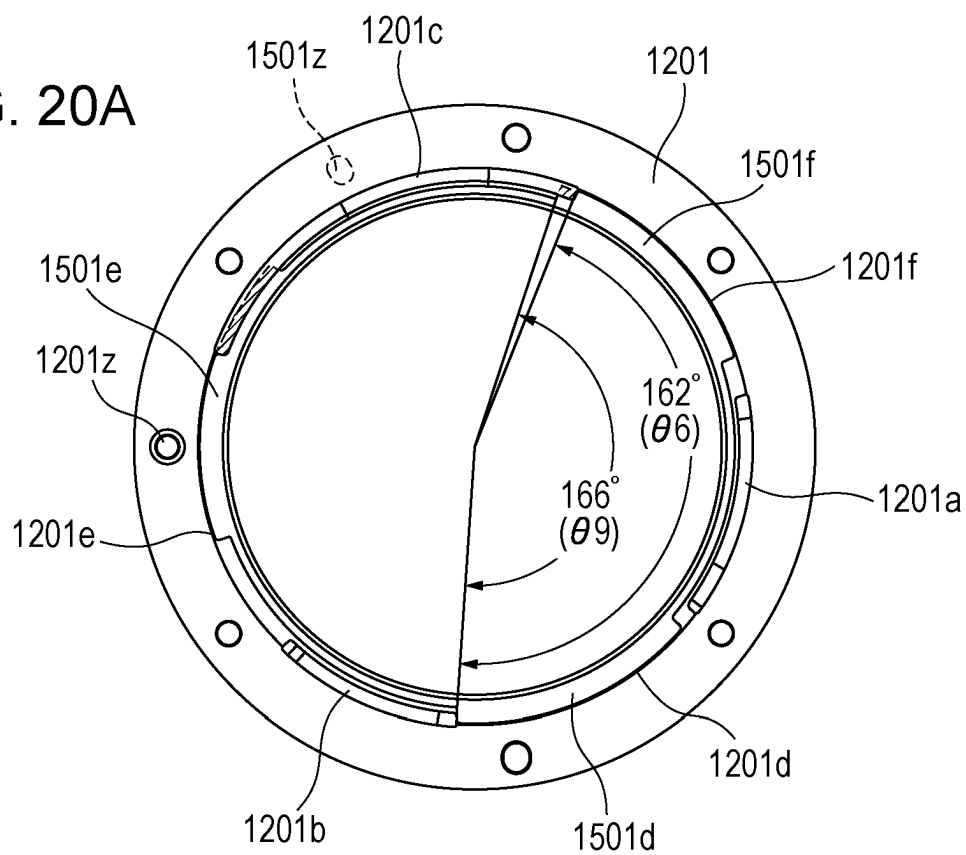


FIG. 20B

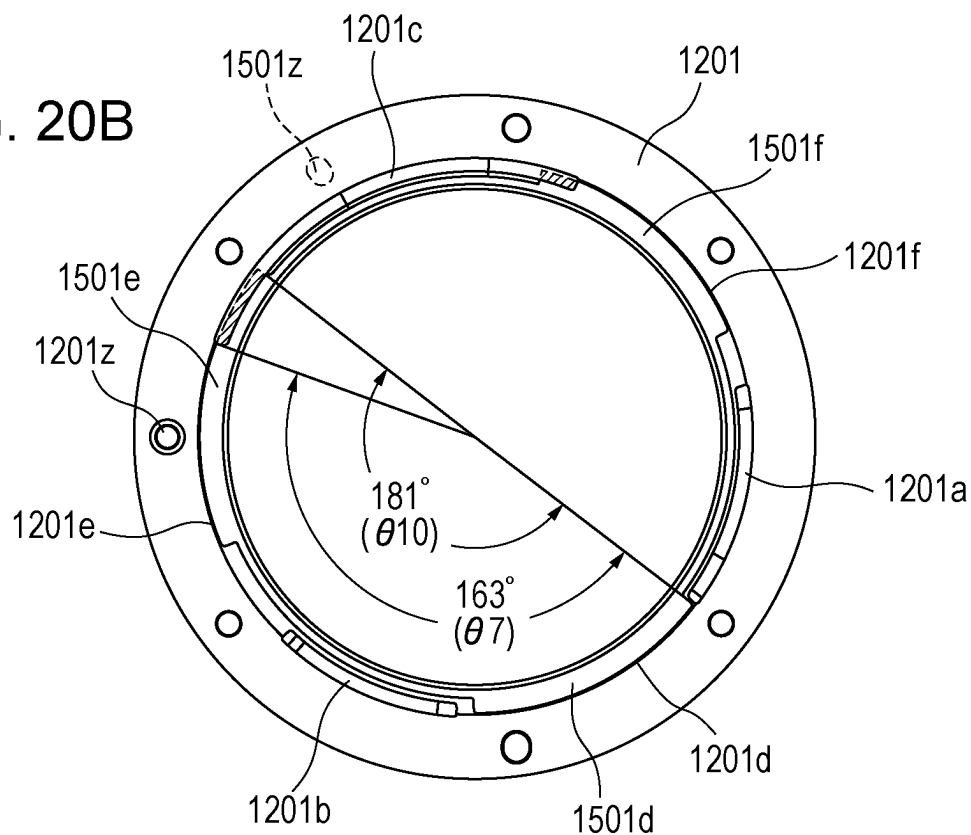


FIG. 21A

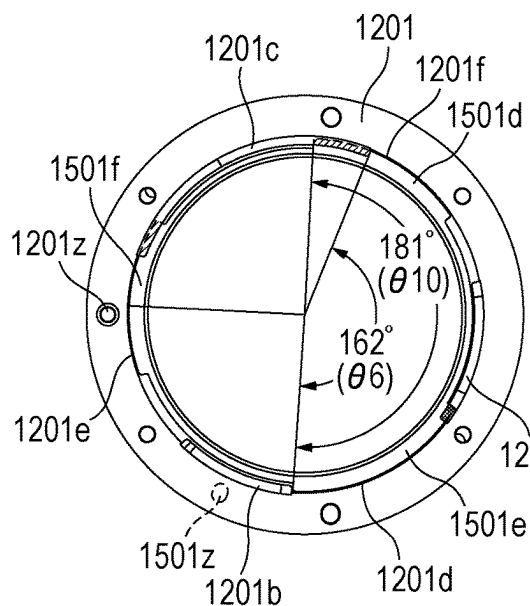


FIG. 21C

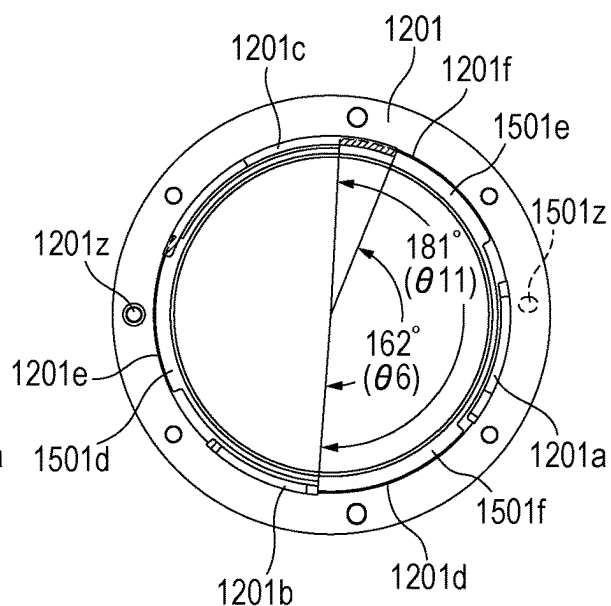


FIG. 21B

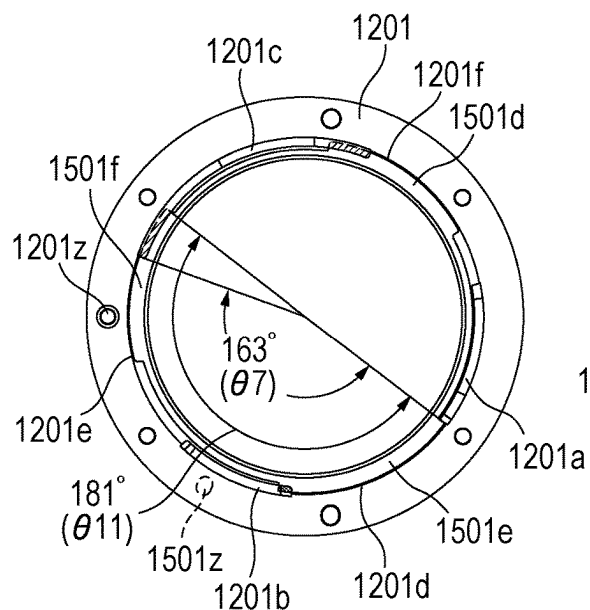


FIG. 21D

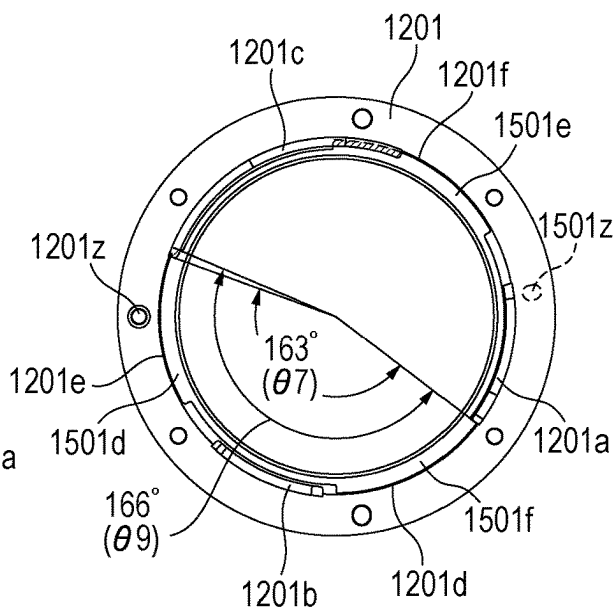


FIG. 22

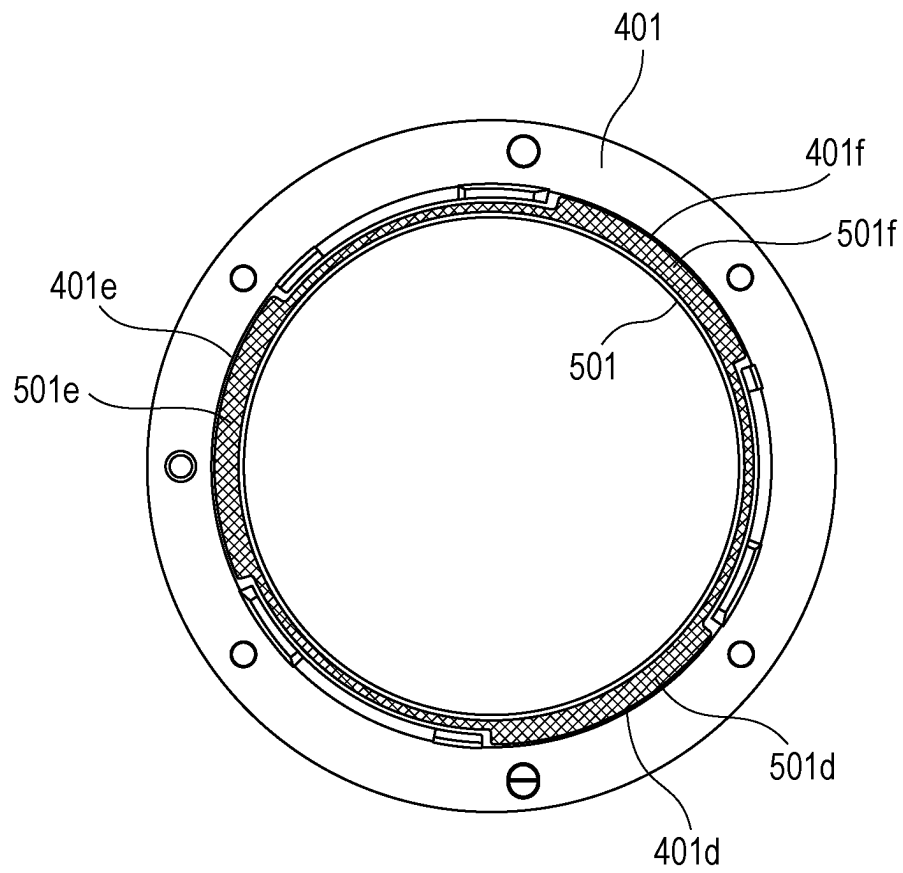


FIG. 23

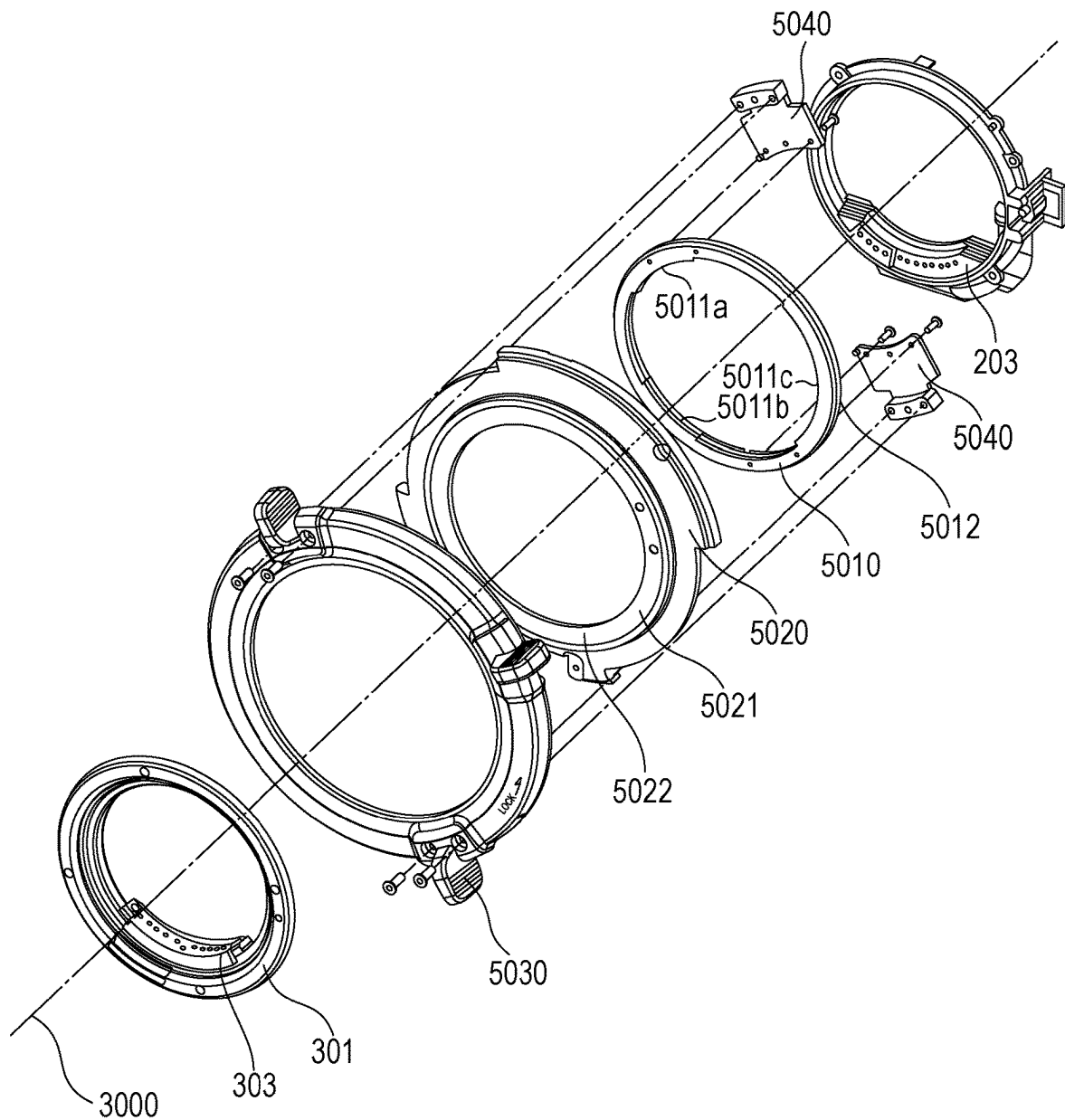


FIG. 24A

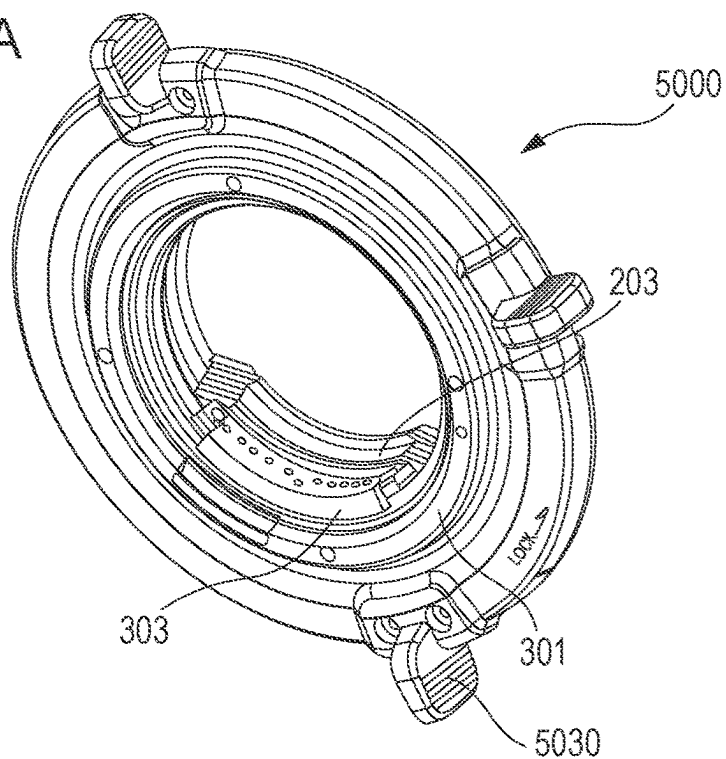


FIG. 24B

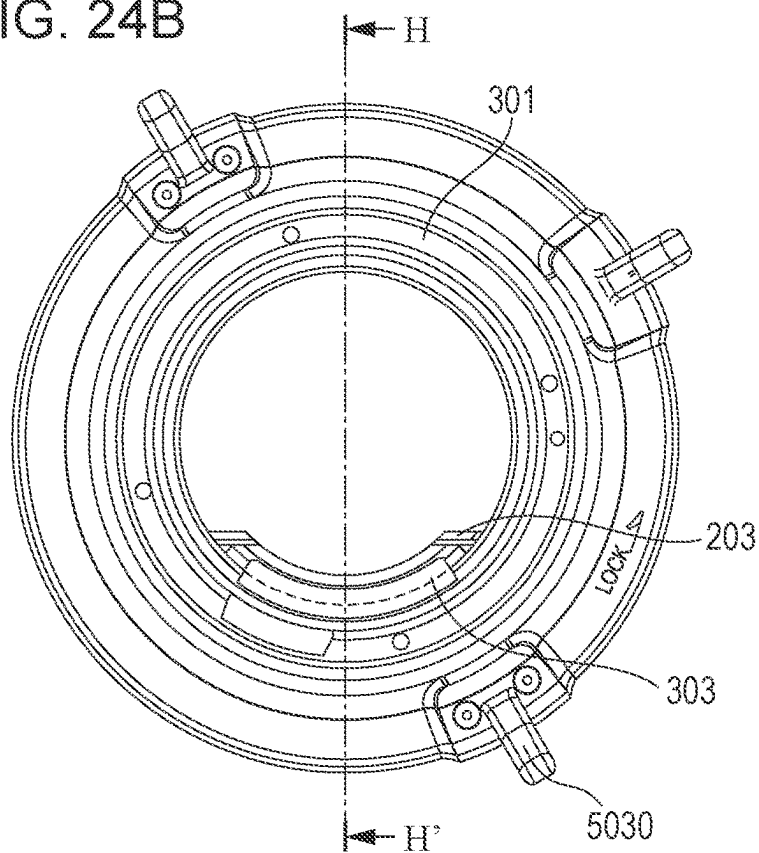


FIG. 24C

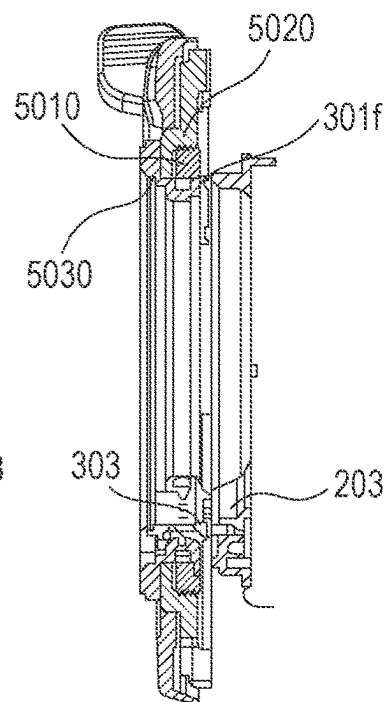


FIG. 25A

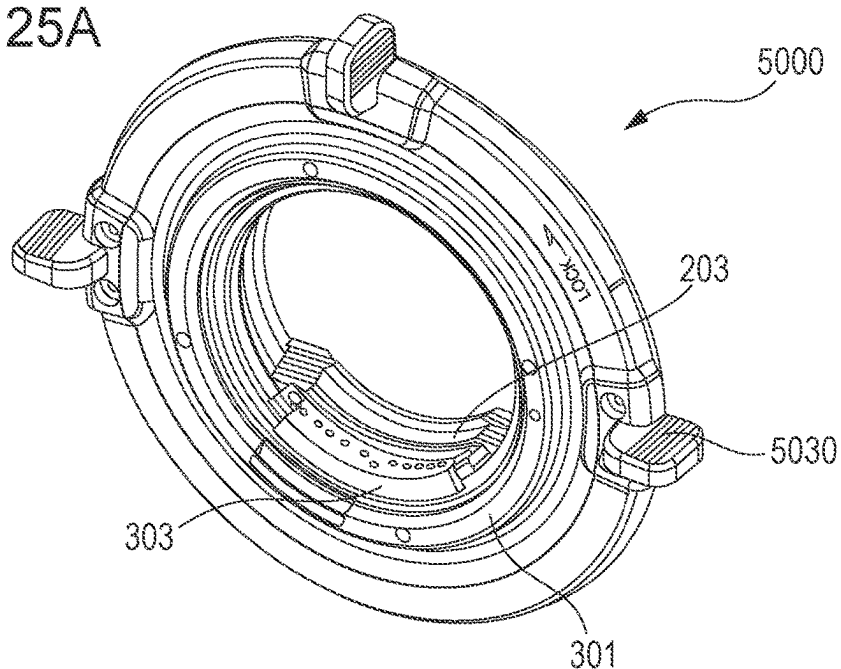


FIG. 25B

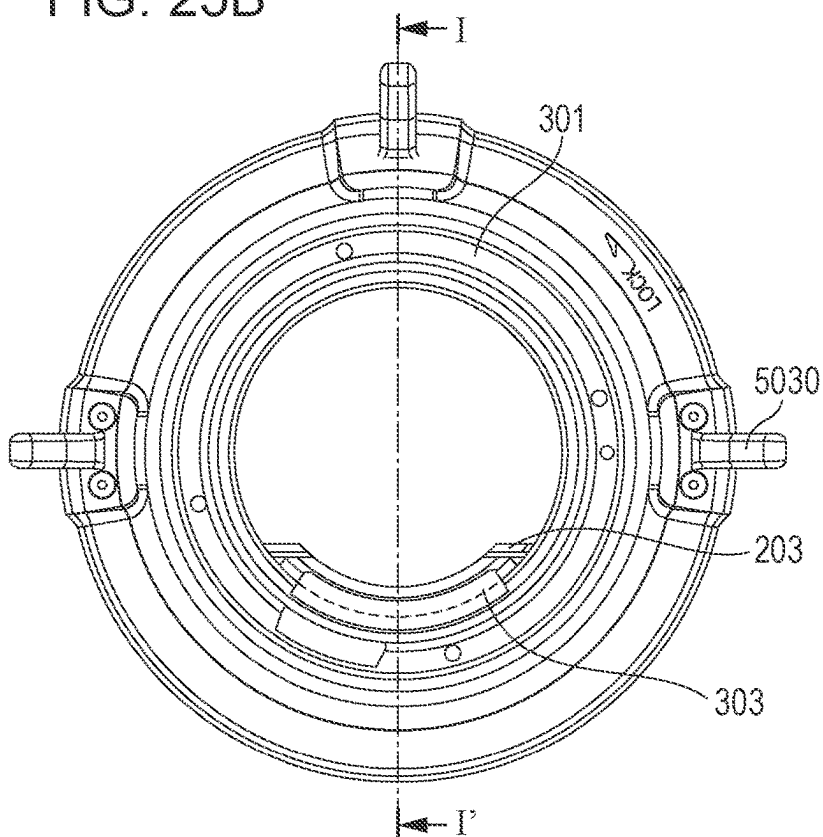
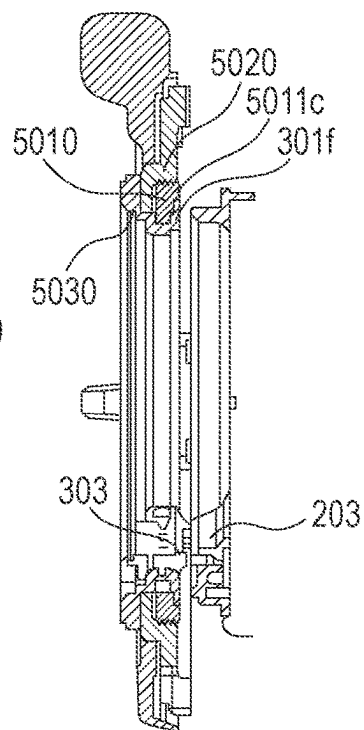


FIG. 25C



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**ADAPTER DEVICE, MOUNT APPARATUS,
AND ACCESSORY****CROSS-REFERENCE TO RELATED
APPLICATIONS**

This application is a Divisional of U.S. patent application Ser. No. 17/184,362, filed on Feb. 24, 2021, which is a Continuation of International Patent Application No. PCT/JP2018/031995, filed Aug. 29, 2018, all of which are hereby incorporated by reference herein in their entireties.

BACKGROUND OF THE INVENTION**Field of the Invention**

The present invention relates to an adapter device, a mount apparatus, and an accessory.

Background Art

Conventionally, an accessory (camera accessory), such as an interchangeable lens, which can receive power supply from a digital camera (herein after, referred to as a camera) and communicate commands, data, and so forth with the camera in a state where the accessory is mounted to the camera is known. It is known that multiple contacts (terminals) capable of electric connection, by coming into contact, are provided on both the camera and a mounting portion of the accessory, to enable the power supply and communication.

There is also known a mounting system for mounting (coupling) the accessory to the camera employing so-called bayonet coupling. In the bayonet coupling, mounts of both the camera and accessory are relatively rotated, and bayonet claws provided on each of the camera and accessory engage with each other.

Corresponding combinations of cameras and accessories employing a mounting system described above are predetermined. For example, corresponding accessories (e.g., interchangeable lenses, etc.) differ depending on the length of the flange focal distance of the camera. In other words, some accessories are not compatible with the flange focal distance of a camera.

In a case where an accessory not compatible with a camera is mounted, problems may occur. For example, in a case where an interchangeable lens not compatible with a camera having a certain flange focal distance is mounted to the camera, an optical image of a subject entering via the interchangeable lens may not be focused on the image plane of the camera. Thus, it may be necessary to restrict connection of a camera and an accessory in an incompatible combination.

In particular, in a case where a corresponding camera has approximately the same diameter as the diameter a plurality of accessories have, it may be difficult to judge from the external appearance whether the camera and the accessories have a corresponding combination.

PTL1 discusses a technique where a protrusion of a lens, which has the same flange focal distance but a short back focus, abuts a wall portion provided to a camera body, thereby preventing erroneous mounting of the lens to a camera body that cannot be mounted with a lens with a short back focus.

CITATION LIST**Patent Literature**

PTL1: Japanese Patent Laid-Open No. 2005-70712

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The technique described in PTL1 only discusses a technique that prevents erroneous mounting of a lens to a camera that cannot operate correctly with a lens that uses the same mounting system and has a short back focus. Thus, the technique discussed in PTL1 cannot prevent erroneous mounting of an accessory that has a different flange focal distance to a camera not compatible with the accessory.

Further, in the technique discussed in PTL1, a wall portion is provided on the camera body such that a protrusion of the lens is abutted to the camera body. Another configuration is thereby needed, and thus the number of parts or manufacturing cost may be increased.

Further, for example, in a case where a conversion lens having substantially the same mount diameter and a long flange focal distance is mounted to a camera having a short flange focal distance, a conversion adapter is necessary to be inserted between the camera and the conversion lens. In this case, the conversion lens has substantially the same diameter on both ends, and thus there is a problem that erroneous mounting may easily occur on the camera and the accessory with respect to the conversion adapter.

The present invention is directed to prevent erroneous mounting of an imaging apparatus and an accessory having mounts that can be mutually engaged with each other.

SUMMARY OF THE INVENTION

The present invention in its first aspect provides an adapter device according to the present invention includes a first mount on one side, and a second mount on another side, the first mount being attachable to and detachable from an imaging apparatus, and the second mount being attachable to and detachable from an accessory, the adapter device being characterized in that the first mount is fastened to the adapter device by a plurality of first screws, and a first mount recess, a first mount claw, a second mount recess, a second mount claw, a third mount recess, and a third mount claw are disposed following a circumferential direction of the first mount, the second mount is fastened to the adapter device by a plurality of second screws, and a fourth mount claw, a fourth mount recess, a fifth mount claw, a fifth mount recess, a sixth mount claw, and a sixth mount recess are disposed following a circumferential direction of the second mount, the first, second, and third mount claws are insertable into mount recesses provided to the imaging apparatus, the fourth, fifth, and sixth mount claws are insertable into mount recesses provided to the accessory, and a smallest angle range out of angle ranges in a circumferential direction of the first mount out of the first, second, and third mount recesses is equal to or smaller than a smallest angle range out of angle ranges in a circumferential direction of the second mount out of the fourth, fifth, and sixth mount claws.

The present invention in its second aspect provides an adapter device comprising a first mount on one side and a second mount on another side, a first mount being attachable to and detachable from an imaging apparatus, and the second mount being attachable to and detachable from an accessory, wherein the first mount is fastened to the adapter device by a plurality of first screws, and a first mount recess, a first mount claw, a second mount recess, a second mount claw, a third mount recess, and a third mount claw are disposed following a circumferential direction of the first mount, the second mount is fastened to the adapter device by a plurality of second screws, and a fourth mount claw, a fourth mount recess, a fifth mount claw, a fifth mount recess, a sixth mount claw, and a sixth mount recess are disposed following a circumferential direction of the second mount, the first,

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second, and third mount claws are insertable into mount recesses provided to the imaging apparatus, the fourth, fifth, and sixth mount claws are insertable to mount recesses provided to the accessory, and a largest angle range out of angle ranges in a circumferential direction of the second mount out of the fourth, fifth, and sixth mount claws is smaller than a largest angle range out of angle ranges in a circumferential direction of the first mount out of the first, second, and third mount recesses.

The present invention in its third aspect provides a mount apparatus being attachable to and detachable from an accessory having a first mount, wherein the mount apparatus has a second mount having a plurality of mount claws that can be engaged with a plurality of claws provided to the first mount, the plurality of mount claws includes a first mount claw, and a second mount claw and a third mount claw each having an angle range in a circumferential direction smaller than the angle range of the first mount claw, disposed following a circumferential direction of the second mount, the mount apparatus has a plurality of biasing members biasing the plurality of mount claws provided to the first mount in a center axis direction of the second mount, the first, second and third mount claws respectively have a first, second and third biasing member arrangement portions on which each of the plurality of biasing members is arranged, the second mount is fastened to the imaging apparatus by a plurality of screws, and in a case where a line extending from a center of the second mount in a gravitational direction and a direction opposite to the gravitational direction when viewing the mount apparatus in a normal position from the center axis direction of the second mount is taken as a first mount center line, and a line orthogonal to the first mount center line is taken as a second mount center line, and when viewing the second mount in the center axis direction of the second mount: the first mount claw overlaps the first mount center line, and at least one of the second mount claw and the third mount claw overlap the second mount center line, in a radial direction of the second mount, the first biasing member arrangement portion overlaps the first mount center line in the radial direction of the second mount, and the second biasing member arrangement portion overlaps the second mount center line in the radial direction of the second mount.

The present invention in its fourth aspect provides an accessory attachable to and detachable from a mount apparatus having a first mount, wherein the accessory has a second mount having a plurality of mount claws and a plurality of mount recesses that can be engaged with a plurality of claws provided to the first mount, the plurality of mount recesses include a first mount recess, and a second mount recess and a third mount recess that have an angle range in the circumferential direction smaller than the angle range of the first mount recess, disposed following a circumferential direction of the second mount, the second mount is fastened to the imaging apparatus by a plurality of screws, in a state that the accessory is attached to the mount apparatus, the plurality of mount claws are biased in a center axis direction of the second mount by each of a plurality of biasing members arranged in each space of the plurality of claws in the mount apparatus, the plurality of mount claws include a first mount claw, a second mount claw and a third mount claw, and in a case where a line extending from a center of the second mount in a gravitational direction and a direction opposite to the gravitational direction when viewing from the center axis direction of the second mount with the mount apparatus to which the accessory is mounted in a normal position, is taken as a first mount center line, and

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when viewing the second mount in the center axis direction of the second mount: the first mount center line does not overlap any of the plurality of mount recesses, and at least the biasing member which biases the first mount claw overlaps the first mount center line in a state that the accessory is attached to the mount apparatus.

Further features of the present invention will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1A is a block diagram of a camera system including a first lens unit **100** and a camera body **10** as a first imaging apparatus to/from which the first lens **100** can be directly attached/detached according to an embodiment of the present invention.

FIG. 1B is a block diagram illustrating a configuration of a mount portion **1** included in the first lens **100** and the camera body **10** according to an embodiment of the present invention.

FIG. 2A is an external perspective view of a camera body **10** in a state where a first lens unit **100** is mounted to the camera body **10** according to an embodiment of the present invention.

FIG. 2B is an external perspective view of a camera body **10** in a state where the first lens unit **100** is detached from the camera body **10** according to an embodiment of the present invention.

FIG. 3A is a disassembled perspective view of the camera body **10** according to an embodiment of the present invention.

FIG. 3B is a disassembled perspective view of the first lens unit **100** according to an embodiment of the present invention.

FIG. 4A is a diagram exemplarily describing angle ranges that camera mount claws and camera mount recesses having a lock pin **202** as a reference occupy in a case of viewing the camera mount **201** according to an embodiment of the present invention from the photographer side (rear face side of the camera body **10**) of the camera body **10**.

FIG. 4B is a diagram exemplarily describing angle ranges that a plurality of camera claws **201a** to **201c** occupies in a case of viewing the camera mount **201** according to an embodiment of the present invention from the photographer side (rear face side of the camera body **10**) of the camera body **10**.

FIG. 4C is a cross-sectional diagram taken along a cross-section A-A' illustrated in FIG. 4B.

FIG. 5A is a diagram exemplarily describing engagement by camera claws at a normal position of the camera body **10** according to an embodiment of the present invention in a state where the first lens unit **100** is mounted to the camera body **10**.

FIG. 5B is a partial cross-sectional view of the camera body **10** and the first lens **100** taken along a cross-section B-B' illustrated in FIG. 5A.

FIG. 6A is a diagram exemplarily describing engagement by lens claws at a vertical position of the camera body **10** according to an embodiment of the present invention in a state where the first lens unit **100** is mounted to the camera body **10**.

FIG. 6B is a partial cross-sectional view of the camera body **10** and first lens **100** taken along a cross-section C-C' illustrated in FIG. 6A.

FIG. 7A is a diagram exemplarily describing angles of the claws and recesses in a circumferential direction at a lens

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mount **301** side in a case of viewing a lens mount **301** according to an embodiment of the present invention from the rear face side (side where the camera body is mounted) in a state where the first lens unit **100** is mounted to the camera body **10**.

FIG. **7B** is a cross-sectional view taken along a cross-section D-D' illustrated in FIG. **7A**.

FIG. **8A** is an external perspective view of the camera mount **201** according to an embodiment of the present invention as viewed from the front face side (subject side).

FIG. **8B** is an external perspective view of the camera mount **201** according to an embodiment of the present invention as viewed from above.

FIG. **9A** is an external perspective view of the lens mount **301** according to an embodiment of the present invention as viewed from the side (rear face side) to which the camera mount **201** is mounted.

FIG. **9B** is an external perspective view of the camera mount **301** according to an embodiment of the present invention as viewed from above.

FIG. **10A** is a diagram exemplarily describing a contact state between terminals on the camera body **10** and the first lens unit **100** according to an embodiment of the present invention in a mounting start state where mounting of the first lens **100** to the camera body **10** has been started.

FIG. **10B** is a diagram exemplarily describing a contact state between terminals on the camera body **10** and the first lens unit **100** according to an embodiment of the present invention in a mounting completed state where mounting of the first lens **100** to the camera body **10** has been completed.

FIG. **11A** illustrates an external perspective view of a second lens unit **50** having been mounted to the camera body **10** via a first conversion adapter **40**.

FIG. **11B** illustrates an external perspective view of a state where the camera body **10**, first conversion adapter **40**, and second lens unit **50** have each been detached.

FIG. **12A** illustrates an external perspective view of a state where the first lens **100** is mounted to a camera body **60** via a second conversion adapter **70**.

FIG. **12B** illustrates an external perspective view where the camera body **60**, second conversion adapter **70**, and the first lens **100** have each been detached.

FIG. **13A** is a diagram exemplarily illustrating angle ranges that camera claws and camera recesses occupy in the circumferential direction of a camera mount **1401** with a lock pin **1401z** as a reference, as viewed from the rear face side (camera body **10** side).

FIG. **13B** is a diagram exemplarily illustrating angle ranges that multiple camera claws **1401a** through **1401c** occupy in the circumferential direction of the camera mount **1401**, as viewed from the rear face side (camera body **10** side).

FIG. **13C** is a cross-sectional diagram taken along a cross-section E-E' illustrated in FIG. **13B**.

FIG. **14A** is a diagram exemplarily illustrating angle ranges that lens claws and lens recesses occupy in the circumferential direction of a lens mount **1301** with a lock groove **1301z** as a reference, as viewed from the rear face side.

FIG. **14B** is a diagram exemplarily illustrating angle ranges that multiple lens recesses **1301a** through **1301c** occupy in the circumferential direction of the lens mount **1301**, as viewed from the rear face side.

FIG. **15A** is a diagram exemplarily describing a frontal view of partway through mounting a predetermined lens unit to a predetermined imaging apparatus that have claws and recesses that interfere with each other.

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FIG. **15B** is a cross-sectional view taken along a cross-section F-F' illustrated in FIG. **15A**.

FIG. **16A** illustrates a way in which a third lens claw and a first camera claw interfere when the lens mount **301** side is attempted to mount to the camera mount **401** side according to the embodiment of the present invention.

FIG. **16B** illustrates a way in which the second lens claw and the second camera claw interfere when the lens mount **301** side is attempted to mount to the camera mount **401** side according to the embodiment of the present invention.

FIG. **17A** is a diagram exemplarily describing a way in which the third lens claw and the third camera claw interfere in a case of attempting to insert the first lens claw into the third camera recess, in a case of attempting to insert incompatible claws into recesses at a lens mount **301** side and a camera mount **401** side according to the embodiment of the present invention.

FIG. **17B** is a diagram exemplarily describing a way in which the second lens claw and the first camera claw interfere in a case of attempting to insert the first lens claw into the third camera recess, in a case of attempting to insert incompatible claws into recesses at a lens mount **301** side and a camera mount **401** side according to the embodiment of the present invention.

FIG. **17C** is a diagram exemplarily describing a way in which the third lens claw and the second camera claw interfere in a case of attempting to insert the first lens claw into the second camera recess, in a case of attempting to insert incompatible claws into recesses at a lens mount **301** side and a camera mount **401** side according to the embodiment of the present invention.

FIG. **17D** is a diagram exemplarily describing a way in which the second lens claw and the third camera claw interfere in a case of attempting to insert the first lens claw into the second camera recess, in a case of attempting to insert incompatible claws into recesses at a lens mount **301** side and a camera mount **401** side according to the embodiment of the present invention.

FIG. **18A** is a diagram exemplarily illustrating angle ranges that camera claws and camera recesses occupy in the circumferential direction of the camera mount **1201** with a lock pin **1202** as a reference, as viewed from the rear face side.

FIG. **18B** is a diagram exemplarily illustrating angle ranges that, regarding multiple camera claws **1201a** through **1201c**, the claws occupy in the circumferential direction of the camera mount **1201**, as viewed from the rear face side.

FIG. **19A** is a diagram exemplarily illustrating angle ranges that lens claws and lens recesses occupy in the circumferential direction of the lens mount **1501** with a lock pin **1501z** as a reference, as viewed from the rear face side.

FIG. **19B** is a diagram exemplarily illustrating angle ranges where recesses are provided regarding the multiple lens recesses **1501a** through **1501c** in the circumferential direction of the lens mount **1501**, as viewed from the rear face side.

FIG. **19C** is a cross-sectional diagram taken along a cross-section G-G' illustrated in FIG. **19B**.

FIG. **20A** is a diagram exemplarily describing a way in which the third lens claw interfere with the third camera claw, when attempting to mount a reference claw at the lens mount **501** side to a reference recess at the camera mount **201** side according to an embodiment of the present invention.

FIG. **20B** is a diagram exemplarily describing a way in which the second lens claw interfere with the third camera claw, when attempting to mount a reference claw at the lens

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mount **501** side to a reference recess at the camera mount **201** side according to an embodiment of the present invention.

FIG. **21A** illustrates a way in which the first lens claw and the third camera claw interfere in a case of attempting to insert the second lens claw into the first camera recess, when attempting to mount a claw other than the reference claw at the lens mount **501** side to a reference recess at the camera mount **201** side according to an embodiment of the present invention.

FIG. **21B** illustrates a way in which the third lens claw and the third camera claw interfere in a case of attempting to insert the second lens claw into the first camera recess, when attempting to mount a claw other than the reference claw at the lens mount **501** side to a reference recess at the camera mount **201** side according to an embodiment of the present invention.

FIG. **21C** illustrates a way in which the second lens claw and the third camera claw interfere in a case of attempting to insert the third lens claw into the first camera recess, when attempting to mount a claw other than the reference claw at the lens mount **501** side to a reference recess at the camera mount **201** side according to an embodiment of the present invention.

FIG. **21D** illustrates a way in which the first lens claw and the third camera claw interfere in a case of attempting to insert the third lens claw into the first camera recess, when attempting to mount a claw other than the reference claw at the lens mount **501** side to a reference recess at the camera mount **201** side according to an embodiment of the present invention.

FIG. **22** is a diagram exemplarily describing a state in which claws provided to the camera mount **401** side and lens mount **501** side according to an embodiment of the present invention are engaged.

FIG. **23** is a disassembled perspective view of a mount mechanism **5000** according to a modification of the present invention.

FIG. **24A** is an external perspective view of a mount mechanism **5000** exemplarily illustrating a non-coupled state of the mount mechanism **5000** according to a modification of the present invention.

FIG. **24B** is a frontal view of the mount mechanism **5000** exemplarily illustrating a non-coupled state of the mount mechanism **5000** according to the modification of the present invention.

FIG. **24C** is a cross-sectional diagram taken along a cross-section H-H' illustrated in FIG. **24B**.

FIG. **25A** is an external perspective view of a mount mechanism **5000** exemplarily illustrating a coupled state of the mount mechanism **5000** according to a modification of the present invention.

FIG. **25B** is a frontal view of the mount mechanism **5000** exemplarily illustrating a coupled state of the mount mechanism **5000** according to the modification of the present invention.

FIG. **25C** is a cross-sectional diagram taken along a cross-section I-I' illustrated in FIG. **25B**.

DESCRIPTION OF THE EMBODIMENTS

FIGS. **1A** and **1B** are block diagrams of a camera accessory and imaging apparatus according to an embodiment of the present invention. FIG. **1A** is a diagram exemplarily describing a camera system including a first lens unit (hereinafter referred to as first lens) **100**, serving as an accessory according to the embodiment of the present inven-

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tion, and a camera body **10** serving as a first imaging apparatus to/from which the first lens **100** can be directly attached/detached. FIG. **1B** is a diagram describing the configuration of a mount portion **1** included in the first lens **100** and the camera body **10**. That is in FIG. **1**, the mounts provided on the first lens **100** and the camera body **10** are collectively referred to as the mount portion **1**. Each mount provided to the first lens **100** and camera body **10** will be described below.

As illustrated in FIG. **1A**, the camera body **10** has a charge-accumulation type solid-state imaging device (hereinafter referred to simply as sensor) **11**, such as a complementary metal-oxide semiconductor (CMOS) sensor or the like, that performs photoelectric conversion of an optical image of a subject guided by a photography lens group **101** provided on the inner side of the first lens **100** and outputs electric signals. The camera body **10** also has an analogue to digital (A/D) conversion unit **12** that converts analog electric signals output from the imaging sensor **11**, and an image processing unit **13** that generates image signals by performing various types of image processing on the digital signals. The image signals (still image and moving image) generated at the image processing unit **13** can be displayed on a display unit **14**, and recorded in a recording medium **15**.

The camera body **10** has memory **16**. This memory **16** serves as a buffer for performing processing on image signals, and also stores operation programs (computer programs) that a below-described camera control unit **18** uses.

The camera body **10** has a camera operating input unit **17** including a power source switch for turning the power source on/off, a photography switch (release switch) for starting recording of image signals, and selection/settings switches for performing various types of menu settings. The camera body **10** has the camera control unit **18** including a microprocessor (central processing unit (CPU) serving as a computer) that centrally controls the operations of the camera body **10** and first lens **100**. For example, the camera control unit **18** performs various types of settings based on signals input from the camera operating input unit **17**, or controls communication with a lens control unit **103** provided to the first lens **100** via the mount portion **1**.

On the other hand, the first lens **100** has the photography lens group **101** an optical member having multiple lens groups, such as a zoom lens, shift lens, and focus lens, and light amount adjustment members, such as a diaphragm. The first lens **100** has actuators for moving or operating optical members, such as the multiple lens groups and diaphragm, and a lens drive unit **102** that drives the actuators. The first lens **100** includes the lens control unit **103** including a microprocessor (logical CPU (LCPU)) for lenses, which centrally controls operations of the first lens **100**. The lens control unit **103** controls the lens drive unit **102** by communicating with the camera control unit **18** via, for example, the mount portion **1**.

(Basic Configuration of Mount Portion **1**)

Next, the configuration of the mount portion **1** including a camera mount face A provided on the camera body **10** side, and an accessory mount face B provided on the first lens **100** side will be described with reference to FIG. **1B**. The camera mount and accessory mount include a locking mechanism, a mount holding mechanism, and multiple electric terminals. The mounts will be described in detail below.

The mount portion **1** has multiple terminals that are capable of electrically connecting the camera body **10** and the first lens **100** to each other, as illustrated in FIG. **1B**. The multiple terminals are, at the camera mount face A, exposed to the outside of the camera body **10** as multiple camera-side

contact pins provided to a contact holding member **203**. The multiple terminals are, at the accessory mount face B, exposed to the outside of the camera accessory (e.g., first lens **100**) as multiple accessory-side contact faces provided to a contact face holding member **303**. The contact pins and the contact faces of the camera body **10** side and the camera accessory (e.g., first lens **100**) side are electrically connected to each other with mutually corresponding contacts in a state where the camera accessory is mounted to the camera body **10**.

A camera power source unit **19** generates a communication control power source (VDD) as a power source to be supplied to a mounted camera accessory via a VDD terminal, or power source to be supplied to a first communication I/F unit **21a** via a power source switching unit **20**. The camera power source unit **19** also generates a driving power source (VBAT) as a power source to be supplied to the mounted camera accessory via a VBAT terminal.

The camera power source unit **19** generates a 3.3 V power source as a power source to be supplied to the camera control unit **18**, first communication I/F unit **21a**, and a second/third communication I/F unit **21b**. The camera power source unit **19** also generates 3.0 V power source as a power source to be supplied to the first communication I/F unit **21a** and second/third communication I/F unit **21b** via the power source switching unit **20**.

The power source switching unit **20** is connected to the camera power source unit **19**. The power source switching unit **20** supplies only one of the VDD and 3.0 V power sources generated at the camera power source unit **19**, to the first communication I/F unit **21a** as a power source Vs for communication interface. Switching of power source voltage is executed following instructions from the camera control unit **18**.

Next, a lens power source **104** generates a 3.0 V power source, as a power source voltage for supply to the lens control unit **103** and a lens-side I/F unit **106**, based on VDD supplied from the camera body **10** side.

In a state where the first lens **100** is mounted to the camera body **10**, a driving power source (VBAT) is supplied to a drive circuit unit **105** at the first lens **100** side, via a VBAT terminal from the above-described camera power source unit **19**.

Although the power source voltage for the lens control unit **103** and lens-side I/F unit **106** is the same (3.0 V) in the present embodiment, a configuration may be made where the voltage level that the lens control unit **103** exhibits is 3.3 V. In this case, there is the need to supply power source of a voltage level of 3.0 V and 3.3 V to the lens-side I/F unit **106**, the lens power source **104** therefore generates power sources of 3.0 V and 3.3 V.

Hereafter, the functions of the terminals common to both the camera mount face A and accessory mount face B will be described. VDD terminals **203a** and **303a** are terminals that supply communication control power source (VDD) as communication power primarily used for communication control, to the camera accessory (e.g., the first lens **100**) from the camera body **10**. Note that while the voltage of power source supplied to the first lens **100** is 5.0 V, the voltage of supplied power source will change depending on the type of accessory mounted to the camera body **10**.

VBAT terminals **203b** and **303b** are terminals that supply driving voltage source (VBAT), which is driving power used for operating the mechanical driving units of the actuators used for driving the diaphragm and focus lens, from the camera side to the camera accessory side. In other words, the VBAT terminals **203b** and **303b** are terminals used to supply

power source other than the above-described communication power. Note that the voltage of the power source serving as driving power supplied to the first lens **100** is 4.25 V. The VDD terminals **203a** and **303a** and VBAT terminals **203b** and **303b** described above are power source terminals for supplying power source from the camera body **10** to the camera accessory, for example.

DGND terminals **203m** and **303m** are grounding terminals (GND terminals) corresponding to the communication control power source VDD. Note that grounding in the present embodiment means to set the voltage level of the grounding terminals to approximately the same level as the negative pole side of the power source, such as a battery.

PGND terminals **203d** and **303d** are grounding terminals for connecting the camera body **10**, and a mechanical drive system including motors (actuators) provided to a camera accessory (e.g., the first lens **100**), to ground level. That is to say, the PGND terminals **203d** and **303d** are grounding terminals (GND terminals) corresponding to the driving power source VBAT. The DGND terminals **203m** and **303m** and PGND terminals **203d** and **303d** described above are grounding terminals for grounding various types of power source systems in the camera body **10** and accessory to the ground level.

MIF terminals **203e** and **303e** are terminals for detecting that the camera accessory (e.g., the first lens **100**) has been mounted to the camera body **10**. In the present embodiment, the MIF terminals **203e** and **303e** detect that the first lens **100** has been mounted to the camera body **10**.

The camera control unit **18** detects that the camera accessory has been mounted to or detached from the camera body **10**, by detecting the voltage level that the MIF terminals **203e** and **303e** indicated. Based on this detection, the camera control unit **18** starts supply of power source to power source terminals after having detected mounting of, for example, the camera accessory, and effects control to start communication between the camera body **10** and camera accessory.

TYPE terminals **203c** and **303c** are terminals for distinguishing the type of camera accessory (e.g., the first lens **100**) mounted to the camera body **10**. The camera control unit **18** detects the value of voltage of signals indicated by the TYPE terminals **203c** and **303c**, and distinguishes the type of camera accessory mounted to the camera body **10** based on this value. The first lens **100** is pull-down connected to the DGND terminal at a predetermined resistance value. This resistance value differs depending on the type of the camera accessory.

Next, description will be made regarding various types of communication terminals between the camera body **10** and the camera accessory. The multiple communication terminals provided to the mount portion **1** are divided into multiple communication systems (groups), and each communication system is capable of performing communication independently. In the present embodiment, LCLK terminals **203h** and **303h**, DCL terminals **203f** and **303f**, and DLC terminals **203g** and **303g**, are a first communication system that performs first communication. DLC2 terminals **203i** and **303i** are a second communication system that performs second communication, which is independent from the first communication. Further, CS terminals **203k** and **303k**, and DCA terminals **203j** and **303j** are a third communication system that performs third communication, which is independent from the first and second communication. In the present embodiment, the camera control unit **18** and lens

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control unit **103** can perform communication independently with the first through third, via the above-described multiple communication terminals.

LCLK terminals **203h** and **303h** are terminals of the first communication system, and are terminals for communication clock signals output from the camera body **10** to the camera accessory (e.g., the first lens **100**), and terminals for the camera body **10** to monitor the busy state of the accessory.

DCL terminals **203f** and **303f** are terminals of the first communication system, and are communication data terminals for performing bidirectional communication between the camera body **10** and the camera accessory (e.g., the first lens **100**).

DLC terminals **203g** and **303g** are terminals of the first communication system, and are terminals for communication data output from the camera accessory (e.g., the first lens **100**) to the camera body **10**.

The above-described LCLK terminals, DCL terminals, and DLC terminals, corresponding to the first communication system, can switch the signal output format between so-called CMOS output type and open type. In the present embodiment, the CMOS output type has switch output of voltage at both H (High) and L (Low). In comparison with this, the open type has switch output at only the L side. Although open type used in the present embodiment is the so-called open drain type, the open type may be the open collector type.

DLC2 terminals **203i** and **303i** are terminals of the second communication system, and are terminals for communication data output from the camera accessory (e.g., the first lens **100**) to the camera body **10**.

DCA terminals **203j** and **303j** are terminals of the third communication system, and are terminals for communication data for performing bidirectional communication between the camera body **10** and the camera accessory (e.g., the first lens **100**).

CS terminals **203k** and **303k** are terminals of the third communication system, and are signal terminals for communication requests between the camera body **10** and camera accessory (e.g., the first lens **100**). In the present embodiment, in a case where the first lens **100** is mounted to the camera body **10**, the communication voltage at the corresponding terminals in the first through third communication systems is 3.0 V.

(Configuration of Camera Mount and Accessory Mount)

The configuration of the camera mount **201** and the lens mount (accessory mount) **301** according to the present embodiment will be described with reference to FIGS. **2A** through **3B**. In the following description, the mount provided to the imaging apparatus side will be referred to as camera mount, and the mount provided to the lens unit side will be referred to as lens mount. Concerning mounts of the below-described conversion adapter, the side mounted to the imaging apparatus will be referred to as lens mount, and the side mounted to the lens unit will be referred to as camera mount. In the following description, a second conversion adapter **70** has a camera mount **1201** that is the same as the camera mount **201** provided on the camera body **10**. In the following description, a first conversion adapter **40** has a lens mount **1301** that is the same as the lens mount **301** provided on the first lens **100**.

FIGS. **2A** and **2B** are external perspective views of the camera body **10** and first lens **100** according to the embodiment of the present invention. FIG. **2A** illustrates a state in which the first lens **100** is mounted to the camera body **10**,

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and FIG. **2B** illustrates a state in which the first lens **100** has been removed from the camera body **10**.

As illustrated in FIGS. **2A** and **2B**, the camera body **10** and first lens **100** have a ring-shaped camera mount **201** and ring-shaped lens mount **301**, which each have contact faces that are parallel in a direction orthogonal to the optical axis. The camera mount **201** and the lens mount **301** do not necessarily have a ring shape, and the shape may be an arc-shape. For example, a part of the ring shaped contact face of the camera mount **201** illustrated in FIGS. **2A** and **2B** may be notched out. In the present embodiment, the center axis of the camera mount **201** and the center axis of the lens mount **301** are same as each optical axis of the camera mount **201** or lens mount **301**. A configuration may be employed where the optical axis is different from the center axis. By rotating the camera body **10** and first lens **100** relative to each other, in a state where the reference faces of each of the camera mount **201** and lens mount **301** are in contact with each other, from an unlocked position to a locked position, the first lens **100** is mounted to the camera body **10**. This will be described in detail below.

FIG. **3** illustrates disassembled perspective views of the camera body **10** and first lens **100** according to the embodiment of the present invention. FIG. **3A** illustrates a disassembled perspective view of the camera body **10**, and FIG. **3B** illustrates a disassembled perspective view of the first lens **100**.

As illustrated in FIG. **3A**, a first optical axis (a first center axis) **3000** indicates a light flux passing through the center of the camera mount **201** of the camera body **10**. As illustrated in FIG. **3B**, an optical axis **3001** is an axis for exemplarily describing a light flux representing an optical image of a subject guided by the first lens **100**. When the first lens **100** is mounted to the camera body **10**, the first optical axis **3000** and optical axis **3001** described above substantially match, and optical fluxes corresponding to these optical axes enter the imaging sensor **11** and are imaged, thereby imaging a subject image.

A lock pin **202** is a first restricting member (lock member) for restricting (locking) the mounted state of the camera body **10** and camera accessory as illustrated in FIG. **3A**. The lock pin **202** is capable of advancing and retreating in a direction parallel to the first optical axis **3000**. Specifically, when the first lens **100** is mounted to the camera body **10**, the lock pin **202** enters a lock groove **301z** a second restricting member provided to the lens mount **301**, and locks the state of the first lens **100** being mounted to the camera body **10**. The lock pin **202** is retracted from the lock groove **301z** by the user operating a lock disengaging member (not illustrated) connected to the lock pin **202**, thereby the locking of the first lens **100** mounted to the camera body **10** can be disengaged. Even in a case of further rotating the first lens **100** and camera body **10** relative to each other in the mounting direction with the lock disengaging member operated, rotation beyond a lock abutting face **301y** is restricted (see FIG. **7A**).

The contact holding member **203** is a holding unit for holding below-described (see FIG. **8**) multiple contact pins (electrical communication contact group) **203a** through **203k** and **203m**, provided on the camera body **10** side. In the present embodiment, the contact pins and contact faces corresponding to the above-described terminals are denoted by the same part numerals as the terminals for the sake of description.

A camera body member **204** is a camera housing that holds the members of the camera body **10**. Specifically, the above-described camera mount **201** and contact holding

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member **203** are fastened to the camera body member **204** by camera mount fastening screws **205a** through **205d** and contact holding member fastening screws that are not illustrated. Camera mount fastening screws **205a** through **205d** are positioned where none of multiple leaf springs **206a**, **206b**, and **206c** are positioned in the camera mount **201**. The camera mount fastening screws **205a** through **205d** are exposed at the contact face of the camera mount **201**, the contact face being contacted with the lens mount **301**. Although the camera body member **204** holds an unillustrated imaging unit to which the aforementioned imaging sensor **11** is mounted and an unillustrated shutter unit, description will be omitted.

A lens mount biasing member **206** is a biasing unit at the imaging apparatus side to pull lens claws **301d** and **301e**, which are multiple bayonet claws provided to the below-described lens mount **301**, toward the camera mount **201** side. Specifically, the lens mount biasing member **206** includes multiple leaf springs **206a**, **206b**, and **206c**, and the bayonet claws provided to the first lens **100** are biased in the optical axis direction by the leaf springs. The lens mount biasing member **206** is held by both the camera mount **201** and camera body member **204** in a space formed therebetween.

A front lens **101a** and rear lens **101b** are optical members making up the photography lens group **101**, as illustrated in FIG. 3B. While the photography lens group **101** has multiple lenses, only the front lens **101a** disposed at the subject-side end and the rear lens **101b** disposed at the camera body **10** side end are illustrated in the present embodiment to simplify description.

A lens barrel **302** is a holding member that holds the photography lens group **101**. The lens mount **301** is fastened to the lens barrel **302** by lens mount fastening screws **304a** through **304d**. The lens mount fastening screws **304a** through **304d** are exposed at the contact face of the lens mount **301**, the contact face being contacted with the camera mount **201**.

A contact face holding member **303** is a holding unit for holding multiple contact faces (electrical communication contact group) **303a** through **303k** and **303m** provided to a below-described (see FIG. 9) first lens **100** side. The contact face holding member **303** is fastened to the lens mount **301** by lens contact portion holding portion fastening screws **305a** and **305b**.

(Configuration of Bayonet Claws at Camera Body **10** Side)

Next, the multiple bayonet claws provided to the camera mount **201** at the camera body **10** side will be described with reference to FIGS. 4 through 6. FIG. 4 illustrates diagrams exemplarily describing a case of viewing the camera mount **201** according to the embodiment of the present invention from the photographer side (rear face side of the camera body **10**) of the camera body **10**. FIG. 4A is a diagram illustrating angle ranges that camera mount claws (hereinafter referred to simply as camera claws) and camera mount recesses (hereinafter referred to simply as camera recesses) having the lock pin **202** as a reference occupy in the circumferential direction of the camera mount **201**. The camera recesses of the camera mount **201** are notch portions between adjacent claws. Camera recesses of the mount are therefore formed further toward the outside in the radial direction with respect to the center axis of the mount than the claws of the camera mount **201**. Lens recesses of the lens mount **301** are formed further toward the inside in the radial direction with respect to the center axis (parallel to the optical axis) of the mount than the lens claws. FIG. 4B is a diagram illustrating angle ranges that multiple camera claws

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201a through **201c** occupy in the circumferential direction of the camera mount **201**. FIG. 4C is a cross-sectional diagram taken along a cross-section A-A' illustrated in FIG. 4B.

In the following description, protrusions protruding from recesses in the inner radial direction at the camera mount side will be referred to as camera claws, and protrusions protruding from recesses in the outer diameter direction at the below-described lens mount (accessory mount) side will be referred to as lens mount claws.

In the following description, the side of the camera mount **201** as viewed from the side of the photographer (rear face of the camera body **10**) of the camera body **10** will be referred to as rear face side, and the opposite side will be referred to as front face side (or camera accessory side). It should be noted that in the following description, description will be made with a case of viewing the camera mount **201** from the rear face side as a reference, but in a case of viewing the camera mount **201** from the front face side, angles and directions in the following description stipulating angles (clockwise, etc.) will be reversed.

As illustrated in FIGS. 4A and 4B, a first camera claw **201a**, second camera claw **201b**, and third camera claw **201c** are provided in order to the camera mount **201** in the circumferential direction (inner radial direction). When viewing the camera mount **201** from the rear face side as illustrated in FIG. 4, the camera claw provided at a position farthest from the lock pin **202** is the first camera claw **201a**. The second camera claw **201b** and third camera claw **201c** are consecutively provided in order from the first camera claw **201a** in a clockwise direction.

Recesses, which are a first camera recess **201d**, second camera recess **201e**, and third camera recess **201f**, are provided in order to the camera mount **201** in the circumferential direction (inner radial direction). When viewing the camera mount **201** from the rear face side as illustrated in FIG. 4, the recess that is provided at a position nearest to the lock pin **202** is the second camera recess **201e**. The third camera recess **201f** and first camera recess **201d** are consecutively provided in order from the second camera recess **201e** in a clockwise direction.

As illustrated in FIG. 4C, a fitting member **201x**, which restricts movement of the camera accessory in a direction parallel to the optical axis when the camera accessory is mounted, is provided to the camera mount **201** side in the circumferential direction. In the present embodiment, the diameter (i.e., the inner diameter of the fitting member **201x**) in a direction orthogonal to the optical axis (approximately the same as the center axis) of the fitting member **201x** at the camera mount **201** side is the mount diameter.

To mount the first lens **100** to the camera body **10**, the camera claws of the camera body **10** are first inserted into the lens mount recesses of the first lens **100**, and the lens mount claws at the first lens **100** side are inserted into the camera recesses at the camera body **10** side. This state is a mounting start state (first state). In the following description, accessory mount recesses will be referred to simply as accessory recesses, and accessory mount claws will be reference to simply as accessory claws. For example, in the case of the lens mount **301**, this will be lens recesses and lens claws.

The camera mount **201** and lens mount **301** are then rotated relative to each other from the mounting start state, whereby the camera claws and lens claws engage in the radial direction orthogonal to the optical axis, and transition to a state where the lens claws are biased by the above-described lens mount biasing member **206**. In this state,

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relative positional relation between the camera body **10** and the first lens **100** transitions to a locked position where the lock pin **202** has fit into the lock groove **301z** as described above. This state is a mounting complete state (second state). In the mounting complete state, contact of corresponding terminals at the camera mount **201** and lens mount **301** is complete.

As described above, the camera claws and accessory claws are so-called bayonet claws, and as described above, have shapes such that the first lens **100** can be mounted (can be coupled) to the camera body **10** by the so-called bayonet coupling method by engaging of the camera claws and accessory claws.

For the description, the end portions of the camera claws **201a**, **201b**, and **201c**, in the circumferential direction, will be referred to as first end portion **201a1**, second end portion **201a2**, third end portion **201b1**, fourth end portion **201b2**, fifth end portion **201c1**, and sixth end portion **201c2**, respectively. Each of the end portions are denoted with part numerals in order from the first camera claw **201a** in the clockwise direction, when viewing the camera mount **201** from the rear face side, as described above. The camera claws **201a**, **201b**, and **201c** include first, second, and third leaf spring disposition portions **201a3**, **201b3**, and **201c3** in which the above-described plurality of leaf springs **206a**, **206b**, and **206c** of the lens mount biasing member **206** are disposed, respectively. In the present embodiment, the first, second, and third leaf spring disposition portions **201a3**, **201b3**, and **201c3** each correspond to the notch portions provided between the above-described end portions of each camera claw. The size of the notch portions is a size at which each of the plurality of leaf springs can be arranged. For example, the first leaf spring disposition portions **201a3** is a notch (separation portion) provided in the camera claw **201a** between the first end portion **201a1** and the second end portion **201a2** of the camera claw **201a**.

As illustrated in FIG. 4A, the angles (angle ranges) that the camera claws and camera recesses occupy in the circumferential direction of the camera mount **201** are stipulated as follows in the present embodiment. For the first camera claw **201a**, the angle $\theta a = 48^\circ$. For the second camera claw **201b**, the angle $\theta b = 40^\circ$. For the third camera claw **201c**, the angle $\theta c = 92^\circ$. Further, for the first camera recess **201d**, the angle is 57° . For the second camera recess **201e**, the angle is 66° . For the third camera recess **201f**, the angle is 57° . In here, the angle range of the claw (bayonet claw) and the recess indicates an angle range in which each bayonet claw and the recess (notch portion) the bayonet claw can be inserted occupy in each mount. For example, if the camera mount **201** has a ring shape, the angle range occupied by the first camera claw **201a** with respect to the entire camera mount **201** (360°) becomes 40° . Even in a case where the mount does not have a complete ring shape, the mount may be considered ring-shaped (360°) to define the angular range of the claws and recesses.

When viewing the camera claws from the rear face side of the camera body **10**, the angles in the clockwise direction where the camera claws are disposed in the circumferential direction of the camera mount **201** with the position (referred to as reference position) of the lock pin **202** as a reference are stipulated as follows. The first camera claw **201a** is disposed between 169° and 217° with the reference position as a start point. The second camera claw **201b** is disposed between 274° and 314° with the reference position as a start point. The third camera claw **201c** is disposed between 20° and 112° with the reference position as a start point.

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When viewing the camera mount **201** from the optical axis direction in the present embodiment, of the multiple bayonet claws, the third camera claw **201c** overlaps a first camera mount center line **3003** that extends in the vertical direction from the center axis, which is parallel to the optical axis, of the camera mount **201** in the radial direction of the camera mount **201**. The first camera mount center line **3003** is a line that, in a normal position of the camera body **10** where the lock pin **202** is situated at the left side when viewing the camera mount **201** from the front face side, extends in the gravitational direction and the opposite direction from the gravitational direction, from the center (optical axis) of the camera mount **201**. That is to say, the first camera mount center line **3003** is a vertical line that passes through the optical axis in a normal position of the camera. In other words, the first camera mount center line **3003** is a vertical line that passes through the center of the camera mount **201** and orthogonal to the center axis of the camera mount **201** when the camera body **10** is placed on a horizontal plane.

In contrast, a line that, in a vertical position of the camera body **10** where the lock pin **202** is situated at the top side when viewing the camera mount **201** from the front face side, extends in the gravitational direction and the opposite direction from the gravitational direction, from the center (optical axis) of the camera mount **201**, is a second camera mount center line **3002**. That is to say, the second lens mount center line **3002** is a horizontal line that passes through the optical axis when the camera body **10** is in the normal position.

The second camera mount center line **3002** overlaps the lock pin **202** and first camera claw **201a** in the radial direction of the camera mount **201**. The first camera mount center line **3003** and the second camera mount center line **3002** are substantially orthogonal.

FIG. 5 illustrates diagrams exemplarily describing engagement of the camera body **10** and first lens **100** by camera claws at a normal position of the camera body **10** according to the embodiment of the present invention. FIG. 5A is a diagram illustrating a state where the first lens **100** is mounted to the camera body **10**, when viewing from the front face side. FIG. 5B is a partial cross-sectional view of the camera body **10** and first lens **100** taken along a cross-section B-B' illustrated in FIG. 5A. In FIG. 5, the camera body **10** is positioned in the above-described normal position, and, in this state, a camera grip **204a** provided to the camera body member **204** is situated to the left side when viewing the camera body **10** from the front face side.

FIG. 6 illustrates diagrams exemplarily describing engagement of the camera body **10** and first lens **100** by bayonet claws at a vertical position of the camera body **10** according to the embodiment of the present invention. FIG. 6A is a diagram illustrating a state where the first lens **100** is mounted to the camera body **10**, when viewing from the front face side. FIG. 6B is a partial cross-sectional view of the camera body **10** and first lens **100** taken along a cross-section C-C' illustrated in FIG. 6A. In FIG. 6, the camera body **10** is positioned in the above-described vertical position, and the camera grip **204a** in this state is situated to the top side when viewing the camera body **10** from the front face side.

In general, in a case of mounting an interchangeable lens to a camera, a gap occurs between the camera and interchangeable lens in a direction orthogonal to the optical axis, due to dimensional tolerance of the two, and looseness of the lens as to the camera increases due to this gap. Particularly, in a case of coupling a camera and interchangeable lens by

bayonet coupling, the above-described looseness of the lens becomes greater the farther away from positions where the bayonet claws engage each other in the circumferential direction of the lens. Further, the looseness (bowing) in the gravitational direction of the interchangeable lens as to the camera becomes greater due to the weight of the interchangeable lens itself. Thus, if a subject is imaged in a case where the looseness of the interchangeable lens as to the camera is great, an unnatural image may be obtained, which is unintended by the user, due to the deviation of optical axis of the camera and optical axis of the lens.

In the present embodiment, the third camera claw **201c**, which has the widest angle of the camera-side bayonet claws, overlaps the first camera mount center line **3003** in the radial direction of the camera mount **201**, as illustrated in FIG. 5B. According to this configuration, when the first lens **100** is mounted to the camera body **10**, the first camera mount center line **3003** overlaps the position where the third camera claw **201c** and a later-described third lens claw **301f** are engaged. In this case, looseness (bowing) of the camera accessory, such as the first lens **100** mounted to the camera body **10**, in the gravitational direction can be suppressed in the normal position of which the frequency of usage is highest for operating the imaging apparatus, for example. Due to the above-described configuration, the gap **41** illustrated in FIG. 5B can be kept from becoming large, so looseness of the first lens **100** as to the camera body **10** in the direction indicated by the arrow illustrated in FIG. 5B can be suppressed.

In the present embodiment, the first camera claw **201a** overlaps the second camera mount center line **3002** in the radial direction of the camera mount **201**, as illustrated in FIG. 6B. According to this configuration, when the first lens **100** is mounted to the camera body **10**, the second camera mount center line **3002** overlaps the position where the first camera claw **201a** and a later-described first lens claw **301d** are engaged. In this case, for example, looseness (bowing) of the camera accessory, such as the first lens **100** mounted to the camera body **10**, in the gravitational direction can be suppressed even in the vertical position of the imaging apparatus. Due to the above-described configuration, since the gap **42** illustrated in FIG. 6B can be kept from becoming large, looseness of the first lens **100** as to the camera body **10** in the direction indicated by the arrow illustrated in FIG. 6B can be suppressed.

As illustrated in FIG. 4B, the angle of the third camera claw **201c** in the circumferential direction of the camera mount **201**, situated at the top side in a case where the camera body **10** is at the normal position, is greater than the sum of angles of the first claw **201a** and second claw **201b** situated at the lower side. More precisely, the total sum of angles in the circumferential direction of the camera claws provided to the camera mount **201** is greater at the upper side of the second camera mount center line as a reference as compared to the lower side. Specifically, the sum of angle θc and the angle $\theta a1$ of the first camera claw **201a** on the upper side of the second camera mount center line **3002** is larger than the sum of angle θb and the angle $\theta a2$ of the first camera claw **201a** on the lower side of the second camera mount center line **3002**. That is to say, the layout of the camera claws and recesses is determined to satisfy the two following expressions:

$$\theta c \geq \theta a + \theta b \quad (1), \text{ and}$$

$$\theta a1 + \theta c > \theta a2 + \theta b \quad (2).$$

According to this configuration, in the normal position of the camera body **10** regarding which the frequency of usage is highest, the strength of camera claws at the upper side (opposite side from the gravitational direction) of the second camera mount center line **3002** that indicates the horizontal direction of the camera mount **201** can be made to be greater than the strength of the camera claws at the lower side (gravitational direction). Thus, in the normal position of the camera body **10** regarding which the frequency of usage is highest in a state where the first lens **100** is mounted, the camera body **10** according to the present embodiment can reduce looseness (bowing) of the first lens **100** as to the camera body **10** in the gravitational direction. According to the camera body **10** of the present embodiment, in the normal position of the camera body **10** regarding which the frequency of usage is highest in a state where the first lens **100** is mounted, deformation of camera claws and lens claws due to the weight of the first lens **100** itself can be suppressed.

(Configuration of Bayonet Claws at First Lens **100** Side)

Next, multiple lens claws (accessory claws) provided to the lens mount **301** at the first lens **100** side will be described with reference to FIG. 7. FIG. 7 is a diagram exemplarily describing a case of viewing the lens mount **301** according to the embodiment of the present invention from the rear face side (side where the camera body **10** is attached) in a state where the first lens **100** is mounted to the camera body **10**. FIG. 7A exemplarily describes angles of the claws and recesses in the circumferential direction at the lens mount **301** side. FIG. 7B is a cross-sectional view taken along a cross-section D-D' illustrated in FIG. 7A. In the following description, the camera body **10** is positioned in the normal position in the state illustrated in FIG. 7.

As illustrated in FIG. 7A, a first lens claw **301d**, second lens claw **301e**, and third lens claw **301f** are provided in order to the lens mount **301** in the circumferential direction (inner radial direction). When viewing the lens mount **301** from the rear face side as illustrated in FIG. 7A, the lens claw provided at a position farthest from the lock groove **301z** is the first lens claw **301d**. The second lens claw **301e** and third lens claw **301f** are consecutively provided in order from the first lens claw **301d** in a clockwise direction.

Recesses, which are a first lens recess **301a**, second lens recess **301b**, and third lens recess **301c**, are provided in order, to the lens mount **301** in the circumferential direction (inner radial direction). When viewing the lens mount **301** from the rear face side, the recess provided at a position nearest to the lock groove **301z** is the third lens recess **301c** as illustrated in FIG. 7. The first lens recess **301a** and second lens recess **301b** are then consecutively provided in order from the third lens recess **301c** in a clockwise direction.

As illustrated in FIG. 7B, a fitting member **301x**, which restricts movement in a direction parallel to the optical axis of the imaging apparatus when mounted on the imaging apparatus, is provided to the lens mount **301** side in the circumferential direction. In the present embodiment, the diameter (i.e., the inner diameter of the fitting member **301x**) in a direction orthogonal to the optical axis (approximately the same as the center axis) of the fitting member **301x** at the lens mount **301** side is the mount diameter.

For the sake of description, the end portions of the lens claws **301d**, **301e**, and **301f**, in the circumferential direction, will be referred to as first end portion **301d1**, second end portion **301d2**, third end portion **301e1**, fourth end portion **301e2**, fifth end portion **301f1**, and sixth end portion **301f2**. The end portions are denoted with part numerals in order

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from the first lens claw **301d** in the clockwise direction, when viewing the lens mount **301** from the rear face side, as described above.

As illustrated in FIG. 7, the angles (angle ranges) that the lens claws and lens recesses occupy in the circumferential direction of the lens mount **301** are stipulated as follows in the present embodiment. For the first lens claw **301d**, the angle $\theta d=53^\circ$. For the second lens claw **301e**, the angle $\theta e=62^\circ$. For the third lens claw **301f**, the angle $\theta f=53^\circ$. Further, for the first lens recess **301a**, the angle is 52° . For the second lens recess **301b**, the angle is 44° . For the third lens recess **301c**, the angle is 96° .

When viewing the lens claws from the rear face side, the angles where the lens claws are disposed in the circumferential direction of the lens mount **301** with the position (referred to as reference position) of the lock groove **301z** as a reference in the clockwise direction are stipulated as follows. The first lens claw **301d** is disposed between 159° and 212° with the reference position as a start point. The second lens claw **301e** is disposed between 256° and 318° with the reference position as a start point. The third lens claw **301f** is disposed between 54° and 107° with the reference position as a start point.

When viewing the lens mount **301** from the optical axis direction in the present embodiment, of the multiple bayonet claws, the second lens claw **301e** overlaps a first lens mount center line **3005** that extends in the vertical direction of the lens mount **301**, in the radial direction of the lens mount **301**. When viewing the lens mount **301** from the rear face side and front face side in the present embodiment, of the multiple bayonet claws, the third lens claw **301f** overlaps the first lens mount center line **3005** in the radial direction of the lens mount **301**.

The first lens mount center line **3005** is a line that extends in the gravitational direction and the opposite direction from the gravitational direction, from the center (optical axis) of the camera mount **301**, in a normal position of the camera body **10** to which the first lens **100** has been mounted. In other words, the first lens mount center line **3005** is a vertical line that passes through a center of the lens mount **301** and is orthogonal to the center axis of the lens mount **301** when the camera body **10** to which the first lens **100** is mounted is placed on a horizontal plane. In contrast, in a vertical position of the camera body **10** to which the first lens **100** has been mounted, a line that extends in the gravitational direction and the opposite direction from the gravitational direction, from the center (optical axis) of the camera mount **301**, is a second lens mount center line **3006**. This second lens mount center line **3006** overlaps the lock groove **301z** and first lens claw **301d** in the radial direction of the lens mount **301**. The first lens mount center line **3005** and second lens mount center line **3006** are mutually orthogonal.

In the present embodiment, when the first lens **100** is mounted to the camera body **10**, the second lens claw **301e** and third lens claw **301f** overlap the first lens mount center line **3005** in the radial direction of the lens mount **301**, as illustrated in FIG. 7. According to this configuration, the first lens mount center line **3005** overlaps two positions of engaging positions between lens claws and camera claws, when the first lens **100** is mounted to the camera body **10**. In this case, for example, in the normal position of the camera body **10** regarding which the frequency of usage is highest, looseness (bowing) of the first lens **100** mounted to the camera body **10** in the gravitational direction can be reduced. In a case where the first lens **100** is mounted to the camera body **10**, the third leaf spring arrangement portion **201c3** described above in FIGS. 4A and 4B overlaps with

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the first lens mount center line **3005** in the radial direction of the lens mount **301**. Specifically, in the case where the first lens **100** is mounted to the camera body **10**, the region where the third lens claw **301f** is biased by the leaf spring **206c** overlaps with the first lens mount center line **3005**, and thus the rattling of the first lens **100** can be suppressed more effectively.

In the present embodiment, when the first lens **100** is mounted to the camera body **10**, the first lens claw **301d** overlaps the second lens mount center line **3006** in the radial direction of the lens mount **301**, as illustrated in FIG. 7A. According to this configuration, when the first lens **100** is mounted to the camera body **10**, the second lens mount center line **3006** overlaps the position where the first lens claw **301d** and the first camera claw **201a** engage. In this case for example, even in the vertical position of the imaging apparatus, looseness (bowing) of the first lens **100** mounted to the camera body **10** in the gravitational direction can be reduced. In a case where the first lens **100** is mounted to the camera body **10**, the first leaf spring arrangement portion **201a3** described above in FIGS. 4A and 4B overlaps with the second lens mount center line **3006** in the radial direction of the lens mount **301**. Specifically, in the case where the first lens **100** is mounted to the camera body **10**, the region where the first lens claw **301d** is biased by the leaf spring **206a** overlaps with the first lens mount center line **3006**, and thus the rattling of the first lens **100** can be suppressed more effectively.

In a state where the first lens **100** is mounted to the camera body **10**, the second camera mount center line **3002** and second lens mount center line **3006** overlap a lock region where the lock pin **202** and lock groove **301z** are fit, as described above. According to this configuration, in the vertical position of the camera body **10**, the engaging position of the first camera claw **201a** and first lens claw **301d** and the above-described lock region overlap the mount center lines extending in the gravitational direction and the opposite direction thereof. According to this configuration, in the vertical position of the imaging apparatus, for example, looseness (bowing) of the first lens **100** mounted to the camera body **10** in the gravitational direction can be suppressed even more effectively.

(Configuration of Contact Pins in Camera Body **10**)

The configuration of the contact pins provided to the camera body **10** side will be described with reference to FIG. 8 below. FIG. 8 illustrates external perspective views viewing a camera mount **201** according to the embodiment of the present invention from the front face side (subject side). FIG. 8A is an external view of the camera mount **201** from the optical axis direction. FIG. 8B is an external perspective view of the camera mount **201** from above.

The contact holding member **203**, and contact pins **203a** through **203k** and **203m** held by the contact holding member **203** are disposed following the circumferential direction of the camera mount **201** on the inner side of the camera mount **201**, as illustrated in FIG. 8A. FIG. 8A illustrates an array line on which the contact points are disposed as an array line **3007**.

The contact pins **203a** through **203k** and **203m** are movable pins that can advance and retreat (protrude and retract) in a direction parallel to the optical axis **3000**, and are biased from behind toward the first lens **100** side by leaf springs (not illustrated). As described above, the contact pins **203a** through **203k** and **203m** have the functions of the above-described respective terminals, and the part numerals by which the contact pins are denoted are the same as those of the terminals, to facilitate description.

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As illustrated in FIG. 8B, out of the contact pins **203a** through **203k** and **203m**, the contact pins **203a**, **203b**, **203c**, and **203d** are higher in the direction (toward the front face) parallel to the optical axis **3000** as compared to the other contact pins. The above-described configuration is realized by differing the amount of protrusion from the contact holding member **203** in the optical axis direction in the present embodiment. Alternatively, a configuration may be employed where this is realized by differing the amount of protrusion of the contact pins from the contact holding member **203**.

In general, in a case of employing the bayonet coupling method and attaching/detaching an interchangeable lens to/from a camera, the contact pins at the camera side and the contact face on the lens side slide over each other. For example, at the camera side, contact pins other than the contact pin situated at the edge in the rotational direction to complete mounting of the interchangeable lens slide over one or another contact face provided to the lens side. Thus, the more times the camera accessory is attached to and detached from the camera, the more the contact pins and contact faces are worn.

The greater the number of terminals there is, the more pronounced this problem is, and the number of times of sliding between contact pins and contact faces increases. Voltage drop increases due to the contact impedance of the terminals rising due to wearing of the contact pins and contact faces, and the interchangeable lens may malfunction, for example, due to the voltage dropping below the allowable voltage range for operation of the electric circuits.

To decrease the number of times of sliding of the contact pins, the contact height of the camera-side contact pins and the interchangeable-lens-side contact faces are differed in a direction parallel to the optical axis **3000**, between an upper tier and a lower tier, in the present embodiment. Specifically, the contact holding member **203** has a step in a direction parallel to the optical axis **3000**, with the contact pins **203a** through **203d** being provided to the upper tier, and the contact pins **203e** through **203k** and **203m** being provided to the lower tier. The contact face holding member **303** also is stepped in a direction parallel to the optical axis **3001**, with the contact faces **303a** through **303d** provided to the lower tier, and the contact faces **303e** through **303k** and **303m** provided to the upper tier.

The upper tier of the contact holding member **203** at the camera mount **201** side is a tier protruding toward the front face side (subject side) of the camera body **10** namely the side the lens mount **301** is coupled when the first lens **100** is mounted to the camera body **10**. The lower tier of the contact holding member **203** is a tier recessed toward the rear face side (imaging sensor **11** side) of the camera body **10**. The upper tier of the contact face holding member **303** at the lens mount **301** side is a tier protruding toward the rear face side (imaging sensor **11** side) of the camera body **10** namely the side the camera mount **201** is coupled when the first lens **100** is mounted to the camera body **10**. The lower tier of the contact face holding member **303** is a tier recessed to the front face side (subject side) of the camera body **10**.

In a case where the lens mount **301** is mounted to the camera mount **201**, the contact pins provided to the upper tier side of the contact holding member **203** and the contact faces provided to the lower tier side of the contact face holding member **303** are electrically in contact among corresponding terminals. In a case where the lens mount **301** is mounted to the camera mount **201**, the contact pins provided to the lower tier side of the contact holding member **203** and the contact faces provided to the upper tier

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side of the contact face holding member **303** are electrically in contact among corresponding terminals. Thus, the steps at the camera mount **201** side and lens mount **301** side have a mutually engageable shape, and the contact pins at the camera side and contact faces at the lens side that are provided to different steps do not come into contact when mounting the lens mount **301** to the camera mount face **A**.

According to this configuration, the contact pins **203e** through **203k** and **203m** do not come into contact with the contact faces **303a** through **303d**, when relatively rotating the first lens **100** as to the camera body **10** from the mounting start state to the mounting completed state. Even when relatively rotating the first lens **100** as to the camera body **10** from the mounting completed state to the mounting start state, the contact pins **203e** through **203k** and **203m** do not come into contact with the contact faces **303a** through **303d**. That is to say, the number of times of sliding between contact pins and contact faces can be reduced on both the camera mount **201** side and lens mount **301** side.

The contact pins at the camera body **10** side and the contact faces at the first lens **100** side are all in a non-contact state in the mounting start state of the first lens **100** as to the camera body **10**, and thereby short-circuiting among terminals before completion of mounting of the first lens **100** can be prevented.

(Configuration of Contact Faces in First Lens **100**)

The following is a description regarding the configuration of the contact faces provided to the first lens **100** side, with reference to FIG. 9. FIG. 9 illustrates external perspective views of the lens mount **301** according to the embodiment of the present invention, as viewed from the side (rear face side) to which the camera mount **201** is mounted. FIG. 9A is an external view of the lens mount **301** from the optical axis direction. FIG. 9B is an external perspective view of the lens mount **301** from above.

The contact face holding member **303**, and contact faces **303a** through **303k** and **303m** held by the contact face holding member **303** are disposed on the inner side of the lens mount **301**, following the circumferential direction of the lens mount **301**, as illustrated in FIG. 9A. An array line on which the contact points are disposed is illustrated in FIG. 9A as an array line **3008**. As described above, the contact faces **303a** through **303k** and **303m** have the functions of the above-described respective terminals, and the part numerals by which the contact faces are denoted are the same as those of the terminals, to facilitate description.

As illustrated in FIG. 9B, out of the contact faces **303a** through **303k** and **303m**, the contact faces **303a**, **303b**, **303c**, and **303d** are lower in height in the direction parallel to the optical axis **3001** as compared to the other contact faces. In the present embodiment, the above-described configuration is realized by differing the amount of protrusion of the contact face holding member **303** in the optical axis direction.

According to the configuration described above, the contact faces **303a** through **303d** do not come into contact with the contact pins **203e** through **203k** and **203m**, when relatively rotating the first lens **100** as to the camera body **10** from the mounting start state to the mounting completed state. Even when relatively rotating the first lens **100** as to the camera body **10** from the mounting completed state to the mounting start state, the contact faces **303a** through **303d** do not come into contact with the contact pins **203e** through **203k** and **203m**. That is to say, the number of times of sliding between contact pins and contact faces can be reduced at both the camera mount **201** side and lens mount **301** side.

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The contact face holding member **303** has a first inclined guiding face **303n** and a second inclined guiding face **303p** for drawing corresponding contact pins toward the rear face direction (imaging sensor **11** side) when mounting the first lens **100** to the camera body **10**, as illustrated in FIG. 9B. According to this configuration, the contact pressure of the contact pins provided to the camera body **10** side as to the contact face holding member **303** gradually changes, whereby deformation and wear of the contact pins provided to the camera body **10** can be reduced.

(Operations of Attaching First Lens **100** to Camera Body **10**)

Next, the operations at the time of mounting the first lens **100** to the camera body **10** will be described with reference to FIG. 10.

FIG. 10 illustrates diagrams exemplarily describing a contact state between terminals on the camera body **10** and first lens **100** according to the embodiment of the present invention. FIG. 10A is a diagram for describing the contact state between each terminal in the mounting start state where mounting of the first lens **100** to the camera body **10** has been started. FIG. 10B is a diagram for describing the contact state between each terminal in the mounting completed state where mounting of the first lens **100** to the camera body **10** has been completed. In the state illustrated in FIG. 10A, the contact pins **203a** and **203b** at the camera mount **201** side in the optical axis direction overlap the contact faces **303a** and **303b** at the lens mount face B side. However, since the height of the contact holding member **203** and the contact face holding member **303** in the direction parallel with the optical axis is made to differ for each region, as described above, none of the contact pins and contact faces come into contact in the state illustrated in FIG. 10A.

The state illustrated in FIG. 10B is a state where the first lens **100** has been rotated by approximately 60° as to the camera body **10** from the state illustrated in FIG. 10A toward the direction indicated by the arrow (see FIG. 10A). That is to say, in the present embodiment, the relative rotational angle of the camera body **10** and first lens **100** from the mounting start state to the mounting completed state is approximately 60°. In the state illustrated in FIG. 10B, the lock pin **202** is in the state of fitting (locked) the lock groove **301z**.

Rotating the first lens **100** by approximately 60° as to the camera body **10** guides the first lens claw **301d** into the rear face side (imaging sensor **11** side) of the first camera claw **201a**, and these are engaged with each other in the optical axis direction. In this state, the second lens claw **301e** is guided into the rear face side (imaging sensor **11** side) of the second camera claw **201b**, and these are engaged with each other in the optical axis direction. In the is state, the third lens claw **301f** is guided into the rear face side (imaging sensor **11** side) of the third camera claw **201c**, and these are engaged with each other in the optical axis direction. Further, the positions of the lens claws provided to the first lens **100** are positioned by being biased toward the rear face side by the lens mount biasing member **206** provided to the camera mount **201** side, thereby coupling the camera body **10** and the first lens **100**.

The contact pins at the camera body **10** side are pressed toward the rear face side (imaging sensor **11** side) by the first guiding face **303n** and second guiding face **303p**. The contact pins of the camera body **10** side come into contact with corresponding places on the contact faces **303a** through **303k** and **303m** at the lens mount **301** side in a state of being pressed, and come into contact with the respectively corresponding contact faces in the mounting completed state. At

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this time, the contact pin **203m** provided to the camera mount **201** side and the contact face **303e** provided to the lens mount **301** firstly start coming into contact.

(Basic Configuration of Conversion Adapter)

Next, the basic configuration of a conversion adapter (adapter device) a camera accessory mountable to the camera mount **201** of the above-described camera body **10** will be described with reference to FIGS. 11 through 12.

FIG. 11 illustrates diagrams describing the first conversion adapter **40** mountable to the camera body **10**, and a second lens unit **50**. FIG. 11A illustrates an external perspective view of the second lens unit **50** having been mounted to the camera body **10** via the first conversion adapter **40**. FIG. 11B illustrates an external perspective view of a state where the camera body **10**, first conversion adapter **40**, and second lens unit **50** have each been detached. The second lens unit (hereinafter referred to as second lens) **50** has a lens mount **501** that is short in flange focal distance, but has the same mount diameter as the camera mount **201**, as to the camera body **10**. That is to say, the second lens **50** has the same mount diameter as the above-described first lens **100**, but unlike the first lens **100**, is a camera accessory not compatible with direct mounting to the camera body **10**.

FIG. 12 illustrates diagrams for describing the second conversion adapter **70** mountable to a camera body **60** and the first lens **100**. FIG. 12A illustrates an external perspective view of a state where the first lens **100** is mounted to the camera body **60** via the second conversion adapter **70**. FIG. 12B illustrates an external perspective view where the camera body **60**, second conversion adapter **70**, and first lens **100** have each been detached.

In a case where a lens unit with a long flange focal distance is directly mounted to an imaging apparatus with a short flange focal distance, trouble, such as the focal point not being formed at an accurate position, will occur, and imaging a subject in this way will result in an unnatural image being obtained. For example, the above problem will occur if the second lens unit **50** having a long flange focal distance is mounted to the above-described camera body **10**.

In the same way, in a case where a lens unit with a short flange focal distance is directly mounted to an imaging apparatus with a long flange focal distance, trouble, such as the focal point not being formed at an accurate position, will occur, and imaging a subject in this way will result in an unnatural image being obtained. For example, the above problem will occur if the first lens **100** having a short flange focal distance is mounted to the camera body **60** serving as a second imaging apparatus and having a longer flange focal distance than the above-described camera body **10**.

Particularly, the camera body **10** and camera body **60**, and the first lens **100** and second lens **50**, have the same mount diameters, and therefore it is difficult for a user to judge which imaging apparatuses and which lens units have flange focal distances that are compatible for direct mounting.

It is thereby preferable that only lens units that are compatible can be directly mounted to a certain imaging apparatus, so that imaging apparatuses and lens units that have mutually incompatible flange focal distances are not erroneously directly mounted.

In a case of mounting an incompatible lens unit to an imaging apparatus, a conversion adapter needs to be interposed between the two to adjust the flange focal distance. However, in a state where one side of the conversion adapter is mounted to the imaging apparatus, if a lens unit compatible with direct mounting to the imaging apparatus is mounted to the other side of the conversion adapter, the focal point may not be accurately formed, as described above.

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Thus, the one side and other side of the conversion adapter preferably have configurations to restrict imaging apparatuses and camera accessories that are directly mountable.

Specifically, in a case of mounting a lens unit having a long flange focal distance to an imaging apparatus having a short flange focal distance, a conversion adapter is preferable where one end side is only mountable to this imaging apparatus, and the other end side is only mountable to this lens unit. In a case of mounting a lens unit having a short flange focal distance to an imaging apparatus having a long flange focal distance, a conversion adapter is preferable where one end side is only mountable to this imaging apparatus, and the other end side is only mountable to this lens unit.

The first conversion adapter **40** has the lens mount **1301** attached to an adapter barrel **40a** by a plurality of fastening screws (not illustrated), at one end side in the optical axis direction. The plurality of fastening screws are the same as the lens mount fastening screws **304a** to **304d** in the lens mount **301** of the first lens unit **100** described above. This lens mount **1301** is an accessory mount attachable to and detachable from the camera mount **201** provided to the camera body **10** described above.

A camera mount **1401** is attached to the adapter barrel **40a** by fastening screws (not illustrated), at the other end side of the first conversion adapter **40** in the optical axis direction. This camera mount **1401** is a camera mount that is attachable to and detachable from the lens mount **501** of the second lens **50**. The camera mount **1401** of the first conversion adapter **40** is attached such that the imaging plane of the imaging sensor **11** of the camera body **10** will be situated at a position corresponding to the flange focal distance of the second lens **50**.

The second conversion adapter **70** has the lens mount **1501** attached to an adapter barrel **70a** by fastening screws (not illustrated), at one end side in the optical axis direction. This lens mount **1501** is an accessory mount that is attachable to and detachable from the camera mount **401** provided to the camera body **60**.

The second conversion adapter **70** has the camera mount **1201** attached to the adapter barrel **70a** by a plurality of fastening screws (not illustrated), at the other end side in the optical axis direction. The plurality of fastening screws are the same as the camera mount fastening screws **205a** to **205d** in the camera mount **201** of the camera body **10** described above. This camera mount **1201** is a camera mount that is attachable to and detachable from the lens mount **301** of the first lens **100**, as described above. As described above, the members having the same mount are the camera mount **201** of the camera body **10** and the camera mount **1201** of the second conversion adapter **70**, the lens mount **301** of the first lens **100** and the lens mount **1301** of the first conversion adapter **40**, the camera mount **401** of the camera body **60** and the camera mount **1401** of the first conversion adapter, and the lens mount **501** of the second lens **50** and the lens mount **1501** of the second conversion adapter **70**. In the following description, assume that each mount has the same configuration, and thus duplicated description will be omitted.

A first optical member **701a** and a second optical member **701b** are provided to the second conversion adapter **70**, between the adapter barrel **70a** and the lens mount **1501** in the optical axis direction. The first optical member **701a** and second optical member **701b** enable the second conversion adapter **70** to extend length of the flange focal distance of the first lens **100** in accordance with the imaging plane of the imaging sensor disposed in the camera body **60**. While the

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optical members have been illustrated as two lenses for the sake of convenience, this is not restrictive.

(Configuration of First Conversion Adapter **40**)

Next, the angles (phases) at which bayonet claws are disposed on the circumferential direction of the camera mount and lens mount of the first conversion adapter **40** will be described with reference to FIG. **13** through FIG. **17**. The description of the lens mount **1301** described below is synonymous with the description of the lens mount **301**. Further, the description of the camera mount **1401** described below is synonymous with the description of the camera mount **401**.

FIG. **13** illustrates diagrams for exemplarily describing displacement angles of bayonet claws in the camera mount **1401** provided on one end of the first conversion adapter **40**. FIG. **13A** is a diagram illustrating angle ranges that camera claws and camera recesses occupy in the circumferential direction of the camera mount **1401** with the lock pin **1401z** as a reference, as viewed from the rear face side (camera body **10** side). FIG. **13B** is a diagram illustrating angle ranges that multiple camera claws **1401a** through **1401c** occupy in the circumferential direction of the camera mount **1401**, as viewed from the rear face side (camera body **10** side). FIG. **13C** is a cross-sectional diagram taken along a cross-section E-E' illustrated in FIG. **13B**.

The first conversion adapter **40** is a mount adapter used for mounting the second lens unit **50** having a long flange focal distance to the camera body **10** having a short flange focal distance. Thus, it is preferable for the first conversion adapter **40** to be configured such that the camera body **60** having a long flange focal distance cannot be directly mounted to the lens mount **1301**, and the first lens **100** having a short flange focal distance cannot be directly mounted to the camera mount **1401**. According to this configuration, the positional relation of claws and recesses can be realized such that the lens mount **1301** provided to one end (first end) of the first conversion adapter **40** and the camera mount **1401** provided to the other end (second end) cannot each be directly mounted.

A first camera claw **1401a**, second camera claw **1401b**, and third camera claw **1401c**, are provided in order, to the camera mount **1401** in the circumferential direction (inner radial direction). When viewing the camera mount **1401** from the rear face side as illustrated in FIG. **13A**, the camera claw provided at a position farthest from the lock pin **1401z** is the first camera claw **1401a**. The second camera claw **1401b** and third camera claw **1401c** are consecutively provided in order from the first camera claw **1401a** in a clockwise direction.

A first camera recess **1401d**, second camera recess **1401e**, and third camera recess **1401f** are provided in order, to the camera mount **1401** in the circumferential direction (inner radial direction). When viewing the camera mount **1401** from the rear face side as illustrated in FIG. **13A**, the recess provided at a position nearest to the lock pin **1401z** is the second camera recess **1401e**. The third camera recess **1401f** and first camera recess **1401d** are consecutively provided in order from the second camera recess **1401e** in a clockwise direction.

As illustrated in FIG. **13C**, a fitting member **401x** that restricts movement of the camera accessory in a direction parallel to the optical axis when the camera accessory is mounted, is provided to the camera mount **1401** side. In the present embodiment, the diameter (i.e., the inner diameter of the fitting member **401x**) in a direction orthogonal to the

optical axis (approximately the same as the center axis) of the fitting member **401x** at the camera mount **401** side is the mount diameter.

The way of bayonet coupling of the first conversion adapter **40** and second lens **50** is approximately the same as the way of bayonet coupling of the camera body **10** and first lens **100** described above, and thus description will be omitted.

For the sake of description, each of the end portions of the camera claws **1401a**, **1401b**, and **1401c** in the circumferential direction will be referred to as a first end portion **1401a1**, second end portion **1401a2**, third end portion **1401b1**, fourth end portion **1401b2**, fifth end portion **1401c1**, and end portion **1401c2**. Each of the end portions are denoted with part numerals in order from the first camera claw **1401a** in the clockwise direction, when viewing the camera mount **1401** from the rear face side, as described above.

As illustrated in FIG. 13A, the angles (angle ranges) that the camera claws and camera recesses occupy in the circumferential direction of the camera mount **1401** in the first conversion adapter **40** according to the present embodiment are stipulated as follows. For the first camera claw **1401a**, the angle $\theta 1 = 56^\circ$. For the second camera claw **1401b**, the angle $\theta 2 = 62^\circ$. For the third camera claw **1401c**, the angle $\theta 3 = 62^\circ$. Further, for the first camera recess **1401d**, the angle is 57° . For the second camera recess **1401e**, the angle is 66° . For the third camera recess **1401f**, the angle is 57° . That is to say, the camera mount **1401** has different angles for the camera claws with respect to the above-described camera mount **201** of the camera body **10**, but the angles of the camera recesses are the same.

When viewing the camera claws from the rear face side of the first conversion adapter **40**, the angles where the camera claws are disposed on the circumferential direction of the camera mount **1401** with the position (referred to as reference position) of the lock pin **1401z** as a reference are stipulated as follows. The first camera claw **1401a** is disposed between 159° and 215° with the reference position as a start point. The second camera claw **1401b** is disposed between 272° and 334° with the reference position as a start point. The third camera claw **1401c** is disposed between 40° and 102° with the reference position as a start point.

FIG. 14 illustrates diagrams exemplarily describing angles of disposing the bayonet claws on the lens mount **1301** provided to the other end of the first conversion adapter **40**. FIG. 14A is a diagram illustrating angle ranges that lens claws and lens recesses occupy in the circumferential direction of the lens mount **1301** with the lock groove **1301z** as a reference, as viewed from the rear face side. FIG. 14B is a diagram illustrating angle ranges that multiple lens recesses **1301a** through **1301c** occupy in the circumferential direction of the lens mount **1301**, as viewed from the rear face side.

The angles (angle ranges) that the lens recesses occupy in the circumferential direction of the lens mount **1301** are, represented by $\theta 4$ as the angle of the first lens recess **1301a** and $\theta 5$ as the angle of the second lens recess **1301b**, as illustrated in FIG. 14B. The angle of disposing the claws and recesses on the lens mount **1301** of the first conversion adapter **40** is the same as the lens mount **301** of the first lens **100** described above, and therefore description will be omitted.

The angles of the claws and recesses in the circumferential direction, on the lens mount **1301** and camera mount **1401** provided to the first conversion adapter **40**, will be compared. For example, at the lens mount **1301** side, the angle $\theta 5$ of the second lens recess **1301b** having the smallest

angle is 44° . Whereas, at the camera mount **1401** side, the angle $\theta 1$ of first camera claw **1401a** having the smallest angle is 56° . That is to say, the claw having the smallest angle at the camera mount **1401** side is larger than the recess having the smallest angle at the lens mount **1301** side ($\theta 5 < \theta 1$). In this case, of the recesses on the lens mount **1301** side and the claws on the camera mount **1401** side, at least one or more lens recess and camera claws will interfere with each other. Thus, even if an attempt is made to mount the lens unit to the camera body such that the optical axes at the camera mount **1401** side and lens mount **1301** side are approximately parallel, the claws and recesses interfere, and thereby the lens unit cannot be mounted to the camera body.

However, if only one claw and recess are interfering, there may be cases where claws of the camera mount side can be inserted into recesses of the lens mount side. FIG. 15 illustrates diagrams exemplarily describing a mounting method of a predetermined imaging apparatus **1000** and a predetermined lens unit **2000** having claws and recesses that interfere with each other. FIG. 15A is a diagram exemplarily describing a frontal view of partway through mounting a predetermined lens unit to a predetermined imaging apparatus having claws and recesses that interfere with each other. FIG. 15B is a cross-sectional view taken along a cross-section F-F' illustrated in FIG. 15A.

For example, in a state where the optical axis of the lens mount side is obliquely inclined as to the optical axis of the camera mount side, one end of a camera claw is inserted into a lens recess, and from this state, the lens mount and camera mount are rotated relatively to each other, as illustrated in FIGS. 15A and 15B. In this case, even if the camera claws and lens recesses originally interfere with each other, this interference with each other is resolved during the relative rotation of the lens mount and camera mount, and transition can be made to a state where the camera claw is inserted into this lens recess. In this case, if there are no other camera claws and lens recesses interfering, the lens unit can be mounted to the camera body.

In the present embodiment, the claws and recesses on the lens mount **1301** side and camera mount **1401** side are disposed such that the angle of a predetermined lens recess adjacent to a reference lens claw is smaller than the angle of two camera claws adjacent to a reference camera recess.

Specifically, in the present embodiment, the angles $\theta 4$ and $\theta 5$ of the first lens recess **1301a** and second lens recess **1301b** adjacent to the first lens claw **1301d** are smaller than the angles $\theta 1$ and $\theta 2$ of the first camera claw **1401a** and second claw **1401b** adjacent to the first camera recess **1401d**. That is to say, $\theta 4$ (52°) is smaller than $\theta 1$ (56°), and $\theta 5$ (44°) is smaller than $\theta 2$ (62°) ($\theta 4 < \theta 1$, $\theta 5 < \theta 2$).

Thus, even if an attempt is made to insert the first lens claw **1301d** (**301d**) into the first camera recess **401d** (**1401d**), the second lens claw **1301e** (**301e**) interferes with the second camera claw **401b** (**1401b**) in a sure manner, as well does the third lens claw **1301f** (**301f**) with the first camera claw **401a** (**1401a**), as illustrated in FIG. 16.

FIG. 16 illustrates diagrams exemplarily describing the way in which claws interfere with each other when attempting to mount the lens mount **301** (**1301**) side to the camera mount **1401** (**401**) side according to the embodiment of the present invention. FIG. 16A illustrates the way in which the third lens claw **1301f** (**301f**) and the first camera claw **401a** (**1401a**) interfere. FIG. 16B illustrates the way in which the second lens claw **1301e** (**301e**) and the second camera claw **401b** (**1401b**) interfere.

As described above, incompatible lens units and imaging apparatuses, and the mount portions of conversion adapters

are configured such that two claws of each other out of the claws that an incompatible lens unit and imaging apparatus have interfere with each other, in the present embodiment. According to this configuration, the risk of an incompatible lens unit being erroneously mounted to an imaging apparatus, or an incompatible lens unit and imaging apparatus being erroneously mounted to mount units provided to both ends of a conversion adapter, can be reduced.

However, even in a case where the above conditions are satisfied, there may be cases where, when attempting to insert three claws into incompatible recesses, for example, two claws are inserted into recesses depending on the angles of the claws and recesses.

In the present embodiment further takes a configuration where the angles of at least two adjacent lens recesses are smaller than the angles of all camera claws. Specifically, in the present embodiment, the angles $\theta 4$ and $\theta 5$ of the first lens recess **1301a** and second lens recess **1301b** in the circumferential direction are smaller than the angles $\theta 1$ through $\theta 3$ of the first through third camera claws **1401a** through **1401c** in the circumferential direction. That is to say, the relation between the claws and recesses at the lens mount **1301** side and the camera mount **1401** side according to the present embodiment satisfy $\theta 4 < \theta 1$, $\theta 4 < \theta 2$, $\theta 4 < \theta 3$, $\theta 5 < \theta 1$, $\theta 5 < \theta 2$, and $\theta 5 < \theta 3$.

FIG. 17 illustrates diagrams exemplarily describing a case of attempting to insert incompatible claws into recesses at the lens mount **1301** side and camera mount **1401** side according to the embodiment of the present invention. FIG. 17A illustrates the way in which the third lens claw **1301f** (**301f**) and third camera claw **401c** (**1401c**) interfere in a case of attempting to insert the first lens claw **1301d** (**301d**) into the third camera recess **401f** (**1401f**). FIG. 17B illustrates the way in which second lens claw **1301e** (**301e**) and first camera claw **401a** (**1401a**) interfere in a case of attempting to insert the first lens claw **1301d** (**301d**) into the third camera recess **401f** (**1401f**). FIG. 17C illustrates the way in which the third lens claw **1301f** (**301f**) and second camera claw **401b** (**1401b**) interfere in a case of attempting to insert the first lens claw **1301d** (**301d**) into the second camera recess **401e** (**1401e**). FIG. 17D illustrates the way in which the second lens claw **1301e** (**301e**) and third camera claw **401c** (**1401c**) interfere in a case of attempting to insert the first lens claw **1301d** (**301d**) into the second camera recess **401e** (**1401e**).

As illustrated in FIGS. 17A through 17D, the lens claws and camera claws interfere in at least two places in the present embodiment, regardless of the relative rotational angle of the lens mount **1301** and camera mount **1401**. According to this configuration, erroneous mounting of a lens unit and conversion adapter having the lens mount **301** or the lens mount **1301** to an imaging apparatus and conversion adapter having the camera mount **401** or the camera mount **1401** can be prevented even more effectively. (Configuration of Second Conversion Adapter **70**)

Next, the angles (phases) at which bayonet claws are disposed in the circumferential direction of the camera mount and lens mount of the second conversion adapter **70** will be described with reference to FIGS. 18 through 21. The description of the camera mount **1201** described below is synonymous with the description of the camera mount **201**. The description of the lens mount **1501** described below is also synonymous with the description of the lens mount **501**.

The second conversion adapter **70** is a mount adapter used for mounting the first lens **100**, which has a short flange focal distance to the camera body **60** having a long flange focal distance. The lens mount **1501** is provided at one end (third

end) side of the second conversion adapter **70**, and the camera mount **1201** is provided at the other end (fourth end). The camera mount **1201** of the second conversion adapter **70** has the same configuration as the camera mount **201** of the camera body **10** the first imaging apparatus described above. The lens mount **1501** of the second conversion adapter **70** has the same configuration as the lens mount **501** of the second lens **50** described above.

It is therefore preferable for the second conversion adapter **70** to be configured such that the camera body **10** having a short flange focal distance cannot be directly mounted to the lens mount **1501**, and the second lens **50** having a long flange focal distance cannot be directly mounted to the camera mount **1201**. This configuration can be realized by satisfying the positional relation of claws and recesses such that the lens mount **1501** provided to one end (third end) of the second conversion adapter **70** and the camera mount **1201** provided to the other end (fourth end) each cannot be directly mounted.

FIG. 18 illustrates diagrams exemplarily describing angles of disposing bayonet claws in the camera mount **1201** provided on one end of the second conversion adapter **70**. FIG. 18A is a diagram illustrating angle ranges that camera claws and camera recesses occupy in the circumferential direction of the camera mount **1201** with the lock pin **202** as a reference, as viewed from the rear face side. FIG. 18B is a diagram illustrating angle ranges that, regarding multiple camera claws **1201a** through **1201c**, the claws occupy in the circumferential direction of the camera mount **1201**, as viewed from the rear face side. The angles of disposing the claws and recesses on the camera mount **1201** of the second conversion adapter **70** are the same as the camera mount **1201** of the camera body **10** described earlier, and thus description will be omitted.

As illustrated in FIG. 18B, the angle ranging from the sixth end **1201c2** of the third camera claw **1201c** to the third end **1201b1** of the second camera claw **1201b**, in the circumferential direction of the camera mount **1201** including the first camera claw **1201a**, is $\theta 6$ (162°). The angle ranging from the second end **1201a2** of the first camera claw **1201a** to the fifth end **1201c1** of the third camera claw **1201c**, in the circumferential direction of the camera mount **1201** including the second camera claw **1201b**, is $\theta 7$ (163°). The angle ranging from the fourth end **1201b2** of the second camera claw **1201b** to the first end **1201a1** of the first camera claw **1201a**, in the circumferential direction of the camera mount **1201** including the third camera claw **1201c**, is $\theta 8$ (215°).

FIG. 19 illustrates diagrams exemplarily describing angle of disposing bayonet claws in a lens mount **1501** provided on the other end of the second conversion adapter **70**. FIG. 19A is a diagram illustrating angle ranges that lens claws and lens recesses occupy in the circumferential direction of the lens mount **1501** with the lock pin **1501z** as a reference, as viewed from the rear face side. FIG. 19B is a diagram illustrating angle ranges where recesses are provided regarding the multiple lens recesses **1501a** through **1501c** in the circumferential direction of the lens mount **1501**, as viewed from the rear face side. FIG. 19C is a cross-sectional diagram taken along a cross-section G-G' illustrated in FIG. 19B.

A first lens claw **1501d**, second lens claw **1501e**, and third lens claw **1501f** are provided in order to the lens mount **1501** in the circumferential direction (inner radial direction). In a case of viewing the lens mount **1501** from the rear face side as illustrated in FIG. 19A, the lens claw provided at a position farthest from the lock groove **1501z** is the first lens

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claw **1501d**. The second lens claw **1501e** and third lens claw **1501f** are consecutively provided in order from the first lens claw **1501d** in a clockwise direction.

A first lens recess **1501a**, second lens recess **1501b**, and third lens recess **1501c** are provided in order, to the lens mount **1501** in the circumferential direction (inner radial direction). In a case of viewing the lens mount **1501** from the rear face side as illustrated in FIG. **19A**, the recess provided at a position nearest to the lock groove **1501z** is the third lens recess **1501c**. The first lens recess **1501a** and second lens recess **1501b** are consecutively provided in order from the third lens recess **1501c** in a clockwise direction.

For the sake of description, the end portions of the lens claws **1501d**, **1501e**, and **1501f**, in the circumferential direction, will be referred to as first end portion **1501d1**, second end portion **1501d2**, third end portion **1501e1**, fourth end portion **1501e2**, fifth end portion **1501f1**, and sixth end portion **1501f2**. The end portions are denoted with part numerals in order from the first lens claw **1501d** in the clockwise direction, when viewing the lens mount **1501** from the rear face side, as described above.

As illustrated in FIG. **19A**, the angles (angle ranges) that the lens claws and lens recesses occupy in the circumferential direction of the lens mount **1501** are stipulated as follows. The angle of the first lens claw **1501d** is 53° , the angle of the second lens claw **1501e** is 62° , and the angle of the third lens claw **1501f** is 53° . That is to say, the angles of the lens claws on the lens mount **1501** side are the same as the angles of the lens claws on the lens mount **1301** side described above.

In contrast, the angles of the lens recesses at the lens mount **1501** side differ from the angles of the lens recesses at the lens mount **1301** (301) side described above. Specifically, the angle of the first lens recess **1501a** is 60° , the angle of the second lens recess **1501b** is 66° , and the angle of the third lens recess **1501c** is 66° .

When viewing the lens claws from the rear face side, the angles in the clockwise direction where the lens claws are disposed on the circumferential direction of the lens mount **1501** with the position (referred to as reference position) of the lock groove **1501z** as a reference are stipulated as follows. The first lens claw **1501d** is disposed between 157° and 210° with the reference position as a start point. The second lens claw **1501e** is disposed between 276° and 338° with the reference position as a start point. The third lens claw **1501f** is disposed between 44° and 97° with the reference position as a start point.

As illustrated in FIG. **19C**, a fitting member **1501x**, which restricts movement of the imaging apparatus in a direction parallel to the optical axis when mounted to the imaging apparatus, is provided to the lens mount **1501** side. In the present embodiment, the diameter (i.e., the inner diameter of the fitting member **1501x**) in a direction orthogonal to the optical axis (approximately the same as the center axis) of the fitting member **1501x** at the lens mount **1501** side is the mount diameter. The mount diameters of the camera mounts and lens mounts described above are approximately the same length.

As illustrated in FIG. **19B**, the angle ranging from the fifth end **1501f1** of the third lens claw **1501f** to the second end **1501d2** of the first lens claw **1501d**, in the circumferential direction of the lens mount **1501** including the first lens recess **1501a**, is 09° (166°). The angle ranging from the first end **1501d1** of the first lens claw **1501d** to the fourth end **1501e2** of the second lens claw **1501e**, in the circumferential direction of the lens mount **1501** including the second lens recess **1501b**, is 010° (181°). The angle ranging from the third

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end **1501e1** of the second lens claw **1501e** to the sixth end **1501f2** of the third lens claw **1501f**, in the circumferential direction of the lens mount **1501** including the third lens recess **1501c**, is 011° (181°).

The angles of the claws and recesses in the circumferential direction will be compared between the camera mount **1201** and lens mount **1501** provided to the second conversion adapter **70**. For example, the angle (92°) of the third camera claw at the camera mount **1201** side is larger than the angles (66°) of the second and third lens recesses **1501b** and **1501c** that are the largest angle of the lens recesses at the lens mount **1501** side. That is to say, the angle of at least one camera claw at the camera mount **1201** side is larger than the angle of the lens recess having the largest angle at the lens mount **1501** side.

In this case, of the claws at the camera mount **1201** side and recesses at the lens mount **1501** side, at least one or more camera claws and lens recesses interfere with each other. Thus, even if an attempt is made to mount the lens unit to the camera body such that the optical axis at the camera mount **1201** side and lens mount **1501** side are approximately parallel to each other, the claws and recesses interfere, and therefore the lens unit cannot be mounted to the camera body.

However, if only one claw and recess are interfering, there may be cases where claws of the camera mount side can be inserted into recesses of the lens mount side, in the same way as the above description of the first conversion adapter. For example, there are cases where, in a state where the optical axis of the lens mount side is obliquely inclined as to the optical axis of the camera mount side, one end of a camera claw is inserted into a lens recess, and the lens mount and camera mount are rotated relative to each other, and the lens unit can be mounted to the camera body.

In the present embodiment, the claws and recesses at the camera mount **1201** side and the lens mount **1501** side are disposed to satisfy $06 < 09$ and $07 < 010$ in the circumferential direction. FIG. **20** illustrates diagrams exemplarily describing the way in which claws interfere with each other, when attempting to mount a reference claw at the lens mount **1501** (501) side to a reference recess at the camera mount **201** (1201) side according to the embodiment of the present invention. FIG. **20A** illustrates the way in which the third lens claw **1501f** (501f) and the third camera claw **201c** (1201c) interfere. FIG. **20B** illustrates the way in which the second lens claw **1501e** (501e) and the third camera claw **201c** (1201c) interfere.

As illustrated in FIGS. **20A** and **20B**, in a case where, for example, the reference first lens claw **1501d** (501d) is attempted to be inserted into the reference camera recess **201d** (1201d), the third camera claw **201c** (1201c) interferes with the second lens claw **1501e** (501e) and third lens claw **1501f** (501f).

That is to say, as viewed from the rear face side, the total sum of the angles of one reference lens claw, another lens claw adjacent thereto in the clockwise direction, and a lens recess situated between these lens claws, is taken as a first angle. As viewed from the rear face side, the total sum of the angles of one reference camera recess, another camera recess adjacent thereto in the clockwise direction, and a camera claw portion situated therebetween, is taken as a second angle. As viewed from the rear face side, the total sum of the angles of one reference lens claw, another lens claw adjacent thereto in the counterclockwise direction, and a lens recess situated between these lens claws, is taken as a third angle. As viewed from the rear face side, the total sum of the angles of one reference camera recess, another camera

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recess adjacent thereto in the counterclockwise direction, and a camera claw portion situated therebetween, is taken as a fourth angle. According to the present embodiment, it is sufficient to layout the claws and recesses at the camera mount **1201** side and lens mount **1501** side such that the first angle described above is larger than the second angle, and the third angle is larger than the fourth angle.

According to this configuration, at least two lens claws and one camera claw interfere with each other. Thus, the risk of an incompatible lens unit being erroneously mounted to an imaging apparatus, or an incompatible lens unit and imaging apparatus being erroneously mounted to mount units provided to both ends of a conversion adapter, can be reduced.

However, even in a case where, for example, the above conditions are satisfied, there may be cases where, when attempting to insert three claws into incompatible recesses two claws are inserted into recesses depending on the angles of the claws and recesses.

The claws and recesses at the camera mount **1201** side and lens mount **1501** side are therefore disposed in the circumferential direction so as to satisfy $06 < 09$, $06 < 010$, $06 < 011$, $07 < 09$, $07 < 010$, and $07 < 011$. That is to say, as viewed from the rear face side, the total sum of the angles of two claws other than the reference lens claw described above, and a lens recess situated therebetween, is taken as a fifth angle. According to the present embodiment, it is sufficient to layout the claws and recesses at the camera mount **1201** side and lens mount **1501** side such that the second angle and fourth angle described above are smaller than any of the first angle, third angle, and fifth angle.

FIG. 21 illustrates diagrams exemplarily describing the way in which claws interfere with each other, when attempting to mount a claw other than the reference claw at the lens mount **1501** (**501**) side to a reference recess at the camera mount **201** (**1201**) side according to the embodiment of the present invention. FIG. 21A illustrates the way in which the first lens claw **1501d** (**501d**) and the third camera claw **201c** (**1201c**) interfere in a case of attempting to insert the second lens claw **1501e** (**501e**) into the first camera recess **201d** (**1201d**). FIG. 21B illustrates the way in which the third lens claw **1501f** (**501f**) and the third camera claw **201c** (**1201c**) interfere in a case of attempting to insert the second lens claw **1501e** (**501e**) into the first camera recess **201d** (**1201d**). FIG. 21C illustrates the way in which the second lens claw **1501e** (**501e**) and the third camera claw **201c** (**1201c**) interfere in a case of attempting to insert the third lens claw **1501f** (**501f**) into the first camera recess **201d** (**1201d**). FIG. 21D illustrates the way in which the first lens claw **1501d** (**501d**) and the third camera claw **201c** (**1201c**) interfere in a case of attempting to insert the third lens claw **1501f** (**501f**) into the first camera recess **201d** (**1201d**).

As illustrated in FIGS. 21A through 21D, at least two lens claws and one camera claw interfere, regardless of the relative rotation angle of the lens mount **1501** and camera mount **1201**, according to the present embodiment. According to this configuration, erroneous mounting of a lens unit and conversion adapter having the lens mount **501** and the lens mount **501** to an imaging apparatus and conversion adapter having the camera mount **201** and the camera mount **1201** can be prevented even more effectively.

FIG. 22 is a diagram exemplarily describing a state in which claws provided to the camera mount **401** side and lens mount **501** side according to the embodiment of the present invention are engaged. FIG. 22 illustrates a state in which the camera mount **401** provided to the camera body **60** is engaged to the lens mount **501** provided to the second lens

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50. The first lens claw **501d** can be inserted into the first camera recess **401d**, as illustrated in FIG. 22. The second lens claw **501e** can be inserted into the second camera recess **401e**. Further, the third lens claw **501f** can be inserted into the third camera recess **401f**. That is to say, the camera mount **401** side and the lens mount **501** side are a combination compatible with being directly mounted to each other. The camera mount **201** side and the lens mount **301** side are a combination compatible with being directly mounted to each other, as illustrated in FIG. 10A.

As described above, the camera mount **401** (**1401**) corresponding to an imaging apparatus (e.g., the camera body **60**) having a long flange focal distance, and the lens mount **501** (**1501**) corresponding to a lens unit (e.g., the second lens **50**) having a long flange focal distance can be directly mounted to each other. However, the camera mount **201** (**1201**) corresponding to an imaging apparatus (e.g., the camera body **10**) having a short flange focal distance and the lens mount **501** (**1501**) corresponding to a lens unit (e.g., the second lens **50**) having a long flange focal distance cannot be directly mounted to each other. The camera mount **201** (**1201**) corresponding to an imaging apparatus (e.g., the camera body **10**) having a short flange focal distance and the lens mount **301** (**1301**) corresponding to a lens unit (e.g., the first lens **100**) having a short flange focal distance can be directly mounted to each other. However, the camera mount **401** (**1401**) corresponding to an imaging apparatus (e.g., the camera body **60**) having a long flange focal distance and the lens mount **301** (**1301**) corresponding to a lens unit (e.g., the first lens **100**) having a short flange focal distance cannot be directly mounted to each other.

An imaging apparatus and camera accessory employing the configuration of the above-described embodiment can therefore prevent erroneous mounting of an imaging apparatus and camera accessory having approximately the same mount diameter but are not mutually compatible.

Although a preferable embodiment of the present invention has been described, the present invention is not restricted to this embodiment, and various modifications and alterations can be made within the scope of the essence thereof. For example, a case of employing a digital camera has been described as an example of the imaging apparatus in the above-described embodiment, but imaging apparatuses other than digital cameras, such as digital video cameras, and security cameras, may be employed in the configuration.

Although a case of employing an interchangeable lens and an intermediate accessory, such as a conversion adapter, as an example of the camera accessory carrying out the present invention, has been described in the embodiment above, this is not restrictive. Any sort of camera accessory may be employed as long as it is a device having an accessory mount capable of directly connecting (coupling) to a camera mount provided to the above-described imaging apparatuses.

Although a configuration has been described in the above embodiment where both of the camera mount and accessory mount corresponding to each other have a pair of electric terminals through which electric connection is established in a mutually mounted state, this is not restrictive. For example, a configuration may be made having electric terminals that do not correspond to each other. For example, a configuration may be made where a terminal that the camera body **10** does not have is provided to a camera accessory mountable to the camera body **10**.

Although a configuration has been described in the above embodiment where the electric terminals are disposed in order along a circumferential direction where the diameter

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from the center of the camera mount and accessory mount is approximately the same, this is not restrictive. For example, a configuration may be made where the electric terminals are disposed along the circumferential direction at different diameters from the center of the camera mount and accessory mount. In this case, the dimensions of the mounts in the radial direction will increase, but friction between terminals can be suppressed.

Although a case has been described in the above embodiment where each mount has three claws in the circumferential direction, the number of claws is not restricted to three, as long as the conditions of the above-described embodiment are satisfied.

A configuration may be made where each of the three claws provided to the mounts in the above-described embodiment is further divided. That is to say, a configuration may be made where one claw is one claw group made up of multiple claws. A case where a groove is provided to one claw falls under this, for example. In this case, the ranges of angles of claw groups disposed in the circumferential direction of the mounts is the same as those in the embodiment described above.

Although a case has been described in the above embodiment where the mount diameter of each mount is the inner diameter of a fitting portion provided to each mount, this is not restrictive. For example, at each mount, an inner diameter of which the radius is the distance from the center (substantially the same as optical axis) of each mount to multiple electronic contacts (terminals) may be used as the mount diameter, or an inner diameter of which the radius is the distance from the center of each mount to a claw or recess may be used as the mount diameter.

Although a configuration has been described in the above embodiment where a device having one of a camera mount and an accessory mount is actually rotated as to a device having the other mount, thereby bayonet-coupling the devices to each other, this is not restrictive. For example, a configuration may be employed where a camera mount and accessory mount are relatively rotated, and the camera mount and accessory mount are bayonet-coupled. Specifics of this will be described in detail below.

FIG. 23 is a disassembled perspective view of a mount mechanism 5000 according to a modification of the present invention. FIG. 24 illustrates diagrams for exemplarily describing a non-coupled state of the mount mechanism 5000 according to the modification of the present invention. FIG. 25 illustrates diagrams for exemplarily describing a coupled state of the mount mechanism 5000 according to the modification of the present invention. In FIGS. 23 through 25, the lens mount capable of bayonet-coupling to a movable mount portion 5010 of the mount mechanism 5000 is also illustrated, for the same of description. Members that are the same as those in the above-described embodiment are denoted by the same reference numerals, and description thereof will be omitted.

As illustrated in FIG. 23, the mount mechanism 5000 according to the present embodiment has, in order from the side to which the lens mount is attached, an operation portion 5030, a fixed mount portion 5020, the movable mount portion 5010, and the contact holding member 203, centered on the optical axis 3000. The operation portion 5030 is a ring-shaped operating unit capable of rotating on a center axis, and is fixed to the movable mount portion 5010 by screws, by arm portions 5040. In the present modification, the operation portion 5030 and movable mount portion 5010 are fixed at two positions, using two arm portions 5040 disposed in an orthogonal direction as to the center axis.

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According to this configuration, by the operation portion 5030 being rotationally operated, the movable mount portion 5010 also rotates integrally about the center axis as the center.

Provided on the movable mount portion 5010 are movable mount claws 5011a, 5011b, and 5011c, that are each capable of bayonet-coupling with bayonet claws 301a through 301c provided to the lens mount 301. Also provided to the movable mount portion 5010 is a screw portion 5012 that has been threaded around the center axis, and the screwed state as to a below-described screw portion 5022 of the fixed mount portion 5020 changes in accordance with the rotation of the movable mount portion 5010 around the center axis.

The fixed mount portion 5020 has a camera mount face 5021 that comes into contact with the mount face of the lens mount, and the screw portion 5022 with which the screw portion 5012 of the above-described movable mount portion 5010 is screwed. Unlike the above-described movable mount portion 5010, the fixed mount portion 5020 does not rotate about the center axis in accordance with rotation operations of the operation portion 5030.

Next, the bayonet coupling method of the mount mechanism 5000 according to the present modification will be described with reference to FIGS. 24 and 25. The bayonet claws provided to the lens mount 301 are in a state of being capable to engage the movable mount claws 5011a through 5011c of the movable mount portion 5010, in a state of being passed through an opening portion of the operation portion 5030 and an opening portion of the fixed mount portion 5020. The state illustrated in FIGS. 24A through 24C is a state where the operation portion 5030 is situated at an unlocked position. In this state, the lens mount face of the lens mount and the camera mount face 5021 of the fixed mount portion 5020 are in contact, but the claws of each of the lens mount 301 and movable mount portion 5010 do not engage each other, and do not overlap, as viewed from the center axis direction. FIG. 24C is a cross-sectional diagram taken along a cross-section H-H' illustrated in FIG. 24B. FIG. 25 exemplifies the mount mechanism 5000 in a state where the operation portion 5030 has been rotationally operated from this state.

The state illustrated in FIGS. 25A through 25C is a state where the operation portion 5030 is situated in a locked position. In this state, the claws of each of the lens mount 301 and movable mount portion 5010 overlap each other, and thereby are engaged in the center axis direction. In this state, the screwing state of the screw portion 5022 of the fixed mount portion 5020 and the screw portion of the movable mount portion 5010 changes in accordance with rotational operation of the operation portion 5030, and the movable mount portion 5010 moves toward the imaging device side in the center axis direction. FIG. 25C is a cross-sectional diagram taken along a cross-section I-I' illustrated in FIG. 25B. As illustrated in FIGS. 24C and 25C, the movable mount portion 5010 moves away from the fixed mount portion 5020 in a center axis direction in accordance with a non-locked state of the mount mechanism 5000 changing to a locked state of the mount mechanism 5000. According to this configuration, each of the movable mount claws 5011a through 5011c in a state of being engaged with the bayonet claws at the lens mount 301 side moves toward the imaging apparatus side.

As described above, the mount mechanism 5000 according to the present modification can move the movable mount portion in the center axis direction as to the fixed mount portion, by rotating the movable mount portion that has claws capable of engaging lens-mount-side claws on the

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center axis. According to this configuration, the mount mechanism 5000 according to the present modification can reduce occurrence of gaps (looseness) occurring between the lens mount and camera-side mount in a coupled state.

Although a configuration has been described in the above modification where the mount mechanism 5000 is provided to the imaging apparatus side, this can be applied to a configuration where the mount mechanism 5000 is provided to a camera accessory side, such as an interchangeable lens.

According to the present invention, erroneous mounting of an imaging apparatus and an accessory having mounts that can be mutually engaged with each other can be prevented.

While the present invention has been described with reference to exemplary embodiments, it is to be understood that the invention is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

The invention claimed is:

1. An adapter device comprising:

a first mount disposed on one side of the adapter device, fastened to the adapter device by a plurality of first screws, and configured to be attachable to and detachable from an imaging apparatus; and

a second mount disposed on another side of the adapter device, fastened to the adapter device by a plurality of second screws, and configured to be attachable to and detachable from an accessory,

wherein a first mount recess, a second mount recess, a third mount recess, a first mount claw, a second mount claw, and a third mount claw are provided following a circumferential direction of the first mount, and the first, second, and third mount claws are insertable into imaging apparatus mount recesses provided to the imaging apparatus,

wherein a fourth mount claw, a fifth mount claw, a sixth mount claw, a fourth mount recess, a fifth mount recess, and a sixth mount recess are provided following a circumferential direction of the second mount, and the fourth, fifth, and sixth mount claws are insertable into accessory mount recesses provided to the accessory,

wherein a smallest angle range out of angle ranges in the circumferential direction of the first mount out of the first, second, and third mount recesses is equal to or smaller than a smallest angle range out of angle ranges in the circumferential direction of the second mount out of the fourth, fifth, and sixth mount claws,

wherein the first mount is configured to couple, by a bayonet coupling method, to a third mount on which a plurality of mount recesses and a plurality of mount claws corresponding respectively to the first, second, and third mount recesses and the first, second, and third mount claws are provided, whereby, with the imaging apparatus having the third mount, the adapter device can be attached to and detached from the imaging apparatus having the third mount and, in a state where the adapter device is mounted to the imaging apparatus, the first, second, and third mount recesses engage the plurality of mount claws provided to the third mount in a center axis direction of the first mount,

wherein the first mount has a restricting member configured to restrict, in a state where coupling of the first mount and the third mount by the bayonet coupling method has been completed, relative rotation of the first mount and the third mount in the circumferential direction of the first mount, and

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wherein, in a case of viewing the first mount from a side where the imaging apparatus is mounted, in a state where the first mount is mounted to the imaging apparatus, the first mount claw is disposed in a range of 159° to 212° from the restricting member in the circumferential direction of the first mount, the second mount claw is disposed in a range of 256° to 318° from the restricting member in the circumferential direction of the second mount claw, and the third mount claw is disposed in a range of 54° to 107° from the restricting member in the circumferential direction of the first mount.

2. The adapter device according to claim 1, wherein the first mount and the second mount have a plurality of terminals to enable electrical connection along each of the circumferential direction of the first mount and the circumferential direction of the second mount.

3. The adapter device according to claim 1, wherein the first mount and the second mount have substantially the same mount diameter.

4. The adapter device according to claim 1, wherein an angle range in the circumferential direction of the first mount of two mutually adjacent mount recesses out of the first, second, and third mount recesses provided to the first mount is equal to or smaller than an angle range in the circumferential direction of the second mount of two mutually adjacent mount claws out of the fourth, fifth, and sixth mount claws provided to the second mount.

5. The adapter device according to claim 1, wherein an angle range in the circumferential direction of the first mount of two mutually adjacent mount recesses out of the first, second, and third mount recesses provided to the first mount is equal to or smaller than an angle range in the circumferential direction of the second mount of all mount claws of the fourth, fifth, and sixth mount claws provided to the second mount.

6. The adapter device according to claim 1, wherein, by relative rotation of the first mount and the third mount in the circumferential direction of the first mount, the adapter device can be transitioned between a first state where the first, second, and third mount claws provided to the first mount are inserted into mount recesses corresponding to the plurality of mount recesses provided to the third mount and a second state where the first, second, and third mount claws are engaged with corresponding mount claws out of the plurality of mount claws provided to the third mount in an optical axis direction of the first mount such that the adapter device is mounted to the imaging apparatus.

7. The adapter device according to claim 1, wherein the accessory has a fourth mount and the second mount is configured to couple, by a bayonet coupling method, to the fourth mount on which a plurality of mount recesses and a plurality of mount claws corresponding respectively to the fourth, fifth, and sixth mount recesses and the fourth, fifth, and sixth mount claws are provided, whereby the adapter device can be attached to and detached from the accessory having the fourth mount and, in a state where the adapter device is mounted to the accessory, the fourth, fifth, and sixth mount recesses engage the plurality of mount claws provided to the fourth mount in a center axis direction of the second mount.

8. An adapter device comprising:

a first mount disposed on one side of the adapter device, fastened to the adapter device by a plurality of first screws, and configured to be attachable to and detachable from an imaging apparatus; and

a second mount disposed on another side of the adapter device, fastened to the adapter device by a plurality of second screws, and configured to be attachable to and detachable from an accessory,

wherein a first mount recess, a second mount recess, a third mount recess, a first mount claw, a second mount claw, and a third mount claw are provided following a circumferential direction of the first mount, and the first, second, and third mount claws are insertable into imaging apparatus mount recesses provided to the imaging apparatus,

wherein a fourth mount claw, a fifth mount claw, a sixth mount claw, a fourth mount recess, a fifth mount recess, and a sixth mount recess are provided following a circumferential direction of the second mount, and the fourth, fifth, and sixth mount claws are insertable into accessory mount recesses provided to the accessory,

wherein a largest angle range out of angle ranges in the circumferential direction of the first mount out of the first, second, and third mount recesses is equal to or smaller than a largest angle range out of angle ranges in the circumferential direction of the second mount out of the fourth, fifth, and sixth mount claws,

wherein (i) a total sum of angle ranges in the circumferential direction of the second mount, out of the fourth, fifth, and sixth mount claws and the fourth, fifth, and sixth mount recesses, one mount recess serving as a reference, an adjacent mount recess adjacent to the one mount recess, and a situated mount claw situated between the adjacent mount recess and the one mount recess, in a case of viewing the second mount from a center axis direction of the second mount, is smaller than (ii) a total sum of angle ranges in the circumferential direction of the first mount, out of the first, second, and third mount recesses and the first, second, and third mount claws, one mount claw serving as a reference on an accessory side, an adjacent mount claw adjacent to the one mount claw on the accessory side, and a situated mount recess situated between the one mount claw and the adjacent mount claw, in a case of viewing the first mount from the center axis direction of the second mount, and

wherein (iii) a total sum of angle ranges in a circumferential direction of a camera-side mount recess serving as a reference, another mount recess adjacent to the camera-side mount recess serving as a reference, and a situated mount claw situated between the camera-side mount recess and the other mount recess, in the case of viewing the second mount from the center axis direction of the second mount, is equal to or smaller than (iv) a total sum of angle ranges in a circumferential direction of an accessory reference mount claw serving as a reference on the accessory side, another mount claw adjacent to the accessory reference mount claw, and a situated mount recess situated between the accessory reference mount claw and the other mount claw.

9. The adapter device according to claim 8, wherein the first mount and the second mount have a plurality of terminals to enable electrical connection along each of the circumferential direction of the first mount and the circumferential direction of the second mount.

10. The adapter device according to claim 8, wherein the first mount and the second mount have substantially the same mount diameter.

11. The adapter device according to claim 8, wherein at least (i) (a) a total sum of angle ranges of the sixth mount recess, the fourth mount recess, and the fourth mount claw situated between the sixth mount recess and the fourth mount recess in the circumferential direction of the second mount, and (b) a total sum of angle ranges of the fourth mount recess, the fifth mount recess, and the fifth mount claw situated between the fourth mount recess and the fifth mount recess in the circumferential direction of the second mount in a case of viewing the second mount from a center axis direction of the second mount, are equal to or smaller than (ii) (a) a total sum of angle ranges of the third mount claw, the first mount claw, and the first mount recess situated between the third mount claw and the first mount claw in the circumferential direction of the first mount, (b) a total sum of angle ranges of the first mount claw, the second mount claw, and the second mount recess situated between the second mount claw and the second mount recess in the circumferential direction of the first mount, and (c) a total sum of angle ranges of the second mount claw, the third mount claw, and the third mount recess situated between the second mount claw and the third mount claw in the circumferential direction of the first mount in a case of viewing the first mount from a center axis direction of the first mount.

12. The adapter device according to claim 8, wherein the second mount is configured to couple, by a bayonet coupling method, to a fourth mount on which a plurality of mount recesses and a plurality of mount claws corresponding respectively to the fourth, fifth, and sixth mount recesses and the fourth, fifth, and sixth mount claws are provided, whereby, with the accessory having the fourth mount, the adapter device can be attached to and detached from the accessory having the fourth mount and, in a state where the adapter device is mounted to the accessory, the fourth, fifth, and sixth mount recesses engage the plurality of mount claws provided to the fourth mount in a center axis direction of the second mount.

13. The adapter device according to claim 12, wherein, by relative rotation of the second mount and the fourth mount in the circumferential direction of the second mount, the adapter device can be transitioned between a first state where the fourth, fifth, and sixth mount claws provided to the second mount are inserted into mount recesses corresponding to the plurality of mount recesses provided to the fourth mount and a second state where the fourth, fifth, and sixth mount claws are engaged with corresponding mount claws out of the plurality of mount claws provided to the fourth mount in the center axis direction of the second mount such that the adapter device is mounted to the accessory.

14. The adapter device according to claim 12, wherein the second mount has a restricting member configured to restrict, in a state where coupling of the second mount and the fourth mount by the bayonet coupling method has been completed, relative rotation of the second mount and the fourth mount in the circumferential direction of the second mount, and

wherein, in a case of viewing the second mount from a side to which the fourth mount is provided, the fourth mount claw is disposed in a range of 169° to 217° from the restricting member, the fifth mount claw is disposed in a range of 274° to 314° from the restricting member, and the sixth mount claw is disposed in a range of 20° to 112° from the restricting member.